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Part I

Sets with Structures

Chapter 1

Set Theory

1.1 Overview

Sets are our playgrounds. But plain sets are inaccessible and provide little insight. Interesting things happen when we define build rules and structures the sets.

When we want to talk about the fundamental concepts of analysis, such as convergence and continuity, we need to endow the sets with the topological structure: open sets need to specified. This class of open sets then allow us to describe the properties of a subset of a topological space, notably, compactness and connectedness.

If we further specify a metric on a space, then we can utilize the ordering and completeness of the real numbers. The metric is a generalisation of the idea of distance. Completeness and boundedness are central concepts of metric spaces.

We can impose algebraic structure on a set and set up groups, rings and fields, etc. To capture the idea of linearity, we can consider the vector space and linear transformations. The key concepts are bases and dimension.

In order to analyse a class of functions, it's useful to consider the linearity of functions and the magnitude, encoded in the concept of norm.

1.2 Sets and Mappings

1.2.1 Basic Set Theory

We follow the **Zermelo–Fraenkel** set theory with 6 axioms. One of the axiom is the **axiom of infinity** which states that there exists the natural number $\mathbb{N}$ and we can prove by recursion/induction.

The usual set operations are: union, intersection and complement. One of the important rules is the **de Morgan's Law** which states that:

$$X \setminus \bigcup_{i \in I} A_i = \bigcap_{i \in I} (X \setminus A_i)$$
$$X \setminus \bigcap_{i \in I} A_i = \bigcup_{i \in I} (X \setminus A_i)$$

Some commonly seen sets are: the power set $P(X) = \{\text{All subsets of } X\}$, the empty set.

An important notion is that of **product space**. The product of X_1 and X_2 is defined as the set of ordered pairs of elements from the two sets.

Product Space. A product space is a **Cartesian Product of sets** indexed by I, if I is infinite, it's the set of all mappings from I to X_i

$$X := \prod_{i \in I} X_i := \left\{ f : I \to \bigcup_{i \in I} X_i | f(i) \in X_i \right\}$$

1.2.2 Relations, equivalence class and quotient set

Let X be a set. A **relation** is a collection of subsets $R \subset P(X \times X)$. We say x and y satisfy the relation or $x \sim y \iff (x, y) \in R$

A relation is an **equivalent** relation if it's reflexive, transitive and symmetric. The equivalence class of x **modulo** (such equivalent relation) R is the set $\dot{x} = \{y \in X : y \sim x\}$.

Quotient set. The quotient set of X modulo R is $X/R := \{\dot{x} \mid x \in X\}$

1.2.3 Mappings

A mapping $f : X \to Y$ is a subset of $P(X \times Y)$ with the restriction that for each $x \in X$ there is one and only one $y \in Y$ such that $(x, y) \in f$.

There are some common notions with respect to mappings: surjective, injective, bijective; direct image, inverse image, inverse mapping(if f is bijective).

Notice that the inverse image f^{-1} preserves the set operations(union, intersection, complements), for example: $f^{-1}(B_1 \bigcup B_2) = f^{-1}(B_1) \bigcup f^{-1}(B_2)$ for a mapping $f : X \to Y$

1.2.4 Ordering on a set

Partial Ordering. A relation $\leq$ defined on a set X is a partial order, if it's antisymmetric($x \leq y$ and $y \leq x \implies x = y$), reflexive and transitive.

Totally Ordered. A partially ordered set is totally ordered if it satisfies the **comparability** that **for any** $a, b \in X$, $a \leq b$ or $b \leq a$.

Linear Ordering. $\forall x, y \in X$ either $x \leq y$, $y \leq x$ or $x = y$.

Well-Ordered. For all nonempty subset A of X there is $x \in A$ such that $\forall y \in A, x \leq y$.

For example, $[0, 1]$ is not well-ordered. Induction can be extended to well-ordered sets.

Chain. A chain is a linearly ordered subset of a partially ordered set.

Maximal. $z \in (X, \leq)$ is maximal, if there is no $w \in X$ such that $z < w$.

1.2.5 Section of a Set

Section. Let $E \subset X \times Y$, then fix $x_0 \in X$, the section of E at x_0 is

$$E_{x_0} := \{y \in Y : (x_0, y) \in E\}$$

If $f : X \times Y \to Z$ is a mapping, then the sections of f at x_0 is the mapping $h(y) := f(x_0, y) : Y \to Z$.

1.3 Axiom of Choice and Zorn's Lemma

1.3.1 Axiom of Choice. *If I is not empty, for any i, A_i is not empty, then $\prod_{i \in I} A_i$ is not empty.*

1.3.2 Zorn's Lemma. *Let $(P, \leq)$ be a partially ordered set, if for each chain $L \subset P$ there is an upper bound $y \in P$ such that $x \leq y, \forall x \in L$, then P has a maximal element.*

These axioms of choice are equivalent to **Hausdorff's Maximal Principle** and **Well-ordering Principle** (See Dudley 2018 and Folland). Acemoglu, Ozdaglar, and Tahbaz-Salehi 2015

1.4 Cardinality

1.4.1 Equivalence Theorem. *If A, B are two sets, and there is one-to-one functions: $f : A \to B$ and $g : B \to A$ then A, B have the same **cardinality**.*

1.4.2 Proposition (Countable union of countable sets). *Countable union of countable sets is still countable.*

Continuum. The set $2^{\mathbb{N}}, \mathbb{R}, [0, 1]$ has the cardinality of c, called the **continuum**.

Chapter 2

Point-Set Topology

2.1 Topological Space

2.1.1 Topology Definition

Topology. A topological space is $(X, \mathcal{T})$ where $\mathcal{T} \subset P(X)$ is a **topology** if it is:

1. Closed under set union: for any family of $(U_i)_{i \in I} \subset \mathcal{T}, \bigcup_{i \in I} U_i \in \mathcal{T}$
2. Closed under **finite** intersection;
3. $\varnothing, X \in \mathcal{T}$

Note. Topology and the open set system is what determines continuity and convergence.

Interior and Closure. For $A \subset (X, \mathcal{T})$,

$$\operatorname{int}(A) := \bigcup \{U \in \mathcal{T} : U \subset A\}$$

$$\bar{A} := \bigcap \{F \subset X : A \subset F, F \text{ is closed}\}$$

So the interior is the largest open set contained in A and the closure is the smallest closed set containing A.

Discrete and indiscrete topology. 2^X and $\{X, \varnothing\}$.

Limit Point. x is a limit point of A: for any neighbourhood U of x, $U \bigcap (A \setminus \{x\}) \neq \varnothing$. Notice that here $\{x\}$ is excluded.

Remark. A set $A \subset (X, \mathcal{T})$ is closed iff all the limit points of A are contained in A

Proof. If A is closed and x is a limit point of A, then if $x \in A^c$, A^c is an open set(thus a neighbourhood) that is disjoint with A, contradictory to x being a limit point. On the $\Leftarrow$ part, for any $x \in A^c$, x is not a limit point of A, then there is a neighbourhood $U \subset A^c$, thus A^c is open. □

Perfect Set. A subset $Y \subset (X, \mathcal{T})$ is a **perfect set** if every point of Y is a limit point of Y.

2.1.2 Examples of Topological Spaces

This section gives several examples of topological spaces.

Example. Let X be an **infinite** set, then define the open sets to be the any complement of finite subsets, then this is an example of non-Hausdorff topological space. Any sequence of distinct elements converge to every point; non-metrizable. (See Question)

2.1.3 Base, Neighbourhood Base, Dense and Separability

Base,Neighborhood Base,Subbase. We have the following definitions:

1. For a topological space $(X, \mathcal{T})$, a **base** for the topology is a subset $\mathcal{B} \subset \mathcal{T}$, such that for any $U \in \mathcal{T}$, $U = \bigcup\{B \in \mathcal{B} : B \subset U\}$.

2. A **neighbourhood base** $\mathcal{N} \subset \mathcal{T}$ of $x \in X$ is a collection of neighbourhoods of x, that for any neighbourhood V of x, $x \in N \subset V$ for some $N \in \mathcal{N}$

3. A collection of subsets $\mathcal{S}$ of X is a **subbase** for the topology $\mathcal{T}$ if finite intersection of sets in $\mathcal{S}$ form a base of $\mathcal{T}$.

2.1.1 Proposition (Base and Neighborhood Base). *A base contains a neighbourhood base for $\mathcal{T}$ for each $x \in X$.*

2.1.2 Proposition (Criterion for Base). *If $\mathcal{B} \subset \mathcal{P}(X)$, $\mathcal{B}$ is a base for a topology on X, iff*

1. *$\forall x \in X$, there exists $V \in \mathcal{B}$, $x \in V$.*

2. *For $U, V \in \mathcal{B}$ and $x \in U \bigcap V$, there exists $W \in \mathcal{B}$ such that $x \in W \subset (U \bigcap V)$.*

2.1.3 Proposition (Subbase and Generated Topology). *1. For any set X and $\mathcal{U} \subset 2^X$, there is a smallest topology $\mathcal{T}$ that includes $\mathcal{U}$, and $\mathcal{U}$ is a subbase of $\mathcal{T}$.*

2. Given a topology $\mathcal{T}$, $\mathcal{U} \subset \mathcal{T}$, $\mathcal{U}$ is a subbase iff $\mathcal{T}$ is the smallest topology including $\mathcal{U}$.

Proof. See Dudley 2018. *052402* □

Dense. A set $A \subset X$ is **dense** if $\bar{A} = X$. A is **nowherer dense** if $\left(\bar{A}\right)^o = \varnothing$.

Separable Topological Space. A topological space $(X, \mathcal{T})$ is **separable** if it has a countable dense subset.

First-Countable. A topological space $(X, \mathcal{T})$ is first-countable if for every point x there is a countable neighbourhood base of x.

Second-countable Topological Space. A topological space $(X, \mathcal{T})$ is **second-countable** if it has a countable base.

2.1.4 First Countability and Sequential Limits. *Closure and continuity in first countable space can be characterised by sequential convergence. If X is a first countable space and $A \subset X$, then $x \in \bar{A}$ iff there is a sequence $\{x_i\}$ in A that converges to x.*

2.1.5 Countability and Separability. *Every second countable space is separable.*

2.1.4 Separation Axioms in Topological Spaces

There are some separation axioms about topological spaces.

Separation Axioms. These properties guarantee the existence of open sets that separate points or closed sets from each other in a topological space $(X, \mathcal{T})$.

T_0: if $x \neq y$, then there is an open set containing x but not y *or* an open set that contains y but not x.

T_1: if $x \neq y$, then there is an open set that contains y but not x.

T_2: if $x \neq y$, then there are disjoint open sets U, V such that $x \in U$, $y \in V$. A T_2 space is called **Hausdorff**.

T_3: X is a T_1 space and for any closed set $A \subset X$, and any $x \in A^c$ there are disjoint open set U, V such that $x \in U$ and $A \subset V$. A T_3 space is called **regular**.

$T_{3\frac{1}{2}}$: X is T_1 and for each closed $A \subset X$ and $x \notin A$ there exists $f \in C(X, [0, 1])$ such that $f(x) = 1$ and $f = 0$ on A, also called **completely regular** or **Tychonoff** space,

T_4: X is a T_1 space and for any disjoint closed sets A, B in X there are disjoint open sets U, V such that $A \subset U$ and $B \subset V$. A T_4 space is called **normal**.

Note. Hausdorff condition guarantees the uniqueness of lim of sequences in a topological space.

2.1.6 Proposition (T_1). *X is a T_1 space iff $\{x\}$ is closed for every $x \in X$.*

Corollary 2.1.6.1. *Every normal space is regular and every regular space is Hausdorff.*

2.1.7 Proposition (Normal spaces). *Normal spaces include many interesting spaces that we are interested in :*

1. *Metric spaces are normal*
2. *Compact Hausdorff spaces are normal.*

Proof. To show that a metric space (X, d) is normal, i.e., for any disjoint closed subsets $A, B \subset X$, there are two disjoint open sets U, V that cover each of them.

Fix any $a \in A$, let $B(a, e_a)$ be the open ball with radius $e_a = \frac{1}{3}d(a, B)$. Then let U be $\bigcup_{a \in A} B(a, e_a)$, similarly, $V := \bigcup_{b \in B} B(b, e_b)$.

First, we know $A \subset U$ and $B \subset V$, U, V are open sets. Next, we show that $U \bigcap V = \varnothing$.

Assume $x \in U \bigcap V$, then there exists $a \in A$ and $b \in B$ such that $d(x, a) < e_a$ and $d(x, b) < e_b$.

$$d(a, b) < e_a + e_b = \frac{1}{3}d(a, B) + \frac{1}{3}d(b, A)$$

by triangle inequality, but then

$$d(a, b) < \frac{1}{3}d(a, b) + \frac{1}{3}d(b, a)$$

a contradiction. So $U \bigcap V = \varnothing$.

To show that a compact Hausdorff space $(X, \mathcal{T})$ is normal. Let A, B be disjoint closed subset of X. First we show that the space is **regular**. For a fixed point a in A and any $b \in B$, $a \neq b$ implies there are two disjoint open sets U_b^a, V_b^a such that $a \in U_b^a$ and $b \in V_b^a$. Then V_b^a is an open cover of a closed subset of a compact space, then it's compact as well. So there is finite subcover $(V_{b_i}^a)$, take $V^a := \bigcup_{i=1}^n V_{b_i}^a$ and $U^a := \bigcap_{i=1}^n U_{b_i}^a$, then both are open sets and disjoint. We can separate a point and a closed set.

Now let $(U^a)_{a \in A}$ is an open cover of A, again there is a subcover. Similarly we can find disjoint open cover of A and B □

There is a very useful result of **regular spaces**.

2.1.8 Proposition (Closure of open subset in Regular spaces). *The following statements are equivalent:*

1. *X is a **regular space**, i.e, for any $x \in X$ and a closed subset $F \subset X$ if $x \notin F$ then there are disjoint open sets U, V that separate x, F.*

2. *For any nonempty open set $U \subset \mathcal{T}$ and $x \in U$ there is an open set V such that $x \in V \subset U$ and $V \subset \overline{V} \subset U$.*

Proof. 051401 □

2.1.5 Subspace and Product spaces

Subspace and Relative Topology. Let $(X, \mathcal{T})$ be a topological space, $Y \subset X$, then $\{U \bigcap Y : U \in \mathcal{T}\}$ is called the **relative topology** on the subspace.

Weak Topology. Let $\{f_i : X \to (Y_i, \mathcal{O}_i), i \in I\}$ be a family of maps, the unique **weak topology** $\mathcal{T}$ is the smallest topology that makes all f_i continuous.

Product topology. Let $(X_i, \mathcal{T}_i)$ be topological spaces for $i \in I$, and $X = \prod_i X_i$. The product topology is the smallest topology such that every coordinate projections $p_i : (x_1, x_2, \ldots, x_n) \to x_i$ are continuous.

2.1.9 Proposition (Openness in Product Space). *If I is infinite, then $\prod_{i \in I} U_i$ where $U_i \neq \varnothing$ is open iff $U_i = X_i$ for all but finitely many i.*

2.1.10 Proposition (Topology of Pointwise Convergence). *For $(X, \mathcal{T})$ and a nonempty set A, $\{f_n\}$ is a sequence in X^A, then $f_n \to f$ in the product topology iff $f_n \to f$ pointwise.*

Some questions will help understand the product topology: see Dudley 2018.

Question 2.1.1. *To prove the following statements.*

1. *Product topology of finitely many spaces with discrete topology is also discrete.*
2. *Infinitely many discrete S_i, each having more than one point, the product topology is not discrete.*
3. *Product of countably many separable topological spaces, with product topology, is separable.*

2.2 Mappings on Topological Spaces

2.2.1 Sequence, Nets and Filters

Note. Nets and filters describe "the path of convergence". A sequence is a special case of nets. Nets and filters in topological spaces are like sequences iin metric spaces

Directed set, Nets. A directed set is a set I with partial ordering $\leq$ such that for any i, j in I, there is $k \in I$ such that $i \leq k$ and $j \leq k$.

A **net** is a mapping defined on a directed set.

Examples of nets: Unconditional series convergence; Riemann Integral (see Dudley 2018 page 29)

2.2.1 Proposition (Riemann Integral as a Net). *Consider the Riemann integral on $[a, b]$, then let*

$$I := \left\{(x_i, y_i)_{i=0}^n : n \in \mathbb{N}, a = x_0 < x_1 < \ldots, x_n = b\right\}$$

*with the partial ordering $\leq$ such that $(x_i, y_i)_n \leq (w_j, z_j)_m$ if $n \leq m$ and every x_i is also in $\{w_1, \ldots, w_m\}$, so (w_j) is a **refinement** of (x_i). Define the Riemann integral as $R : I \to \mathbb{R}$, if the following net converges to a real number.*

$$R(x_i, y_i) = \sum_{i=1}^{n} f(y_i)(x_i - x_{i-1})$$

Filter base and filter. A non-empty collection $\mathcal{F}$ of non-empty subsets of X is a **filter base**, if for any $F, G \in \mathcal{F}$, there is an $H \in \mathcal{F}$ such that $H \subset F \bigcap G$. $\mathcal{G}$ is a **filter** if it's a filter base such that for any $F \in \mathcal{G}$, if $F \subset H \subset X$ then $H \in \mathcal{G}$.

Convergence of nets. A net $(x_i)_{i \in I}$ converges to x iff for every neighborhood A of x there is a $j \in I$ such that $\forall k \geq j, x_k \in A$

Convergence of Filter Base. A filter base $\mathcal{F}$ is said to converge to $x \in X$ if any neighborhood of x belongs to the filter $\mathcal{G} := \{G \subset X : \exists F \in \mathcal{F}, F \subset G\}$

2.2.2 Proposition (Characterisation of continuity). *A mapping $f : (X, \mathcal{T}) \to (Y, \mathcal{U})$ is continuous iff for any nets $(x_i)_{i \in I} \to x$ in X, $f(x_i) \to f(x)$ in Y iff for any convergent filter base $\mathcal{F} \to x$ in X, $f[[\mathcal{F}]] \to f(x)$ in Y.*

2.2.3 Proposition (Characterisation of open and closed set). *For a topological space $(X, \mathcal{T})$,*

1. *A set $F \subset X$ is closed if and only if for all nets with $x_i \in F$, $x_i \to x$ in X implies that $x \in F$.*
2. *A set $U \subset X$ is open if and only if for all $x \in U$ and net $x_i \to x$ in X, there is $j \in I$, for all$i \geq j$, $x_i \in U$.*
3. *In metrizable topological spaces, the "nets" can be replaces with sequences.*

Ultrafilter. A filter $\mathcal{F}$ in a set X is called an **ultrafilter** iff for all $Y \subset X$ either $Y \in \mathcal{F}$ or $Y^c \in \mathcal{F}$.

2.2.4 Ultrafilter and Compactness. *A topological space is compact iff every ultrafilter in it converges.*

2.2.2 Continuity in Topological Spaces

Continuity. A mapping $f \in C[(X, \mathcal{T}_X), (Y, \mathcal{T}_Y)]$ if for every $V \in \mathcal{O}_Y$, $f^{-1}(V) \in \mathcal{O}_X$.

Continuity at a point. A mapping $f : (X, \mathcal{T}_X) \to (Y, \mathcal{T}_Y)$ is continuous at point $x \in X$ if for every neighbourhood $\mathcal{N}$ of $f(x)$, $f^{-1}(\mathcal{N})$ is a neighbourhood of x. It's continuous iff it's continuous at every point.

Homeomorphism and Embedding. A homeomorphism of X onto Y is a **bijective mapping** $f : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$, such that $f \in C((X, \mathcal{O}_X), (Y, \mathcal{O}_Y))$ and $f^{-1} \in C((Y, \mathcal{O}_Y), (X, \mathcal{O}_X))$.

An injective but not surjective $f : X \to Y$ is an **embedding** if $f : (X, \mathcal{O}_X) \to \left(f(X), \mathcal{O}_{Y|f(X)}\right)$ is a homeomorphism.

Note. If two topological spaces are homeomorphic then their open set systems are equivalent. So they have the same topological features.

Continuity is determined by the behaviour of mapping on the base and subbase.

2.2.5 Proposition (Subbase determines Continuity). *A mapping $f : (X, \mathcal{T}) \to Y, \mathcal{O}$ is continuous iff for any member U of a subbase $\mathcal{U}$ of $\mathcal{O}$, $f^{-1}(U) \in \mathcal{T}$.*

Proof. Let $\mathcal{U}$ be a subbase of $\mathcal{O}$, then for any open set $V \in \mathcal{O}$,

$$V = \bigcup \left\{ A \subset Y : A = \bigcap_{i=1}^{n} U_i, U_i \in \mathcal{U}, A \subset V \right\}$$

Then it follows because the inverse image f^{-1} preserves set operations. □

2.2.6 Proposition (Continuity in Product Space). *If $X_i, i \in I$ and Y are topological, then $f : Y \to \prod_i X_i$ is continuous iff $\pi_i \circ f : Y \to X_i$ is continuous for all i.*

2.2.3 Extension of Continuous Functions

Even in a regular space, the only continuous function may be the constant function. But in normal space, there must be nonconstant continuous functions.

2.2.7 Urysohn's Lemma. *Let X be a **normal space**, then if A, B are disjoint **closed** sets in X, then there exists $f \in C(X, [0, 1])$ such that $f = 0$ on A and $f = 1$ on B.*

The Tietze-Urysohn Extension Theorem states that under conditions, a continuous function can be extended from a closed subset to the whole topological space.

2.2.8 Tietze Extension Theorem. *Let X be a normal space, if $A \subset X$ is closed and $f \in C(A, [a, b])$ there exists $F \in C(X, [a, b])$ such that $F|A = f$.*

2.3 Compactness

2.3.1 Compactness in Topological Spaces

Open Cover. A collection of open sets $(A_i)_{i\in I} \subset \mathcal{T}$ is an open cover for $K \subset X$ if $K \subset \bigcup A_i$.

Compactness in Topological space (Heine-Borel). A topological space $(X, \mathcal{T})$ is compact if any open cover of X has a finite subcover.

Compactness of subsets. A subset $K \subset (X, \mathcal{T})$ is compact, if it's compact with the relative topology.

Relatively Compact or Precompact. A subset G of a topological space X is **relatively compact** or **precompact** if the closure of G is compact.

Note.

1. Compact is a concept that works for topological space itself as well as the subsets.
2. Compactness is the "second best thing" to finiteness. Continuity + Compactness gives most of the results we would expect from finiteness.
3. In $\mathbb{R}$ compactness implies boundedness.

2.3.1 Proposition (Closed real intervals are compact). *$[a, b] \subset \mathbb{R}$ is compact.*

Proof. See Dudley 2018. □

2.3.2 Proposition (Compact and Closed Sets). *If topological space $(X, \mathcal{T})$ is compact, then any closed subset F of X is also compact.*

Proof. Let (A_i) be an open cover of F, then $\bigcup_i (A_i) \bigcup F^c$ constitute an open cover of X, which then has a subcover $A_1, \ldots, A_n$ of X, then excluding F^c, it's a finite subcover of F. □

Note. Compactness and Hausdorff condition together usually give very good results.

2.3.3 Proposition (Compact and Hausdorff). *Any compact set K in a Hausdorff space X is closed.*

Proof. Consider any point $y \in X \setminus K$. For every x there are open sets U_x, V_x that separates x, y, $\{U_x : x \in K\}$ is an open cover of K and hence has a finite subcover $(U_{x_1}, U_{x_2}, \ldots, U_{x_n})$, corresponding to $(V_{x_1}, V_{x_2}, \ldots, V_{x_n})$, then $\bigcap_{i=1}^n V_{x_i}$ is open, contains y and is disjoint from K. □

On Compactness's relationship with other concepts: Terry's Blog

There is an important theorem about the compactness of product space.

2.3.4 Tychonoff's Theorem. *If $(X_i, \mathcal{T}_i)$ are all compact topological spaces for all i then $\prod_i X_i$ with the product topology is also compact.*

Proof. Use ultrafilter to show. □

2.3.2 Mappings on Compact sets

2.3.5 Compact and Continuity. *If topological space $(X, \mathcal{T})$ is compact, and f is continuous from X **onto** $(Y, \mathcal{U})$, then Y is compact.*

Proof. For any open cover $\mathcal{A}$ of Y, $f^{-1}(\mathcal{A})$ is an open cover of X and there is a finite subcover $(A_i)_{i=1}^n$ such that $X \subset \bigcup_{i=1}^n A_i$ and because f is onto, $f(X) = Y$. So we have found a finite subcover of Y. □

2.3.6 Inverse of Continuous Functions. *Suppose f is a **continuous, 1-1** mapping of a compact topological space X **onto** a topological space Y, then the inverse mapping f^{-1} is a continuous mapping of Y **onto** X.*

2.3.3 Locally Compact Hausdorff Space

2.3.4 Compactness Theorems

2.4 Connectedness

2.4.1 Connected and Simple Connected

These notions have simple characterisation in normed vector spaces.

Connected Set. A topological space $(X, \mathcal{T})$ is connected if it's not a union of two disjoint open sets.

2.4.1 Proposition (Continuity Preserves Connectedness). *Let $f : (X, \mathcal{T}) \to (Y, \mathcal{U})$, if f is* ***continuous*** *and* ***onto*** *and X is connected, then Y is connected.*

Chapter 3

Metric Structure

3.1 Metric Space

3.1.1 Metric

Metric spaces are perhaps the most relevant to studying statistics and econometrics. Metric is the generalisation of the concept "distance".

Metric. A metric is a mapping $d : X \times X \to \mathbb{R}$ such that the following conditions hold:

1. $d(x, y) \geq 0$, and $d(x, y) = 0 \iff x = y$
2. $d(x, y) = d(y, x)$
3. The triangle inequality holds: $d(x, y) + d(y, z) \geq d(x, z)$

(X, d) is called a metric space.

Note. Metric space is more special than topological spaces in the following sense: a metric can always induce a topology and the topological space is Hausdorff and normal.

3.1.1 Proposition (Metric induces Topology). ***Open balls*** $B(x, r) := \{y \in X : d(x, y) < r\}$ *in a metric space* (X, d)*, forms a base of a topology on* X*.* $\{B(x_0, r)\}$ *is a neighbourhood base of* x_0*.*

Proof. Let $\mathcal{T} = \{\bigcup B(x_i, r_i) : x \in X, r > 0\}$. Fix any $r > 0$, $X \subset \bigcup_x B(x, r)$. For any $U_i \in \mathcal{T}, i \in I$, $\bigcup U_i \in \mathcal{T}$. $\bigcup_i U_i \bigcap \bigcup_j V_j = \bigcup_j (\bigcup_i U_i \bigcap V_j) = \bigcup_j \bigcup_i (U_i \bigcap V_j)$. For

any $x \in U_i \bigcap V_j$, $d(x, x_i) < r_i$ and $d(x, x_j) < r_j$. We can find a small $\varepsilon > 0$ such that $B(x, \varepsilon) \subset U_i \bigcap V_j$ and hence it's a union of open balls, thus is an open set. □

3.1.2 Compactness in Metric Space

3.1.2 Characterisation of Compact Metric Space. *A metric space (X, d), TFAE*

(A) (X, d) is compact: every open cover has finite subcover,

(B) (X, d) is complete and totally bounded,

(C) Every infinite subset of X has a limit point,

(D) Every sequence of points of X has a convergent subsequence.

Proof. we follow the procedure: $A \implies B \implies C \implies D \implies A$

(A) $\implies$ (B). Let (X, d) be a compact metric space, for any $\varepsilon > 0$, $\{B(x, \varepsilon) : x \in X\}$ is an open cover of X and hence has finite open cover and so X is totally bounded.

Let $(x_n)_{n=1}^{\infty}$ be a Cauchy sequence. We want to show that it converges to some point in X. For any m, there is a $n(m)$ such that $d(x_n, x_{n(m)} < \frac{1}{m})$ for all $n \geq n(m)$. Then let $U_m := \left\{x : d(x, x_{n(m)} < \frac{1}{m})\right\}$. U_m are open(easy to see). Now we want to show that U_m can't be an open cover of X, if it is, then there is a finite subcover $\left\{U_{m_1}, U_{m_2}, \ldots, U_{m_s}\right\}$ such that $X \subset \bigcup_{i=1}^{s} U_{m_i}$, but apparently, for $n > \max\{n(m_i)\}$, $x_n \notin \bigcup_i U_{m_i}$, so there is $x \in X$ such that $x \notin \bigcup_m U_m$. It's easy to see that $x = \lim x_n$ □

(B) $\implies$ (C). Let (X, d) be totally bounded and complete. Because it's totally bounded, for any $n > 0$ there is a **finite** subset F_n of X such that $\left\{B(y, \frac{1}{n}) : y \in F_n\right\}$ covers X.

Let A be an infinite subset of X, then we know there is $y_1 \in F_1$, $A \bigcap B(y_1, 1)$ is infinite. Similarly, there is $y_2, \ldots, y_m$ such that $A \bigcap (\bigcap_{i=1}^{m} B(y_m, \frac{1}{m}))$ is infinite. The sequence (y_n) is Cauchy(there is some point x in $B(y_m, \frac{1}{m}) \bigcap B(y_n, \frac{1}{n})$, so $d(y_m, y_n) < \frac{1}{m} + \frac{1}{n}$) and by completeness, converges to a point $y \in X$.

It's a limit point of A because for any $\varepsilon > 0$, there is a ball $B(y, \varepsilon)$ that includes some $B(y_n, \frac{1}{n})$ which contains infinite points of A. □

(C) $\implies$ (D). Let (x_n) be a sequence of points of X, $A = \{x_1, \ldots\}$ is finite, then it must converges(there is a x such that $x_n = x$ for infinitely many n). If A is infinite, then it has a limit point a. Then there is a $x_{n_i} \in B(a, \frac{1}{i}) \bigcap A$, and x_{n_i} converges to a. □

(D) $\implies$ (A). Let $\mathcal{U}$ be an open cover of X, and let

$$f(x) := \sup\{r : B(x,r) \subset U, \quad \text{for some} \quad U \in \mathcal{U}\}, \quad \text{for all} \quad x \in X$$

Then we can prove the following lemma:

3.1.3 Lemma. $\inf\{f(x) : x \in X\} > 0$

Proof. If $\inf\{f(x)\} \leq 0$, then for any $n = 1, 2, \ldots$, there is x_n such that $f(x_n) < \frac{1}{n}$. Then it's a sequence and has a convergent subsequence $x_{n(k)} \to x$, $k \to \infty$. First notice for x, there is a $r > 0$ such that $B(x,r) \subset U_0$ for some $U_0 \in \mathcal{U}$, because $\mathcal{U}$ is an open cover. Then we can find k_1 such that $d(x_{n(k)}, x) < \frac{r}{2}$ for all $k \geq k_1$ and k_2 such that $\frac{1}{n_{k_2}} < \frac{r}{2}$. Take $k_3 = \max\{k_1, k_2\}$, then for $k \geq k_3$, we must have $B(x_{n(k)}, \frac{r}{2}) \subset U_0$ and $f(x_{n(k)}) \geq \frac{r}{2}$ which is contradictory. □

Let $c := \min\{2, \inf\{f(x)\}\} > 0$, choose any $x_1 \in X$, choose x_{n+1} if possible such that $d(x_{n+1}, x_j) > \frac{c}{2}$, for all $j = 1, 2, \ldots, n$. If we can do that then it's a sequence with no convergent subsequence. So there must be a finite n such that $X = \bigcup_{j \leq n} B(x_j, \frac{c}{2})$.

By the definition of f, there must be a $U_j \in \mathcal{U}$ such that $B(x_j, \frac{c}{2}) \subset U_j$, then we have found a finite subcover for X. □

Q.E.D □

3.1.4 Intersection of a sequence of Compact Subsets. *Let X be a metric space and K_n be a sequence of compact subsets of X such that $K_n \supset K_{n+1}$ and*

$$\lim_{n\to\infty} \operatorname{diam} K_n = 0$$

then there is $x \in X$ such that $\bigcap_{n=1}^{\infty} K_n = \{x\}$.

3.1.3 Separability of Metric Space

3.1.5 Countability and Separability Condition. *A metric space is **separable** iff it is **second-countable**, that is, if it has a countable base.*

Proof. $\implies$, Let $A \subset (S, d)$ be dense and countable, then we can show that

$$B = \left\{B(y, \frac{1}{n}) : y \in A, n = 1, 2, \ldots\right\}$$

is countable and is a base for $\mathcal{T}$. Because for any open set U in $\mathcal{T}$, there is m such that $B(y, \frac{1}{m}) \subset U$ and by density we can always find x such that $y \in B(x, \frac{1}{2m}) \subset B(y, \frac{1}{m}) \subset U$

$$U = \bigcup \left\{ B(x, \frac{1}{2m}) : y \in U, B(y, \frac{1}{m}) \subset U, d(x, y) < \frac{1}{2m}, x \in A \right\}$$

$\Longleftarrow$, let B be a countable base, $B = \{B_1, B_2, \ldots\}$, then let $y_i \in B_i$, we can show that $A = \{y_1, y_2, \ldots\}$ is countable and dense, because for any $x \in S$, $B(x, r) = \bigcup_{i \in I} B_i$, so there is some $y_i \in B(x, r)$. □

3.1.4 Boundedness in Metric Spaces

Bounded. A subset A of a metric space (X, d) is bounded if its **diameter** $\sup_{x,y \in A} d(x, y) < M < \infty$.

Totally Bounded. A metric space (X, d) is totally bounded iff for any $\varepsilon > 0$ there is a finite set $F \subset X$ such that $d(F, x) < \varepsilon$ for any $x \in X$.

3.1.6 Proposition. *Totally bounded implies bounded. For a counterexample consider a space (X, d) such that $d(x, y) = \mathbf{1}\{x \neq y\}$.*

Description of Total Boundedness. There are some descriptions of how "totally bounded" a metric space is. Let (X, d) be a totally bounded metric space, given $\varepsilon > 0$

1. $N(\varepsilon, X)$ is the smallest n such that $S = \bigcup_{1 \leq i \leq n} A_i$ for some sets A_i with diameters $\operatorname{diam} A_i \leq 2\varepsilon$

2. $D(\varepsilon, X)$ is the largest number m of points $x_i : i = 1, 2, \ldots, m$ such that $d(x_i, x_j) > \varepsilon$ whenever $i \neq j$.

3.2 Complete Metric Space

3.2.1 Completeness in Metric Space

Cauchy Sequence. In a metric space (X, d), a sequence $(x_n)_{n=1}^{\infty}$ is Cauchy if

$$\limsup_{m \geq n} d(x_m, x_n) = 0$$

Complete Metric Space. A metric space (X, d) is complete if every Cauchy sequence converges to a point in X.

3.2.1 Proposition (Closed subsets). *Closed subsets of a complete metric space is complete.*

Proof. Let A be a closed subset of a complete metric space (X, d), for any Cauchy sequence $(a_i)_{i=1}^{\infty}$, $a_i \in A$, it's also a Cauchy sequence in X and hence converges to $a \in X$. Because A is closed, a is a limit point of A, we know $a \in A$ as well. □

3.2.2 Baire's Category Theorem

3.2.2 Baire's Category Theorem. *Let X be a complete metric space. Then the following two equivalent properties hold:*

1. *Let $(F_i)_{i=0}^{\infty}$ be a sequence of closed subsets of X such that $intF_i = \varnothing$ for all $i \geq 0$, then $int(\bigcup_i F_i) = \varnothing$*

2. *Let $(O_i)_{i=0}^{\infty}$ be a sequence of open subsets of X such that $\bar{A}_i = X$ for all $i \geq 0$, then $\bigcap^{-} A_i = X$.*

Proof. First, we show that the two statements are equivalent.

Then we show the first statement is true: 1. Use Cantor's theorem, to show that the there is a point in the countable intersection of closed sets in a complete metric space. 2. To construct such sequence and prove that for any open set, its intersection with complement is nonempty.

Notice that a characterisation of set F that $\text{int}(F) = \varnothing$ in a metric space X is that for any open set $G \subset X$, $G \bigcap F^c \neq \varnothing$ □

3.2.3 Proposition (Cantor's Intersection Theorem). *Let X be a complete metric space, $(A_n)_{n=1}^{\infty}$ be a sequence of* ***nonempty, closed*** *subsets of X such that*

$$A_0 \supset A_1 \supset \ldots, \text{ and } diam A_n \to 0$$

then $\exists x \in X$ such that

$$\{x\} = \bigcap_{n=0}^{\infty} A_n$$

Proof. Pick x_n for each A_n which constitutes a Cauchy sequence, because $\limsup d(x_n, x_m) = 0$, then we know there is $x := \lim x_n \in X$.

Then we show $x \in \bigcap_{n=0}^{\infty} A_n$ and that it's unique. □

A useful corollary from the Baire's Category Theorem:

Corollary 3.2.3.1. *Let X be a metric space, F_n be closed subsets of X and $X = \bigcup_{i=1}^{n} F_n$*

1. *if* $\operatorname{int}(F_n) = \varnothing$ *for all n , then X is* ***not complete****.*
2. *If X is complete, then there is a n_0 s.t.* $\operatorname{int}(F_{n_0}) \neq \varnothing$.

3.3 Mappings on Metric Spaces

3.3.1 Limit of Functions

Limit of Function in Metric Space. Let $f : E \subset (X, d) \to (Y, e)$, let p be a limit point of E and define $\lim_{x\to p} f(x) = q$ if there is a $q \in Y$ such that for any $\varepsilon > 0$, there is a δ, for all $x \in E$, if $0 < d(x, p) < \delta$, then $e(f(x), q) < \varepsilon$.

3.3.1 Proposition (Limit and Sequential Limits). $\lim_{x\to p} f = q$ *iff for every sequence* (x_n) *that converges to p,* $\lim_{n\to\infty} f(x_n) \to q$.

3.3.2 Continuity, Uniform Continuity and Equicontinuity

Continuity. In a metric space, there is a $\varepsilon - \delta$ description of continuity and convergence, i.e., a function $f : (X, d_X) \to (Y, d_Y)$ is continuous at x_0 iff for any $\varepsilon > 0$, there is a $\delta(x_0, \varepsilon)$ such that for any $x \in B(x_0, \delta)$, $f(x) \in B(f(x_0), \varepsilon)$. It's called continuous if it's continuous at every point of X

If it's possible to choose the δ independent of x_0 that is $\delta = \delta(\varepsilon)$, we say the function is uniformly continuous.

Uniform Continuity. A function $f : (X, d_X) \to (Y, d_Y)$ is uniformly continuous iff for any $\varepsilon > 0$, there is a $\delta(\varepsilon)$ such that for any x_1, x_2, $d_X(x_1, x_2) < \delta \implies d_Y(f(x_1), f(x_2)) < \varepsilon$.

While continuity and uniform continuity are about a specific function, the following definitions apply to a family/class of functions.

Equicontinuity. Let $\mathcal{F}$ be a class of functions from a **topological space** S to a metric space (X, d), it is equicontinuous at x if for every $\varepsilon > 0$ there is a neighborhood U of x such that $d(f(x), f(y)) < \varepsilon$ for all $y \in U$ and $f \in \mathcal{F}$.

Notice how U depends on x, again we can make the statement about uniformaity.

Uniform Equicontinuity. If in the above definition, S is a **metric space** with metric e then, $\mathcal{F}$ is uniformly equicontinuous if $\forall \varepsilon > 0$, $\exists \delta > 0$ such that $e(x, y) < \delta$ implies $d(f(x), f(y)) < \varepsilon$ for all $x, y \in S$ and $f \in \mathcal{F}$.

3.3.2 Compactness and Uniform Equicontinuity. *If (K, d) is a **compact** metric space and (Y, e) is a metric space, then any equicontinuous family of functions from K into Y is uniformly equicontinuous.*

First Proof. Let $\mathcal{F} := \{f : K \to Y\}$ be an equicontinuous family of functions. For any $x \in K$ and $\varepsilon > 0$, there is $\delta(x) > 0$ such that $d(y, x) < \delta(x)$ implies $e(f(y), f(x)) < \frac{1}{2}\varepsilon$, for any $y \in K$ and all $f \in \mathcal{F}$.

$\{B(x, \delta(x)) : x \in K\}$ is an open cover of K and hence has a finite subcover

$$\left\{B(x_1, \frac{1}{2}\delta(x_1)), B(x_2, \frac{1}{2}\delta(x_2)), \ldots, B(x_n, \frac{1}{2}\delta(x_n))\right\}$$

Let $\delta = \frac{1}{2}\min\{\delta_1, \delta_2, \delta_3, \ldots, \delta_n\}$, then for any $x, y \in K$ such that $d(x, y) < \delta$, because $x \in B(x_i, \frac{1}{2}\delta_i)$ for some i, $d(x_i, y) \leq d(x_i, x) + d(x, y) \leq \delta(x_i)$, so

$$e(f(x), f(y)) \leq e(f(x), f(x_i)) + e(f(y), f_{x_i}) < \frac{1}{2}\varepsilon + \frac{1}{2}\varepsilon = \varepsilon$$

for any $f \in \mathcal{F}$. □

Second Proof. Dudley 2018 □

Corollary 3.3.2.1 (Compactness and Uniform Continuity)**.** *A continuous function from a **compact** metric space to any metric space is uniformly continuous.*

3.3.3 Isometry

Isometry. Let (X, d) and (Y, e) be two metric space, then a **bijective** mapping $f : X \to Y$ is an **isometry** if it preserves distance:

$$d(f(x), f(y)) = d(x, y), \quad \text{for all} \quad x, y \in X$$

3.3.4 Boundedness of Mappings

Bounded Functions. A function f from an arbitrary set S to a metric space (X, d) is bounded if its direct image $f(S)$ is bounded in (X, d).

Uniform Boundedness. The class of function

$$\mathcal{F} := \{f_i : S \to (X, d), i \in I\}$$

is uniformly bounded if $\bigcup_{i \in I} f(S) \subset X$ is bounded. That is if there is $x \in X$ and $M < \infty$ such that

$$d(f_i(s), x) \leq M, \quad \forall i \in I, \forall s \in S$$

3.3.5 Banach's Fixed Point Theorem; Contraction Principle

Fixed Point. For any $f : X \to X$, a fixed point of f is $x \in X$ such that $f(x) = x$.

Contraction. Let (X, d) be a metric space, the a mapping $f : X \to X$ is a contraction if there is a constant $k \in (0, 1)$ such that:

$$d(f(x), f(y)) \leq kd(x, y), \quad \text{for all} \quad x, y \in X$$

3.3.3 Banach Fixed Point Theorem. *Let (X, d) be a complete metric space, then*

1. *any **contraction** $f : X \to X$ has one and only one **fixed point** $x \in X$.*
2. *given any point $x_0 \in X$, the sequence $(x_i)_{i=1}^{\infty}$ defined by*

$$x_{i+1} = f(x_n), \quad n \geq 0$$

converges to x and the estimate holds:

$$\|x_n - x\| \leq Ck^n, \quad n \geq 0, \quad \textit{with} \quad C := \frac{d(f(x_0), x)}{1-k}$$

Proof. □

Chapter 4

Finite Dimensional Vector Space

4.1 Vector Space

4.1.1 Subspaces

4.1.2 Basis and Dimension

4.2 Linear Transformation

4.2.1 Linear Transformation and Matrix Representation $TA = BM$

Linear transformations are those transformations/mappings that preserve the "linearity" of the vector spaces.

Linear transformation. A linear transformation between two vector spaces is a mapping $T : U \to V$, such that

1. $T(x + y) = T(x) + T(y)$
2. $T(ax) = aT(x)$, for $a \in K$

The set of all linear transformations is denoted $\mathcal{L}(U, V)$

Linear Operator. A linear transformation $T : U \to U$ is called a linear operator.

Kernel and Range. Let $T : U \to V$ be a linear transformation, the **null space** or **kernel** $N(T) := \{x \in U : Tx = 0_V\}$. The **range** or the **column space** of T is

$R(T) := \{T(x) : x \in U\}$. The kernel and range are subspaces of U and V respectively, the dimension of $N(T)$ and $R(T)$ are called nullity and rank.

Linear transformations are determined by their action on the basis.

4.2.1 Dimension Theorem. *If U is n-dimensional, V is m-dimensional vector spaces and $T : U \to V$ is linear, then*

$$\dim N(T) + \operatorname{rank}(T) = \dim(V)$$

If we represent T as a matrix $A : m \times n$, then

$$\operatorname{rank}(A) = \dim C(A) = \dim C(A^T)$$
$$\dim N(A) = n - r \quad \textit{and} \quad \dim N(A^T) = m - r$$

where $C(A)$ is the column space of A. Also $N(A) \perp C(A^T)$.

Following this dimension theorem we can characterise the "one-to-one" and "onto" properties of linear transformations with the rank and nullity.

4.2.2 Nullity and Rank Condition.

1. *A linear transformation $T : U \to V$ is injective if and only if* $\dim N(T) = 0$.
2. *If furthermore,* $\dim(U) = \dim(V)$ *then T is injective $\iff$ T is surjective $\iff$ rank$(T) = \dim(U)$*

Invertible Linear Transformation. A linear transformation $T : U \to V$ is invertible if there exists another linear transformation $\hat{T} : V \to U$ such that $T \circ \hat{T} = I_V; \hat{T} \circ T = I_U$. Correspondingly, the matrices $A, \hat{A}$ will satisfy $A\hat{A} = \hat{A}A = I$. In that case we write $\hat{A} = A^{-1}$.

4.2.3 Proposition (Calculation of Inverse matrix). *Write $(A|I)$, do the same elementary row/column operations on both sides and turn it into $I|B$, if it's possible, then $B = A^{-1}$.*

Isomorphism. Two vector spaces are isomorphic to each other(it's a equivalence relation) if there is an invertible linear transformation $f : U \to V$.

4.2.4 Dimension condition of Isomorphism. *Two finite-dim vector spaces are isomorphic if and only if they are of the same dimension.*

For finite-dimensional vector spaces, this is due to the fact that the Linear transformation if uniquely determined by its effect on the bases.

4.2.5 Matrix ~ Linear transformations. *Let U, V be vector spaces over a field $\mathbb{F}$, the linear transformations $\mathcal{L}(U, V)$ is isomorphic to $\mathcal{M}_{m\times n}(\mathbb{F})$*

4.2.6 Matrix Representaion. *Let A, B be basis matrix of U, V, then there exists a unique matrix determined by T and the bases, $M = M(T, A, B)$ such that*

$$T(A) = BM$$

Column Space and Null Space. The **column space** $C(M)$ is the space of all linear combinations of the columns vectors of $M_{m\times n}$, which is a subspace of F^m. The **null space** $N(M)$ is the space of all vectors x such that $Mx = 0$, which is a subspace of F^n.

As an example, let T be a linear operator, then for two bases of U, the matrices will be **similar**.

Similar matrix. $A, B \in \mathcal{M}_{n\times n}$ are similar (an equivalence relation) if there is an invertible matrix Q such that

$$B = Q^{-1}AQ$$

4.2.2 Invertible Linear Transformation and Isomorphisms

4.2.3 Linear Functional and Dual Spaces

Linear Functional. Linear transformation $f : V \to K$ where V is a vector space over a field K(which is also a vector space) are called **linear functional**.

Dual Space. For a vector space V over K, the dual space $V^* := \mathcal{L}(V, K)$.

4.2.7 Proposition (Dimension of dual space).

$$\dim(V) = \dim(V^*).$$

Proof. Assume $\dim(V) = n$, then we can define $f_i \in V^*, i = 1, 2, \ldots, n$ as $f_i(e_j) = \delta_{ij}$. Then we prove that (f_i) is a basis for V_*. □

4.3 Linear Operator in Inner Product Spaces

4.3.1 Normal and Self-adjoint Operator

4.3.2 Spectral Theorems

4.3.1 Proposition (Normal Operator and Norm). *If T is normal linear operator, then*

$$\|Tx\| = \|T^*x\| \quad \textit{for all} \quad x \in X$$

4.3.2 Complex Spectral Theorem. *Let $T \in \mathcal{L}(X, \mathbb{C})$ be a **normal** linear operator, that is $TT^* = T^*T$, then there exists an orthonormal basis of X consisting of eigenvectors of T.*

4.3.3 Real Spectral Theorem. *Let $T \in \mathcal{L}(X, \mathbb{R})$ be a **self-adjoint** linear operator, then there exists an orthonormal basis of X consisting of eigenvectors of T.*

Remark. Notice that a normal linear operator in a real vector space may not have an eigenvalue to begin with, for example the rotation:

$$T = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Proof. First show that for any $T \in \mathcal{L}(X, \mathbb{C})$ or self-adjoint $T \in \mathcal{L}(X, \mathbb{R})$, there exists one eigenvalue λ. Use the fundamental theorem of algebra.

Then show there exists an upper-triangular representation of T induction, when dimension is 1, it's true. Show that for complex case, $T|_{R(T-\lambda I)}$ and for real case $T|_{N(T-\lambda I)^\perp}$ are still complex linear operator and self-adjoint linear real operator.

Finally, use normality to show that off-diagonal elements can't be nonzero otherwise $\|Te_j\| \neq \|T^*e_j\|$. □

4.3.3 Positive Semidefinite Operator, Isometry and SVD

Positive Semidefinite Operator. A linear operator is p.s.d. if it's self-adjoint and for any $v \in X$:

$$\langle Tv|v\rangle \geq 0$$

4.3.4 Proposition (Properties of PSD Operator). *T is p.s.d iff T is self adjoint with nonnegative real eigenvalues iff there exists a p.s.d. square root R of T such that $R^2 = T$.*

Isometry. An linear operator S is isometry if $\|Sv\| = \|v\|$ for all v.

Remark. It's also called orthogonal when $\mathbb{F} = \mathbb{R}$ and unitary when over $\mathbb{C}$

4.3.5 Proposition (Properties of Isometries). *S is an isometry iff the matrix of S wrt any o.n.b. is orthonormal iff $S^*S = SS^* = I$.*

4.3.6 Polar Decomposition. *For any linear operator T, there exists isometry S such that:*

$$T = S\sqrt{T^*T}$$

Notice that T^*T is always p.s.d.

Singular Values. The eigenvalues of $\sqrt{T^*T}$ are called singular values.

4.3.7 SVD. *We have the following decomposition from the Polar Decomposition:*

$$T(E) = (ES)D = FD \quad or \quad M_T = FDE^{-1}$$

where E is an o.n.b, F is also an o.n.b because S is an isometry.

q

4.4 Linear Operator in General Vector Spaces

Chapter 5

Old Linear Algebraic Structure: Vector Space

5.1 Vector Space

5.1.1 Vector Space, Subspaces Direct Sum

Linear algebra is about the properties of vector spaces, linear transformations and subspaces. It should be viewed from this angle instead of the stacking of numbers in a matrix form.

Vector space. A set X is a vector space over a field K if we impose the algebraic operations: $+ : X \times X \to X$ and $\cdot : K \times X \to X$ (scalar multiplication) that satisfy the following conditions, for all $x, y \in X$ and $a \in K$

1. Addition is commutative: $x + y = y + x$
2. Addition is associative: $(x + y) + z = x + (y + z)$
3. There is **zero vector** $0 \in X$ such that $x + 0 = x$
4. There is **additive inverse** $y \in X$ for every x such that $x + y = 0$
5. Scalar multiplied by $1 \in K$: $1 \cdot x = x$
6. Associativity of scalar multiplication: $(ab)x = a(bx)$

7. Distribution law 1: $a(x + y) = ax + ay$

8. Distribution law 2: $(a + b)x = ax + bx$

Subspace. A subspace Y of vector space X is a subset of X such that it's also a vector space under the $+$ and $\cdot$ defined on X.

5.1.1 Proposition (Verification of Subspace). *Let V be a vector space and $W \subset V$. W is a subspace of V if the following conditions hold:*

1. $0 \in W$

2. $x + y \in W$ *for all* $x, y \in W$

3. $cx \in W$ *for all* $x \in W$ *and* $c \in K$

Remark. The union of two subspaces is not generally a subspace, but an intersection of two subspaces is always a subspace.

Direct sum. A vector space $V = W_1 \oplus W_2$ if W_1 and W_2 are subspaces of V such that $W_1 \bigcap W_2 = 0$ and $W_1 + W_2 = V$.

5.1.2 Proposition (Dimension). *Let W_1, W_2 be subspaces of V, then* $\dim W_1 + \dim W_2 = \dim(W_1 \bigcap W_2) + \dim(W_1 + W_2)$.

Proof. Let W_1, W_2 be the bases matrix, then $W_1 \bigcap W_2 = N\left(\begin{bmatrix} W_1 & W_2 \end{bmatrix}\right)$. □

5.1.2 Linear Dependence, Bases and Dimensionality

Linear Combinition. Let V be a vector space over K and $S \subset V$, then $v \in V$ is a **linear combinition** of vectors of S if there is a **finite** number of vectors $s_1, s_2, \ldots, s_n \in S$ and $a_1, a_2, \ldots, a_n \in K$ such that $v = \sum_{i=1}^{n} a_i s_i$.

Span. Let $S \subset V$ and S is not empty, then $\operatorname{span}(S)$ is the set of all linear combinitions of vectors of S. If $\operatorname{span}(S) = V$ we say that S **generates** V.

Basis. A basis of a vector space V is a **linearly independent** subset that generates V.

5.1.3 Characterisation of Basis. *If B is a basis of n-dimensional vector space V, then for any $v \in V$, there are unique coefficients $a = (a_1, ..., a_n)^\top$ such that*

$$v = \sum_{i=1}^{n} a_i b_i = Ba$$

where $b_i \in B$ for all i.

5.1.4 Replacement Theorem. *Let V be a vector space. Let $G \subset V$ has n vectors and* $\operatorname{span} G = V$. *Let L be a linearly independent subset of V with m vectors. Then we have*

1. *$m \leq n$*
2. *There is a $H \subset G$ with $n - m$ vectors such that $L \bigcup H$ generates V.*

Note. Relationship between **Linear independent** subsets, **Generating subsets** and **Bases**: Generating subsets may not be linearly independent; linearly independent subsets may not span V; bases do both.

5.1.3 Maximal Independent Sets- Hamel Basis

This part is an instance of the Axiom of Choice. Where we assert that every vector space has a base.

5.1.4 Convexity

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Convex subsets of vector space. Let X be a vector space over $\mathbb{R}$, a subset A of X is convex if for any $a, b \in A$, the **segment** $[a, b] := \{x \in X : x = \lambda a + (1 - \lambda)b\} \subset A$.

Convex Hull. The **convex hull** $\operatorname{co}(A)$ of a subset A of a vector space X is the smallest convex subset that includes A.

Closed Convex Hull. $\overline{\operatorname{co}}A$ is the closed convex hull of a subset A which is the smalles **closed** convex subset of X that includes A.

5.1.5 Proposition (Characterisation of $\mathrm{co}(A)$).

$$\mathrm{co}(A) = \left\{ \sum_{i \in I} \lambda_i a_i : I \textit{ finite }, \lambda_i \geq 0, a_i \in A, \sum_{i \in I} \lambda_i = 1 \right\}$$

Example (n-simplex)**.** The convex hull of $n+1$ affinely independent points(they are not in a hyperplane) in $\mathbb{R}^n$ is a n-simplex.

5.1.6 Proposition (Finiteness and Compactness). *Let A be a finite subset of a normed vector space X, then $\mathrm{co}(A)$ is **compact**.*

Proof. We can consider the convergence of sequences. □

5.1.7 Proposition ($\overline{\mathrm{co}}A$ and $\overline{\mathrm{co}\,A}$). *In a normed vector space $\overline{\mathrm{co}}A = \overline{\mathrm{co}\,A}$*

5.1.5 Hyperplane

See Brezis 2010 is a clearer reference than Ciarlet 2013.

Affine Hyperplane.

5.2 Linear Transformations on Vector Spaces

5.2.1 Linear Operator and Invariant Subspaces

Invariant Subspace and Eigenvalue. An **invariant subspace** of the a linear operator T over U is a subspace $V \neq 0$ such that $Tv \in V$ for all $v \in V$. V_1 is an **one-dimensional** invariant subspace of T iff there exists a scalar λ called the **eigenvalue** such that $Tv = \lambda v$ for all $v \in V_1$.

Diagonalisation. If T over n-dimensional U has n invariant subspaces $U_1, \ldots, U_n$ such that $U = U_1 \oplus \cdots \oplus U_n$, then there exists a basis for U such that $TU = UD$ where D is diagonal matrix containing the eigenvalues. So a matrix A is diagonalisable if it is similar to a diagonal matrix, i.e., $A = P^{-1}DP$ where P is invertible.

5.2.2 Bilinear and Quadratic Forms

Chapter 6

Real and Complex Matrices

6.1 Special Matrices

6.1.1 Matrix of Special Features

Let $A := (a_{ij})_{n\times m}$ be a matrix of either real or complex elements. Then there are some matrices that have special shapes.

Special Matrices.

1. If $a_{ij} = a_{ji}$ then A is **symmetric**.
2. If $a_{ij} = \bar{a}_{ji}$ where $\bar{a}$ is the complex conjugate, then A is **Hermitian** or **self-adjoint**.
3. If $n = m$, A is **square**.

Diagonal. For $A = (a_{ij})_{n\times m}$, the vector of all diagonal elements a_{ii} is the **(principal) diagonal**. The vector of $a_{i,m+1-i}$ is the **skew diagonal**.

Diagonal Dominant Matrix. A is row-diagonal dominant if

$$|a_{ii}| > \sum_{j\neq i}^{m} |a_{ij}|, \quad i = 1, 2, \ldots, n$$

Triangular, Band Matrix, Hessenberg Form.

1. A matrix A is **upper-triangular** if all elements $a_{ij} : j < i$ are 0.
2. A **band matrix** is B such that all elements are 0 except $a_{i,i+c_k}$, for some $\{c_k\}$

3. **Tridiagonal matrix** is a band matrix with $c_k \in \{-1, 0, 1\}$.

4. A **Hessenberg matrix** is upper-triangular except it allows the first subdiagonal to be nonzero.

Positive Definite. Let A be a symmetric matrix, if for any **nonzero** vector x that is conformable, $x^T Ax$ is nonnegative, then A is positive semidefinite, if $x^T Ax > 0$ strictly, A is positive definite.

Orthogonal and Unitary Matrix. A $n \times m$ matrix A is orthogonal if $AA^T = I_n$ and $A^T A = I_m$. Usually, we also require the matrix to be **square**, then $Q^T Q = QQ^T = I$ or $Q^T = Q^{-1}$.

6.1.2 Symmetric Matrices and Positive Definite Matrices

6.1.1 Proposition (Properties of Symmetric Matrices). *1. Symmetric matrices have real eigenvalues.*

2. *Symmetric matrix is **orthogonally** diagonalizable.*

3. *Eigenvectors corresponding to distince eigenvalues of a symmetric matrix are **orthogonal**.*

Spectral Decomposition. When A is symmetric and the eigenvectors are **orthonormal**: $I = \sum_i v_i v_i^T$, then

$$A = \sum_i c_i v_i v_i^T$$

6.1.2 Proposition (Positive definite matrices have positive eigenvalues). *Any **real symmetric** matrix is p.d. (p.s.d) iff all the eigenvalues are positive (nonnegative).*

6.1.3 Proposition (Square Root of a Positive Semi-Definite Matrix). *Let A be a nonnegative definite matrix, $A = VCV^T$ be a spectral decomposition, we write*

$$A^{\frac{1}{2}} = VSV^T \quad \textit{where} \quad S_{ij} = C_{ij}^{\frac{1}{2}}$$

In general for a function $f : \mathbb{K} \to \mathbb{K}$,

$$f(A) = VC_f V^T \quad \textit{where} \quad C_f = \big(f(C_{ij})\big)$$

6.1.3 Projection Matrix

See Rao et al. 1973.

6.2 Operations

6.2.1 Transpose and Conjugate Transpose

Transpose. Let A be a $n \times m$ matrix (a_{ij}), then the transpose of A is $A^T = A' = (a_{ji})$. The **conjugate transpose** is $A_H = A^* = (\bar{a}_{ji})$.

6.2.2 Partition

We can partition a matrix A as $\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$, where the submatrices are conformable. If A_{11} and A_{22} are square, they are called **principal submatrices**. A_{11} is the **leading principal submatrix**.

A **block diagonal** matrix is of the form

$$\begin{bmatrix} X & 0 & 0 \\ 0 & Y & 0 \\ 0 & 0 & \ddots \end{bmatrix} = \operatorname{diag}(X, Y, \dots)$$

The **direct sum** of matrices $A_1, \dots, A_k$ is

$$A_1 \oplus \cdots \oplus A_k = \operatorname{diag}(A_1, \dots, A_k)$$

6.2.3 Addition and Multiplication

Matrix addition and scalar multiplication. Let A, B be $n \times m$ matrix, c be a scalar from the same field. Then $A + B = (a_{ij} + b_{ij})$ and $cA = (ca_{ij})$. With these operations, the space of $n \times m$ matrices is vector space.

Cayley Multiplication. Let A, B be conformable for matrix multiplication, then $C = AB$ if $c_{ij} = \sum_k a_{ik} b_{kj}$.

Remark. Important: the product Av for a vector can be thought of as a linear combination of columns of A. Then $A_{n\times k}B_{k\times m}$ is just m linear combinations of columns of A or n linear combinations of rows of B.

Which leads to the representations:

$$AB = A\begin{bmatrix} B_1 & B_2 & \dots & B_m \end{bmatrix} = \begin{bmatrix} AB_1 & \dots AB_m \end{bmatrix} \quad \text{linear combinations of columns of} \quad A \tag{6.1}$$

$$= \begin{bmatrix} A_1^T \\ \vdots \\ A_n^T \end{bmatrix} B = \begin{bmatrix} A_1^T B \\ \vdots \\ A_n^T B \end{bmatrix} \quad \text{linear combinations of rows of} \quad B \tag{6.2}$$

$$= \begin{bmatrix} A_1 & A_2 & \dots & A_k \end{bmatrix} \begin{bmatrix} B_1 \\ \vdots \\ B_k \end{bmatrix} \quad \text{columns of} \quad A \times \quad \text{rows of} \quad B \tag{6.3}$$

$$= \quad \text{block multiplications} \quad . \tag{6.4}$$

6.2.1 Proposition (Preservation of Shape). *The product of diagonal(upper or lower triangular) matrices is still diagonal(upper or lower triangular).*

Power of Square Matrices, Polynomial. Let A be a **square** matrix, then we can define the power of A. If A^k exists for negative integer k, then we define $A^0 = I$. Let the polynomials of A be

$$p(A) = b_0 I + b_1 A + \cdots + b_k A^k$$

We still have the factorisation of **monomials**

$$I - A^k = (I - A)(I + A + \cdots + A^{k-1}) \quad \text{for positive} \quad k$$
$$I + A^k = (I + A)(I - A + \dots A^{k-1}) \quad \text{for } \textbf{odd} \text{ positive integer} \quad k$$

6.2.2 Proposition (Multiplication of Partitioned Matrices). *If the following multiplications are conformable:*

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = \begin{bmatrix} A_{i\cdot} B_{\cdot j} \end{bmatrix}$$

6.2.4 Hadamard Product and Kronecker Product

Hadamard Product. If $A, B \in \mathbb{R}^{n\times m}$, then $C = A. * B$ if $c_{ij} = a_{ij}b_{ij}$.

Kronecker Product. Let $A \in \mathbb{R}^{n\times m}$, $B \in \mathbb{R}^{k\times j}$, then the **Kronecker product** is defined as

$$A \otimes B = \begin{bmatrix} a_{11}B & \cdots & a_{1m}B \\ \vdots & \ddots & \vdots \\ a_{n1}B & \cdots & a_{nm}B \end{bmatrix}$$

6.2.5 Matrix Inverse

Consistent System of Linear Equations. A linear system $A_{n\times m}x = b$ is consistent, i.e., a solution exists iff $\operatorname{rank}([A \mid b]) = \operatorname{rank}(A)$.

Inverse of a Matrix. Let A be a $n \times n$ square nonsingular matrix, then we consider n linear systems:

$$Ax_i = e_i \quad \text{or} \quad AX = I_n$$

Then X is the inverse of A, called A^{-1} which is nonsingular and satisfies

$$AA^{-1} = A^{-1}A = I_n$$

Remark.

1. The linear system interpretation gives a way to find the inverse, we simply solve the n equations.
2. For singular A, there is a linear combination of the columns that gives 0 vector, meaning there is a nonzero x such that $Ax = 0$.

Inverse of a Nonsquare Matrix. Let A be a $n \times m$ matrix, where $\operatorname{rank}(A) = n \leq m$, then we can still solve the n linear systems:

$$Ax_i = e_i \quad \text{or} \quad AX = I_n$$

where X is a $m \times n$ and $\operatorname{rank}(X) = n$, it's the right inverse of A. Similarly we can define left inverse of A, that is $YA = I_m$.

6.2.3 Proposition (Properties). *1.* $\left(A^{-1}\right)^T = \left(A^T\right)^{-1}$

2. *A matrix has inverse iff it's square and of full rank.*

3. $\left|A\right| = \frac{1}{\left|A^{-1}\right|}$ *and* $\left|A\right| = 0$ *iff* A *is* ***singular****, furthermore,* $A^{-1} = \frac{1}{\det(A)}$ *times the adjugate of* A.

4. $(AB)^{-1} = B^{-1}A^{-1}$.

5. $(I - A)^{-1} = I + A + A^2 + A^3 \ldots$ *if* $\|A\| < 1$.

6. $\operatorname{inv}(A \otimes B) = \operatorname{inv}(A) \otimes \operatorname{inv}(B)$

Generalised Inverse. For A $n \times m$ matrix, the generalised inverse G is $m \times n$ matrix such that

$$AGA = A$$

Denoted as A^{-}.

6.3 Characteristics of Matrices

6.3.1 Rank

Rank of Matrix. The maximum number of linearly independent vectors(rows and columns) is called the **rank** of the matrix $\operatorname{rank}(A)$. For $A \in \mathbb{R}^{n\times m}$, $\operatorname{rank} A \leq \min(n, m)$. If the equality holds, we say the matrix is of full rank.

Nonsingular Matrix. A **square, full-rank** matrix is called **nonsingular**.

6.3.1 Proposition (Diagonal Dominance). *A square matrix that is diagonally dominant is nonsingular.*

6.3.2 Proposition (Interaction with other operations).

1. $\operatorname{rank}(cA) = \operatorname{rank}(A)$, $\operatorname{rank}(A^T) = \operatorname{rank}(A)$.

2. $\operatorname{rank}(PAQ) = \operatorname{rank}(A)$ *for* P, Q *that are products of elementary operator matrices.*

3. $\operatorname{rank}(AB) \leq \min(\operatorname{rank}(A), \operatorname{rank}(B))$.

4. *Sylvester's Law of Nullity:* $\operatorname{rank}(AB) \geq \operatorname{rank}(A) + \operatorname{rank}(B) - n$ *for* $A : n \times n$ *and* $B : n \times m$.

5. $|\text{rank}(A) - \text{rank}(B)| \geq \text{rank}(A+B) \leq \text{rank}(A) + \text{rank}(B)$

6. *If A is **nonsingular**(square and full rank), then* $\text{rank}(AB) = \text{rank}(B)$, $\text{rank}(CA) = \text{rank}(C)$. *And hence AA^T is full rank iff A is of full **row** rank, A^TA is full rank iff A is of full **column** rank.*

6.3.3 Proposition (Rank and Positive Definiteness). *Let A be a square p.d. matrix, A be $n \times m$ and of rank m(full column rank), then A^TCA is positive definite. Conversely if A^TCA is p.d. and C is p.d. then A is of full column rank.*

6.3.2 Determinant of Square Matrices

Determinant of a Square Matrix. Let $\pi_j = (j_1, \ldots, j_n)$ be a **permutation** of $(1, 2, \ldots, n)$. Then the determinant of A is defined as

$$\det(A) = \sum_{\text{all permutations}} \sigma(\pi_j) a_{1j_1} \ldots a_{nj_n}$$

Minor and Cofactor. The **minor** or **complementary minor** associated with a_{ij} element is $\det\left(A_{-(i)(j)}\right)$ and the **cofactor** is the scalar $a_{(ij)} := (-1)^{i+j} \det\left(A_{-(i)(j)}\right)$

6.3.4 Proposition (Laplace Expansion).

$$\det(A) = \sum_{j=1}^{n} a_{ij} a_{(ij)}$$

Adjugate. Define the **adjugate** of A as the transposed matrix of cofactors: $\text{adj}(A) = (a_{(ji)}) = (a_{(ij)})^T$. Then we have $A(a_{(ji)}) = (a_{(ji)})A = \left|A\right| I$.

6.3.5 Proposition (Properties of Determinant). *There are some properties of determinants.*

1. $\left|A\right| = \left|A'\right|$

2. $\left|cA\right| = c^n \left|A\right|$

3. *For partitioned matrices:*

$$\det\begin{pmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{pmatrix} = \det\begin{pmatrix} A_{11} & 0 \\ A_{21} & A_{22} \end{pmatrix} = \left|A_{11}\right| \left|A_{22}\right|$$

and

$$\det\begin{pmatrix} 0 & A_{12} \\ -I & A_{22} \end{pmatrix} = \det(A_{12})$$

4. $\left|AB\right| = \left|A\right|\left|B\right|$.

5. *For Kronecker Product:* $\left|A \otimes B\right| = \left|A\right|^m \left|B\right|^n$ *if* A *is* $n \times n$ *and* B *is* $m \times m$.

6. *For orthogonal matrix* Q, $\det(Q) = 1$ *or* -1.

6.3.6 Proposition (Diagonal Expansion). *To evaluate* $\det(A + \operatorname{diag}(d_1, \ldots, d_n))$ *we can do a diagonal expansion.*

Geometrically, the determinant of a matrix is the signed volume of the parallelotope with 0 as a vertex and the columns as edges.

6.3.3 Matrix Rank, Inverse, Determinants

Determinants. For $A \in M^{n\times n}$,

$$\det(A) = \sum_{i=1}^{n} (-1)^{i+j} A_{ij} \det\left(\tilde{A}_{ij}\right).$$

6.3.7 Proposition (Properties of Determinants). *There are a few properties of the determinants.*

1. $|\det(A)|$ *measures the* $n-$*dimensional volume of the parallelepiped.*

2. *If we interchage two rows or two columns of* A*, then* $\det(A') = \det(A)$.

3. *If we multiply one row or one column by* k*, then* $\det(A') = k\det(A)$.

4. *If we add to one row or one column* $k\times$*another row/column, then* $\det(A') = \det(A)$.

5. *Upper triangular matrix has determinants equal to the product of the diagonal entries.*

6. *If two rows/column are the same,* $\det(A) = 0$.

7. $\det(AB) = \det(A)\det(B)$, $\det\left(A^T\right) = \det(A)$, $\det\left(A^{-1}\right) = \det(A)^{-1}$.

6.3.4 Trace of Square Matrices

Trace. Let $A = (a_{ij})$ be a $n \times n$ square matrix, the trace of A is mapping from $\mathbb{R}^{n\times n} \to \mathbb{R}$

$$\operatorname{tr}(A) = \sum_i a_{ii}$$

We have the following properties:

6.3.8 Proposition (Properties of Trace). *1.* $\operatorname{tr}(A') = \operatorname{tr}(A)$

2. $\operatorname{tr}(A + B) = \operatorname{tr}(A) + \operatorname{tr}(B)$ *and* $\operatorname{tr}(cA) = c \operatorname{tr}(A)$

3. $\operatorname{tr}\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} = \operatorname{tr}(A_{11}) + \operatorname{tr}(A_{22})$ *if* A_{11}, A_{22} *are square.*

4. $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ *provided that both multiplications are conformable.*

5. For quadratic form: $x^T Ax = \operatorname{tr}\left(x^T Ax\right) = \operatorname{tr}\left(Axx^T\right)$.

6. $\operatorname{tr}(A \otimes B) = \operatorname{tr}(A) \operatorname{tr}(B)$

6.4 Eigen-analysis

6.4.1 Eigenvalues and Eigenvectors

Eigenvalue and Eigenvector. Let A be a $n \times n$ matrix, if $v \neq 0$ is a vector and λ is a scalar, such that

$$Av = \lambda v$$

Then we say λ is an **eigenvalue** of A and v is the **eigenvector corresponding to** λ.

Eigenspace and Geometric Multiplicity. For a eigenvalue λ of A, the **eigenspace** is the **null space** of $(A - \lambda I)$. The **geometric multiplicity** of λ is the dimension of the eigenspace corresponding to λ.

6.4.2 The Characteristic Polynomial

Characteristic Polynomial . The **Characteristic Polynomial** of $A : n \times n$ is

$$p_A(c) := \left|A - cI\right| = s_0 + s_1 c + \cdots + s_n c^n$$

It can be shown that an eigenvalue of A is a **root of the characteristic polynomial**. So

$$p_A(c) = (-1)^n (c - \lambda_1) \ldots (c - \lambda_n)$$

by the fundamental theorem of algebra.

6.4.1 Cayley-Hamilton.

$$p_A(A) = 0_{n \times n}$$

6.4.2 Eigenvalue and Trace and Determinant. *From the decomposition of $p_A(c)$ we can see*

$$\prod_{i=1}^{n} \lambda_i = \left|A\right|$$
$$\sum_{i=1}^{n} \lambda_i = \mathrm{tr}(A)$$

6.4.3 Symmetric Matrix. *If A is symmetric, then the eigenvalues are real.*

6.4.3 The Spectrum of a Matrix

Spectrum of a Matrix, algebraic multiplicity. Let A be a $n \times n$ matrix, the **spectrum** of A is the set of all $k, k \leq n$ distince eigenvalues, denoted by σA. Hence

$$p_A(c) = (-1)^n (c - \lambda_1)^{m_1} (c - \lambda_2)^{m_2} \ldots (c - \lambda_k)^{m_k}$$

where m_i is the **algebraic multiplicity** of λ_i.

Simple and Semisimple Eigenvalues. If an eigenvalue has algebraic multiplicity 1, it's called **simple**; if it has same number of algebraic and geometric multiplicity, it's called **semisimple**.

Spectral Radius. For a matrix A, the **spectral radius** is the magnitude of the largest eigenvalue:

$$\rho(A) = \max |\lambda_i|$$

The set of complex number $\{x : |x| = \rho(A)\}$ is the **spectral circle** of A.

6.4.4 Linear Independence of Eigenvectors. *The eigenvectors associated with **distinct** eigenvalues are linearly independent.*

6.5 Matrix Transformation and Reduction

6.5.1 Elementary Matrices and Gaussian Elimination $PA = LU$

There are three elementary operations of matrices:

1. Interchange two rows
2. Row multiplication by a scalar
3. Axpy operation

Correspondingly, there are **elementary operator matrices**.

Permutation, Scalar Row Multiplication, Axpy Row Transformation. If we do the elementary operations on an identity matrix I_n,

1. the E_{ij} is the **permutation matrix** that interchange the i, j-th rows of I_n,
2. $E_i(x)$ is the **scalar row multiplication matrix** if it multiplies the i-th row of I_n with x,
3. $E_{ij}(x)$ is the **axpy row transformation matrix** if it adds to the i-th row $x\times$ the j-th row.

For example for I_4:

$$E_{23} = \begin{bmatrix} 1 & & & \\ & 0 & 1 & \\ & 1 & 0 & \\ & & & 1 \end{bmatrix} \quad E_2(x) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & x & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad E_{23}(x) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & x & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

6.5.1 Proposition (Elementary Operations on Matrix). *If we **pre-multiply** a matrix A with these elementary matrices, we are doing the same elementary **row** operations on A.*

*Similarly we have if we do **column** operations on I_n and **post-multiply** the matrices, we are doing the same column operations on A*

6.5.2 Proposition (Properties of Elementary Operator Matrices). *They are invertible, of full rank, with determinant:*

$$\det\left(E_{ij}\right) = -1 \quad \det(E_i(x)) = x \quad \det\left(E_{ij}(x)\right) = 1$$

By Gaussian Elimination procedure, we can repeat the following steps and arrive at $PA = LU$:

1. If the first pivot is nonzero, we can use it to eliminate all other elements in the first column using Axpy $E_{21}, E_{31}, \ldots$.
2. If the pivot position has zero, we can exchange rows so that the pivots are nonzero, by a permutation matrix P.

For any invertibel matrix A, we can transform it into an upper-triangular matrix U by $EPA = U$, then $PA = E^{-1}U$ and $L = E^{-1}$ is a lower-triangular matrix.

Echelon and Reduced Row Echelon Forms. The **reduced row echelon form** has an echelon form with zeros above and below pivots, which are all 1.

6.5.2 Elementary Reduction

Similar Matrix. A, B are **similar** if there is a nonsingular matrix P such that

$$B = P^{-1}AP$$

Similar matrices have the same eigenvalues.

Orthogonally Similar. A, B are **orthogonally similar** if there is an orthogonal matrix Q such that

$$B = Q^T AQ$$

6.5.3 Proposition (Finding the Q). *If $Q^T AQ = I$ then by notice $[Q^T AQ \quad IQ] = [I \quad Q]$. We can find Q.*

Similar Canonical Factorisation. Let Λ be the matrix diag $\{\lambda_1, \ldots, \lambda_n\}$, and V be the matrix whose columns are the eigenvectors corresponding to Λ, then

$$AV = VC \quad \text{or} \quad A = VCV^{-1}$$

provided V is nonsingular. If A can be written in this way, it's called **diagonalizable**.

6.5.4 Diagonalizability Theorem. *A is diagonalizable iff all eigenvalues are **semisimple**.*

6.5.5 Rank Factorisation.

6.5.3 Orthogonalization Reduction

6.5.6 Proposition (QR Decomposition).

6.5.7 Shur's Decomposition. *Let A be a square matrix over $\mathbb{C}$, then there exists unitary matrix Q such that*

$$A = QUQ^*$$

where U is an upper-triangular matrix.

6.5.4 Eigenvalue Reduction

Singular Value Decomposition. An $n \times m$ matrix A can be factorised into

$$A = UDV^T$$

where U is an $n \times n$ **orthogonal matrix**, V is an $m \times m$ **orthogonal** matrix, and D is a $n \times m$ **diagonal matrix** with **nonnegative entries**.

6.5.8 Proposition (Jordan Reduction). *See Rao et al. 1973.*

Chapter 7

Inner Product Spaces

7.1 Inner Product

Inner product spaces are very very important because here we can extend the "geometry" of Euclidean spaces to a space of infinite dimension. In particular, the ideas of projection, orthogonality carry through and we can identify a Hilbert(complete inner product) space with its dual space.

Inner Product. Let X be a vector space over a field K, an **inner product** is a function $\langle \cdot | \cdot \rangle : X \times X \to K$ such that for all $x, y, z \in X$:

$$\begin{aligned} \langle \alpha x + \beta y | z \rangle &= \alpha \langle x | z \rangle + \beta \langle y | z \rangle \\ \langle x | y \rangle &= \overline{\langle y | x \rangle} \\ \langle x | x \rangle \geq 0 \text{ and } \langle x | y \rangle = 0 &\iff x = y \end{aligned}$$

So if $K = \mathbb{R}$ then the inner product is bilinear, symmetric and positive definite. If $K = \mathbb{C}$ then the inner product is *sesqui-linear*, symmetric and positive definite.

Notice that $\langle x | x \rangle$ is always real.

The norm induced by an inner product is $\|x\| : \langle x | x \rangle^{\frac{1}{2}}$

7.1.1 Geometric Properties

Orthogonal. Two elements $x, y \in (X, K, \langle\cdot|\cdot\rangle)$ are said to be orthogonal if

$$\langle x|y\rangle = 0$$

7.1.1 Pythagorean Theorem. *If $u, v \in X$ and $u \perp v$ then*

$$\|u + v\| = \|u\| + \|v\|$$

7.1.2 Cauchy-Schwarz. *Let X be an inner product space over a field K, for $x, y \in X$, we have*

$$|\langle x|y\rangle| \leq \|x\|\|y\|$$

with equality when $x = cy$ for some $c \in K$.

7.1.3 Parallelogram Equality. *For $u, v \in X$,*

$$\|u + v\|^2 + \|u - v\|^2 = 2\left(\|u\|^2 + \|v\|^2\right)$$

7.1.2 Representation Theorem

7.2 Linear Operator

7.2.1 Spectral Theorem

7.2.2 Polar and Singular Value Decomposition

Sigular Value. The eigenvalues of the matrix $\sqrt{T^*T}$ is called the sigular values.

7.3 Hilbert Space, Completion of Inner Product Spaces

Chapter 8

Norm Structure

8.1 Normed Vector Space

8.1.1 Norm

Norm. On a vector space X over K with $+$ and $\cdot$, a norm is a mapping $\|\cdot\| : X \to \mathbb{R}$ such that for all $x, y \in X$ and $\alpha \in K$:

1. $\|x\| \geq 0$ and $\|x\| = 0 \iff x = 0$
2. $\|\alpha x\| = |\alpha|\|x\|$
3. $\|x + y\| \leq \|x\| + \|y\|$

8.1.1 Any vector space can be normed. *By using Zorn's Lemma.*

Equivalence of Norms. Two norms are **equivalent** is they induce the same topology and a simple criterion is that there is $C, C' > 0$

$$\frac{1}{C'}\|x\|_2 \leq \|x\|_1 \leq C\|x\|_2$$

8.1.2 Some Interesting Examples

Finite-dimensional Normed vector space

A finite-dimensional normed vector space $(X, \|\cdot\|)$ over K has basis $\{e_1, e_2, ..., e_n\}$ and we can define the norms such as

$$\|x\|_p := \left[\sum_{i=1}^{n} |x_i|^p\right]^{\frac{1}{p}}$$

$$\|x\|_\infty := \max_i |x_i|$$

where $x = \sum_{i=1}^{n} x_i e_i$, for $1 \leq p < \infty$

Product of normed space

Product of normed space. Let $X := \prod_{i=1}^{n} X_i$ where X_i is endowed with its norm $\|\cdot\|_{X_i}$, then we can give X a norm that is

$$\|(x_1, x_2, \ldots, x_n)\|_X := \sup_i \|x_i\|_{X_i}$$

Quotient Space

Quotient Norm. Recall that Z being a linear subspace of a vector space X, we can define the quotient space to be $X/Z := \{[x] : x \in X\}$ where $[x] := \{y \in X : y - x \in Z\}$. If Z is **closed** and X is **normed** then the quotient norm can be defined as:

$$\|[x]\| := \inf_{y \in [x]} \|y\|_X = \inf_{z \in Z} \|x - z\|_X$$

$C(K, Y)$. Let $C(K, Y)$ be the set of all continuous functions from a compact topological space K to a normed vector space $(Y, \|\cdot\|_Y)$, then we can define the **sup-norm** to be

$$\|f\|_{\sup} = \sup_{x \in K} \|f(x)\|_Y$$

l^p space. Let $1 \leq p < \infty$, and a sequnce $(x_1, x_2, \ldots)$ in K, let $l^p := \{(x_i)_{i=1}^{\infty} : \sum_{i=1}^{\infty} |x_i|^p < \infty\}$ and let the norm being defined as such:

$$\|x\|_p := \left(\sum_{i=1}^{\infty} |x_i|^p\right)^{\frac{1}{p}}$$

8.1.3 Compactness in Normed Vector Space

8.1.2 Proposition (Compactness). *Let $(X, \|\cdot\|)$ be a normed vector space, then a subset K of X is compact iff it's closed and bounded.*

8.1.4 Dimension Property and Riesz Theorem

8.1.3 Dimension and Equivalence of Norms. *Let $(X, \|\cdot\|)$ be a normed vector space,*

1. *If X is finite dimensional, then any norms are equivalent to $\|\cdot\|_1$.*
2. *If X is infinite dimensional, then there are non-equivalent norms.*

First we characterise the compactness in finite-dimensional normed vector space.

8.1.4 Riesz. *A normed vector space $(X, \|\cdot\|)$ is **finite-dimensional** iff the unit sphere(or the unit closed ball) of X is **compact**.*

Proof. We find the finite cover of the unit sphere, and use the points to span a finite-dimensional space Y then show $X \subset \bar{Y}$. $\implies$ part is from the previous propsition. □

8.2 Banach Spaces and Completion of NVS

8.2.1 Complete Normed Vector Space

Banach Space. A normed vector space is Banach Space, if the metric induced by the norm is complete.

8.2.1 Completion of Normed Vector Space. *Defined as the limit of a Cauchy sequence.*

8.2.2 Fun Examples

There are some interesting examples of Banach spaces.

8.2.2 Proposition (Finite-dimensional Normed Vector Space). *Any finite-dimensional normed vector space is complete.*

8.2.3 Proposition (Quotient Space).

8.2.4 Proposition (Lebesgue Spaces).

8.2.3 Convergence of Series and Characterisation

8.2.5 Convergence of Series in Banach Space.

8.2.6 Characterisation. *A n.v.s X is Banach iff every absolutely convergent series is convergent.*

8.3 Linear Operators on Normed Vector Space

8.3.1 Continuous Linear Operator

Continuous Linear Operator. A linear operator is a mapping $A : X \to Y$ where X, Y are **vector spaces** over the same field $\mathbb{K}$ that is linear. If further we endow the X, Y with **norms**, then we can talk about **continuous linear operators** $\mathcal{L}(X, Y)$.

8.3.1 Characterisation of Continuous Linear Operator. *Let $A : X \to Y$ be a linear operator, where X, Y are normed vector spaces, then TFAE:*

1. *A is continuous on X*
2. *A is continuous at 0_X, the origin of X.*
3. *There exists a $C \geq 0$ such that for any $x \in X$, $\|Ax\| \leq C\|x\|$*
4. *For any bounded subset B of X , the direct image $A(B)$ is bounded in Y (**bounded linear operators**).*

Proof. See *050902* □

Consider the set $\mathcal{L}(X, Y)$ of all continuous linear operators from a normed vector space X to normed vector space Y that are defined over the same field. We can prove that it's a normed vector space with the **operator norm**.

Operator Norm. Let X, Y be normed vector space, the mapping:

$$\|\cdot\| : \mathcal{L}(X, Y) \to \mathbb{R} \quad \text{such that} \quad \|A\| := \sup_{x \neq 0} \frac{\|Ax\|}{\|x\|} \tag{8.1}$$

is a norm on $\mathcal{L}(X, Y)$. So for any $x \in X$, $\|Ax\| \leq \|A\|\|x\|$.

Proof. □

8.3.2 Proposition (Composition). *Let X, Y, Z be normed vector spaces, if $A \in \mathcal{L}(X, Y)$, $B \in \mathcal{L}(Y, Z)$, then $B \circ A \in \mathcal{L}(X, Z)$ and*

$$\|BA\| \leq \|A\|\|B\|$$

8.3.3 Eigenvalue and operator norm. *If $A \in \mathcal{L}(X)$, then any eigenvalue of A satisfies $|\lambda| \leq \|A\|$.*

Operator Norm on Dual Space. $X^* := \mathcal{L}(X, \mathbb{K})$ has the norm $\|l\| = \sup_{x \neq 0} \frac{|l(x)|}{\|x\|}$

8.3.4 Uniform Continuity. *1. Any continuous linear operator is uniformly continuous;*

*2. If X is **finite-dimensional**, then any linear operator from X to Y is continuous.*

Example ($\mathbb{R}^n \to \mathbb{R}^m$)**.** Let $A, B : \mathbb{R}^n \to \mathbb{R}^m$ and $C : \mathbb{R}^m \to \mathbb{R}^k$ be a linear operators, then

1. $\|A\| \leq \infty$ and A is uniformly continuous.
2. $\|A + B\| \leq \|A\| + \|B\|$ and $\|cA\| = |c|\|A\|$.
3. $\|CA\| \leq \|C\|\|A\|$.

8.3.5 Invertible Operators. *Let Ω be the subset of $\mathcal{L}(\mathbb{R}^n)$ containing all invertible operators, then if $A \in \Omega$ and $B \in \mathcal{L}(\mathbb{R}^n)$ and*

$$\|B - A\|\left\|A^{-1}\right\| < 1$$

then $B \in \Omega$ and Ω is an open subset. $A \to A^{-1}$ is continuous.

8.3.2 Compact Linear Operators

Compact (Continuous) Linear Operator. A linear operator $A : X \to Y$ is said to be compact if for any bounded $B \subset X$, $A(B)$ is relatively compact(i.e., $\overline{A(B)}$ is compact). Any compact linear operator is continuous.

8.3.6 Proposition (Criterion for Compact Linear Operator). *1. Convergent subsequence. 2. Finite-dimensional space.*

8.3.3 Hahn-Banach in Normed Vector Space

8.3.7 Hahn-Banach in Normed Vector Space. *Let X be a normed vector space, let Y be a subspace of X and let $l : Y \to K$ be a* ***continuous*** *linear functional, then there exists a continuous linear functional $\tilde{l} : X \to K$ s.t.*

$$\tilde{l}(y) = l(y) \quad \textit{for all} \quad y \in Y \quad \textit{and} \quad \|\tilde{l}\|_{X'} = \|l\|_{Y'}$$

8.3.4 Geometric Forms of Hahn-Banach

8.3.8 First Geometric Form.

8.3.9 Second Geometric Form.

8.3.5 Dual Operator and Banach Closed Range Theorem

8.4 Linear Operator on Banach Space

8.4.1 Uniform Boundedness Principle

Uniformly bounded. A family of continuous linear operators $A_i \in \mathcal{L}(X, Y)$ is uniformly bounded if

$$\sup_{i \in I} \|A_i\|_{\mathcal{L}} \leq \inf$$

If $\|A_n\| \leq \infty$ it will imply $\|A_n x\| \leq \infty$. Interestingly for when X is Banach, the converse is true.

8.4.1 Uniform Boundedness Principle. *Let X be a Banach space, Y be a n.v.s, $(A_i)_{i \in I}$ be a family $A_i \in \mathcal{L}(X, Y)$ s.t.*

$$\forall x \in X, \quad \sup_{i \in I} \|A_i x\|_Y \leq \infty$$

then (A_i) is uniformly bounded.

Proof. 051302 □

Corollary 8.4.1.1. *Let X be a Banach space, Y be a n.v.s, (A_n) be a family of continuous linear operators from X to Y s.t. for any $x \in X$, $A_n x$ converges in Y, then*

$$\sup_{n \geq 1} \|A_n\| < \infty$$

Furthermore for A defined as $Ax := \lim A_n x$, $A \in \mathcal{L}(X, Y)$ and $\|A\| \leq \liminf \|A_n\|$

Proof. 051302 □

8.4.2 Banach Open Mapping Theorem

8.4.2 Open Mapping Theorem. *Let X, Y be Banach spaces and $A \in \mathcal{L}(X, Y)$ be* ***surjective****. Then the direct image $A(U)$ under A of any open subset U of X is an open set of Y.*

Proof. □

8.4.3 Banach Closed Graph Theorem

8.4.3 Closed Graph. *Let X, Y be Banach spaces, $A : X \to Y$ be a* ***closed linear operator****, then $A \in \mathcal{L}(X, Y)$.*

Part II

Basic Analysis

Chapter 9

Analysis of Univariate Real Functions

9.1 Structures on the Real Line

9.1.1 Construction of Real and Complex Numbers

We start with natural numbers which is assumed to exist by the axiom of infinity. Then we define $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ by the equivalent relations.

$\mathbb{Z}$ is the quotient set of $\mathbb{N} \times \mathbb{N}/R_1$ where $R_1 := \{(m,n) \times (p,q) : m + q = n + p\}$.

$\mathbb{Q}$ is the quotient set of $\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})/R_2$ where $R_2 := \{(m,n) \times (p,q) : mq = np\}$

And the $\mathbb{R}$ is the quotient set of all rational Cauchy sequences modulo the equivalent relation that the Cauchy sequences are arbitrarily close to each other after N_0.

$\mathbb{C}$ is defined as $\mathbb{R} \times \mathbb{R}$ satisfying the **complex multiplication**.

9.1.2 Structures

$\mathbb{Z}$ is a totally ordered, commutative ring.

$\mathbb{Q}$ is a totally ordered, commutative field and it is Archimedean.

$\mathbb{R}$ is totally ordered, commutative field, Archimedean. $\mathbb{R}$ has the Least upper bound property.

The reals and complex are both complete and have the Bolzano-Weierstrass property.

9.1.1 Existence of $\mathbb{R}$. *There exists an ordered filed $\mathbb{R}$ which has the least-upper-bound property. And it contains $\mathbb{Q}$ as a subfield.*

9.1.2 Least Upper Bound. *$\mathbb{R}$ has the least upper bound property. For any nonempty subset $A \subset \mathbb{R}$, if there is an upper bound of A, then there is $a \in \mathbb{R}$ such that $a = \sup A$*

9.1.3 Archimedean Property and Density of $\mathbb{Q}$.

1. *If $x, y \in \mathbb{R}$ and $x > 0$ then there is a positive integer n such that*

$$nx > y$$

2. *If $x, y \in \mathbb{R}$ and $x < y$, then there is a $q \in \mathbb{Q}$ such that $x < q < y$.*

9.1.4 Theorem. *Any open subset of $\mathbb{R}$ is at most a countable disjoint union of open intervals.*

Proof. Let G be an open set in $\mathbb{R}$, for $x, y \in G$, define an equivalence relation $x \sim y$ if $(\min(x, y), \max(x, y)) \subset G$. It's indeed an equivalence relation because: 1. $x \sim x$, 2. $y \sim x$ if $x \sim y$ and 3. if $x \sim y, y \sim z$ then

$$(\min(x, z), \max(x, z)) \subset ((\min(x, y), \max(x, y))) \bigcup ((\min(z, y), \max(z, y))) \subset G$$

. The quotient set G modulo $\sim$ are pairwise disjoint open intervals that are subset of G, and $G = \bigcup \{G \backslash \sim\}$, the quotient set is countable because there is an injection from $(G \backslash \sim) \to \mathbb{Q}$. □

9.2 Numerical Sequence and Series

9.2.1 Convergence of Sequence

Convergent Numerical Sequence. A numerical sequence $(x_n) : n = 1, 2, \ldots$ in $\mathbb{R}$ is convergent if for any $\varepsilon > 0$ there is an integer N such that for all $n \geq N$, $d(x_n, x) \leq \varepsilon$ for some $x \in \mathbb{R}$.

Cauchy Sequence. A numerical sequence (x_n) in $\mathbb{R}$ is Cauchy if $\lim_{m\to\infty} \sup_{n\geq m} |x_n - x_m| = 0$.

9.2.1 Algebraic Operations. *The algebraic operations on $\mathbb{R}$ are **continuous**. If $s_n \to s$ and $t_n \to t$ in $\mathbb{R}$ then as $n \to \infty$*

$$s_n + t_n \to s + t, \quad s_n t_n \to st, \quad \frac{s_n}{t_n} \to \frac{s}{t} \quad \textit{if defined}$$

9.2.2 Monotone Bounded Convergence. *A monotonic sequence in $\mathbb{R}$ converges iff it's bounded.*

Proof. Let (x_n) be monotonic and bounded, then $x^* := \sup(x_n)$ exists, for any $\varepsilon > 0$ there exists x_N such that $x^* - \varepsilon < x_N \leq \cdots \leq x^*$ and it means that $\lim x_n = x^*$. □

9.2.3 Bolzano-Weierstrass. *If (x_n) is a bounded sequence in $\mathbb{K}$, then it has a convergent subsequence.*

Proof. This follows from the fact that a bounded interval in $\mathbb{R}$ is compact. □

lim sup and lim inf. Let E be the set of all subsequential limits of a sequence of real numbers (x_n) then define

$$\limsup x_n = \sup E$$

It can be shown that $\limsup x_n \in E$. ALternatively $\limsup_{n\to\infty} x_n = \lim_{n\to\infty} \left(\sup_{m\geq n} x_m\right)$.

9.2.4 Proposition (Comparison of Limits). *If $s_n \leq t_n$ for all but finitely many n then:*

$$\limsup s_n \leq \limsup t_n$$
$$\liminf s_n \leq \liminf t_n$$

9.2.2 Series

Convergence of Series. Let (x_i) be a sequence in $\mathbb{R}$ then we can define the **partial sum** as $S_n = \sum_{i=1}^{n} x_i$ and the **series** is the infinite sum $\sum_{i=1}^{\infty} x_i$ and if S_n converges, we say that the series converges, otherwise it diverges.

Absolute Convergence of Series. A series $\sum a_i$ is said to converge absolutely iff $\sum |a_i|$ converges.

9.2.5 Proposition. *If a series $\sum a_n$ converges absolutely then it converges.*

Proof. Let $S_m = \sum_{i=1}^{m} a_i$, then $|S_n - S_m| = \left|\sum_{i=n}^{m} a_i\right| \leq \sum_{i=n}^{m} |a_i| \to 0$, as $m, n \to \infty$ because $\sum_{i=1}^{n} |a_i|$ is Cauchy. □

9.2.6 Cauchy Criterion. *$\sum_{i=1}^{\infty} x_i$ converges iff for every $\varepsilon > 0$ there is an integer N such that*

$$\left|\sum_{i=n}^{m} x_i\right| < \varepsilon, \quad \text{if} \quad m \geq n \geq N$$

A useful corollary to the Cauchy criterion, it's a necessary condition for a series to be convergent.

Corollary 9.2.6.1 (Infinitismal at Infinity)**.** *If* $\sum_{i=1}^{\infty} a_i$ *converges then* $\lim a_i = 0$.

Based on the Cauchy Criterion we can have the following ways to determine if a series is convergent.

9.2.7 Proposition (Comparison Tests).

1. *If* $|a_i| \leq c_i$ *for* $n \geq N_0$, N_0 *fixed, then if* $\sum c_i$ *converges so does* $\sum a_i$.
2. *If* $a_i \geq d_i \geq 0$ *for* $n \geq N_0$, N_0 *fixed, then if* $\sum d_i$ *diverges, so does* $\sum a_i$.

Proof. Let $S_n = \sum_{i=1}^{n} a_i$, then for any $\varepsilon > 0$, $|S_n - S_m| = \left|\sum_{i=n}^{m} a_i\right| \leq \sum_{i=n}^{m} |a_i| \leq \sum_{i=n}^{m} c_i < \varepsilon$ for sufficiently large m, n. The second statement can be proved by contradition. □

9.2.8 Proposition (Root Test). *Put* $\alpha = \limsup_{n\to\infty} (|a_n|)^{\frac{1}{n}}$. *Then*

1. *if* $\alpha < 1$ *then* $\sum a_n$ *converges,*
2. *if* $\alpha > 1$ *then* $\sum a_n$ *diverges,*
3. *if* $\alpha = 1$ *it's not determined.*

9.2.9 Proposition (Ratio Test). *The series* $\sum_{i=1}^{\infty} a_n$

1. *converges if* $\limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| < 1$
2. *diverges if* $\limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| > 1$ *for* $n \geq n_0$ *where* n_0 *is fixed.*

9.2.3 Series of Nonnegative Terms

9.2.10 Proposition (Monotone Bounded). *If* $\sum_{i=1}^{\infty} a_n$ *with* $a_n \geq 0$ *for all* n *then the partial sums are monotonically increasing, and hence the series converges iff* S_n *forms a bounded sequence.*

9.2.11 Proposition (Some Common Series of Nonnegative Terms). *There are some important series of nonnegative terms.*

1. *Geometric series:* $\sum_{i=1}^{\infty} x^n = \frac{1}{1-x}$ *if* $x < 1$ *otherwise it diverges.*
2. *p series:* $\sum_{n=1}^{\infty} \frac{1}{n^p}$ *converges if* $p > 1$*, otherwise it diverges.*
3. $\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p}$ *converges if* $p > 1$*, otherwise it diverges.*

9.2.12 Lemma. *This is a useful lemma due to Cauchy, if* $a_1 \geq a_2 \geq \cdots \geq 0$*, then the series* $\sum a_n$ *converges iff*

$$\sum_{i=0}^{\infty} 2^k a_{2^k} = a_1 + 2a_2 + 4a_4 \ldots$$

converges.

9.2.13 Integral Test of Convergence. *See Tao Analysis I.*

e.

$$e = \sum_{n=0}^{\infty} \frac{1}{n!}$$

It can be shown that $e = \lim \left(1 + \frac{1}{n}\right)^n$ and it's irrational.

9.2.4 Summation by Part, Product and Rearrangement

9.2.14 Proposition (Summation By Part). *Given two sequences* $(a_n), (b_n)$*, define:*

$$A_n = \sum_{k=0}^{n} a_k; \quad A_{-1} = 0$$

Then if $0 \leq p \leq q$ *we have:*

$$\sum_{i=p}^{q} a_i b_i = \sum_{i=p}^{q-1} A_n(b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p$$

Corollary 9.2.14.1 (Alternating Series)**.** *If* $|c_n|$ *is monotonically decreasing,* $c_{2m-1} \leq 0$ *and* $c_{2m} \geq 0$ *and* $\lim c_n = 0 \implies \sum c_n$ *converges.*

9.2.15 Proposition (Addition). *If* $\sum a_n = A$*,* $\sum b_n = B$ *then* $\sum(a_n + b_n) = A + B$ *and* $\sum ca_n = cA$ *for any fixed* c*.*

9.2.16 Proposition (Product). *Let $c_n = \sum_{k=0}^{n} a_k b_{n-k}$ then $\sum c_n = AB$ conveges if $\sum a_n = A$ and $\sum b_n = B$ and at least one converges absolutely.*

9.2.17 Proposition (Rearrangement).

9.3 Continuity

9.3.1 Limits of Functions and Continuous Function

As $\mathbb{R}$ is a metric space with algebraic operations. We use the same definitions of limits of functions and continuity.

Limits of Real Functions. Let $f : E \subset \mathbb{R} \to \mathbb{R}$, then we say $f(x)$ tends to $A \in \mathbb{R}$ as x tends to a **limit point** $p \in E$, or $\lim_{x\to p} f(x) = A$, if for any $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|A - f(x)| < \varepsilon \quad \text{for all} \quad x \in E, 0 < |t - x| < \delta.$$

9.3.1 Proposition (Algebraic Operations). *About the limits and continuity:*

1. *Let $f, g : E \subset (X, d) \to \mathbb{R}$ and $\lim_{x\to p} f(x) = A$, $\lim_{x\to p} g(x) = B$ then*

$$\lim_{x\to p}(f+g)(x) = A + B \quad \lim_{x\to p}(fg)(x) = AB \quad \lim_{x\to p}\frac{f}{g}(x) = \frac{A}{B}$$

2. *Let f, g be continuous from a metric space X to $\mathbb{R}$, then $f + g$, fg are continuous. $\frac{f}{g}$ is continuous if $g \neq 0$ for all x.*

Proof. The intuition is we are combining the topological structure(convergence) and the algebraic structure, they should be compatible: that is the algebraic operations should be continuous. Fix $\varepsilon > 0$, by the definition of $f(x) \to A$ and $g(x) \to B$, as $x \to p$ in (X, d), we have that there exists $\delta_1, \delta_2 > 0$ such that for $y : d(p, y) < \delta_1$, $|A - f(y)| < \frac{1}{2}\varepsilon$, and $z : d(p, z) < \delta_2$, $|B - g(z)| < \frac{1}{2}\varepsilon$. □

9.3.2 Continuous Function with Compact Domain

9.3.2 Attaining Maximum and Minimum. *A continuous function on a compact metric space into $\mathbb{R}$ attains its supremum and infimum.*

Proof. Let $f : K \to \mathbb{R}$ be a continuous function over a compact domain, then first we show f is bounded and there is $a, b \in K$ such that $f(a)$ is the infimum and $f(b)$ is the supremum.

Because $f(K)$ is compact subset of $\mathbb{R}$, it's bounded. By the least upper bound property of $\mathbb{R}$ we know $\sup f(K) \in f(K)$. □

9.3.3 Continuous Function with Connected Domain

Because the image of a connected set under a continuous function is connected and we have the following property of $\mathbb{R}$:

9.3.3 Proposition (Connectedness in $\mathbb{R}$). *Let $E \subset \mathbb{R}$ then E is connected iff for any $a < c < b$ if $a, b \in E$, then $c \in E$.*

9.3.4 Intermediate Value Theorem. *Let f be a continuous function from $[a, b] \to \mathbb{R}$, if $f(a) < f(b)$ and for any c such that $f(a) < c < f(b)$, there exists a point $x \in (a, b)$ such that $f(x) = c$.*

9.4 Differential Calculus

9.4.1 Derivative of a Real-Valued Function

Differential and integral calculus are of fundamental importance in science.

Derivative. Let $f : [a, b] \to \mathbb{R}$, for a point $x_0 \in (a, b)$, define

$$h(t) = \frac{f(t) - f(x_0)}{t - x_0}, \quad \text{for} \quad t \in (a, b), t \neq x_0$$

define, if the limit exists:

$$\frac{\mathrm{d}f}{\mathrm{d}x}(x_0) = \lim_{t \to x_0} h(t)$$

9.4.1 Differentiable implies Continuous. *If $f : [a, b] \to \mathbb{R}$ is differentiable at $x_0 \in (a, b)$ then f is continuous at x_0.*

Proof. Because we have $f'(x_0) = \lim_{t \to x_0} \frac{f(t) - f(x_0)}{t - x_0}$, we have $\lim_{t \to x_0} f(t) - f(x_0) = \lim_{t \to x_0} f'(x_0)(t - x_0) = 0$ □

9.4.2 Algebraic Operations and Chain Rule

We have the following propositions:

9.4.2 Proposition (Algebraic operations). *Let f, g be $[a, b] \to \mathbb{R}$ that are differentiable, then at any $x \in [a, b]$*

$$(f+g)' = f' + g'$$
$$(fg)' = f'g + fg'$$
$$\left[\frac{f}{g}\right]' = \frac{f'g - g'f}{g^2}$$

Proof. We show the one $(fg)'$, for any $x \in [a, b], t \in (a, b), t \neq x$, $f(t)g(t) - f(x)g(x) = f(t)(g(t) - g(x)) + (f(t) - f(x))g(x)$, letting $t \to x$, we have

$$\frac{f(t)g(t) - f(x)g(x)}{t - x} \to f(t)g'(x) + f'(x)g(x) \to f(x)g'(x) + f'(x)g(x)$$

□

9.4.3 Chain Rule. *If f is differentiable at x and g is differentiable at $f(x)$, then we have*

$$(g \circ f)'(x) = g'(f(x))f'(x)$$

Proof. If f is differentiable at x then in a neighborhood U of x we have $f(t) = f(x) + f'(x)(t-x) + o(t-x)$ and V of $f(x)$ such that $g(m) = g(f(x)) + g'(f(x)) + o(m - f(x))$,

$$\begin{aligned} g \circ f(t) - g \circ f(x) &= g'(f(x))(f'(x)(t-x) + o(t-x)) + o(f(t) - f(x)) \\ &= g'(f(x))f'(x)(t-x) + g'(f(x))o(t-x) + o(f(t) - f(x)) \end{aligned}$$

By the continuity of f, we have the desired result. □

9.4.3 Derivative: Monotonicity and Maximum

Local Maximum. Let $f : (X, d) \to \mathbb{R}$, then f has **local maximum** at p if there is a $\delta > 0$ such that for all $q : d(p, q) < \delta$, $f(q) \leq f(p)$.

9.4.4 Local Maximum and Derivative. *Let $f : [a, b] \to \mathbb{R}$, if f has a local maximum at a point $x \in (a, b)$ and if $f'(x)$ exists, then $f'(x) = 0$.*

Proof. Because f' exists at x, we have for $t \in (a, b)$,

$$f(t) = f(x) + f'(x)(t - x) + o(t - x)$$

Because x is a local maximum, if we take $t \to x+$, we have $f'(x) \leq 0$, if we take $t \to x-$ we have $f' \geq 0$, so $f'(x) = 0$. □

9.4.4 Mean Value Theorem and L'Hospital

9.4.5 Mean Value Theorem. *If f and g are continuous real-valued functions on $[a, b]$ which are **differentiable** in (a, b), then there is a point $x \in (a, b)$ such that*

$$[f(b) - f(a)]g'(x) = [g(b) - g(a)]f'(x)$$

if we take $g(x) = x$, then we have

$$[f(b) - f(a)] = (b - a)f'(x)$$

Proof. Define $h(t) = [f(b) - f(a)]g(t) - [g(b) - g(a)]f(t)$, then $h(t)$ is continuous on $[a, b]$ and differentable on (a, b), and $h(a) = h(b)$, then if $h(t)$ is not constant, by theorem 9.3.2, h attains its maximum or infimum at $p \in (a, b)$, so $h'(p) = 0$. □

9.4.6 L'Hospital Theorem. *Suppose $f, g : (a, b) \to \mathbb{R}$ are real-valued and differentable, and $g'(x) \neq 0$ for all $x \in (a, b)$, if*

$$\frac{f'(x)}{g'(x)} \to A \in [-\infty, \infty] \quad as \quad x \to a$$

and either $f(x) \to 0, g(x) \to 0$ or $g(x) \to \infty$ as $x \to a$, then

$$\frac{f(x)}{g(x)} \to A \quad as \quad x \to a$$

9.4.5 Taylor's Expansion

Higher-order derivative. If f' is differentiable then we calculate the drivative of f' called f'', and henceforth $f^{(3)}, f^{(4)}, \ldots$.

9.4.7 Taylor Expansion. *Let $f(t)$ be a real-valued function on $[a, b]$, n be a positive integer, $f^{(n-1)}$ is continuous on $[a, b]$ and $f^{(n)}(t)$ exists for every $t \in (a, b)$. Let x, y be distinct points of $[a, b]$ and define*

$$P(t) = \sum_{k=0}^{n-1} \frac{f^{(k)}(x)}{k!}(t-x)^k$$

then there exists a point $\tilde{x}$ between x and y such that

$$f(y) = P(y) + \frac{f^{(n)}(\tilde{x})}{n!}(y-x)^n \tag{9.1}$$

Proof. Write in full the equation 9.1:

$$\begin{aligned} f(y) = f(x) + f'(x)(y-x) + \frac{1}{2}f''(x)(y-x)^2 + \frac{1}{3!}f^{(3)}(x)(y-x)^3 + \ldots \\ + \frac{1}{n!}f^{(n)}(\tilde{x})(y-x)^n \end{aligned} \tag{9.2}$$

Define $g(t) = f(t) - P(t) - M(t-x)^n$, where $M := \frac{f(y)-P(y)}{(y-x)^n}$. Then $g^{(n)}(t) = f^{(n)} - Mn!$.

We know $g^{(k)}(x) = 0$ for $k = 0, 1, \ldots, n$, and $g(y) = 0$. By iterativaly using mean value theorem 9.4.5, we have for some $\tilde{x} \in [a, b]$, $M = \frac{1}{n!}f^{(n)(\tilde{x})}$. □

9.5 Riemann-Stieltjes Integral

9.5.1 Integral as Limit of Riemann Sums

Riemann and Stieltjes Integral. Let f be a **bounded real-valued** function on $[a, b]$, let $P = (x_0, x_1, \ldots, x_n)$ be a **partition** of $[a, b]$ such that $a = x_0 \leq x_1 \leq \cdots \leq x_n = b$.

Then we define:

$$M_i = \sup f(x) \quad m_i = \inf f(x) \quad x_{i-1} \le x \le x_i$$
$$U(P, f) = \sum_{i=1}^{n} M_i \Delta x_i \quad L(P, f) = \sum_{i=1}^{n} m_i \Delta x_i$$
$$\overline{\int_a^b} f \, \mathrm{d}x = \inf U(P, f) \quad \underline{\int_a^b} f \, \mathrm{d}x = \sup L(P, f)$$

Then if the upper and lower Riemann integrals are the same, we say f is **Riemann integrable** on $[a, b]$ and write $f \in \mathcal{R}$.

Stieltjes Integral. Let α be a monotonically increasing function on $[a, b]$ we write $\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1})$, replace Δx_i with it, then the resulting integral is called **Stieltjes integral**, denoted as $\int f(x) \, \mathrm{d}\alpha(x)$, Stieltjes-integrable functions $f \in \mathcal{R}(\alpha)$.

9.5.2 Criterion for Integrability

A useful criterion for Riemann-Stieltjes integrable functions:

9.5.1 Criterion for $\mathcal{R}(\alpha)$. *Let f be a bounded real-valued function on $[a, b]$ then $f \in \mathcal{R}(\alpha)$ iff for every $\varepsilon > 0$ there exists a partition P_ε such that*

$$U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon$$

Proof. We prove $\Longleftarrow$, for every P

$$L(P, f, \alpha) \le \underline{\int_a^b} f \, \mathrm{d}\alpha \le \overline{\int} f \, \mathrm{d}\alpha \le U(P, f, \alpha)$$

hence because ε is arbitrary, $\underline{\int} f \, \mathrm{d}\alpha = \overline{\int} f \, \mathrm{d}\alpha$. □

Using the criterion we have

9.5.2 Continuous functions are integrable. *Let f be continuous function on $[a, b]$ then $f \in \mathcal{R}(\alpha)$ on $[a, b]$.*

Proof. Let f be continuous on $[a, b]$, then f is **uniformly continuous** and we can control the size of $M_i - m_i$ when the partiation gets finer and finer. Fix $\varepsilon > 0$, choose the maximum size $\eta < \frac{\varepsilon}{\alpha(b)-\alpha(a)}$, then we can find partition fine enough such that for all $i \leq n$, $\Delta x_i \leq \delta$ and by uniform continuity implies $M_i - m_i \leq \eta$ and hence the criterion is satisfied. □

9.5.3 Monotonicity. *Let f be monotonic on $[a, b]$, let α be continuous and monotonic on $[a, b]$ then $f \in \mathcal{R}(\alpha)$.*

9.5.4 Finitely Discontinuity. *Let f be bounded on $[a, b]$ and it has only finitely many points of discontnuity on $[a, b]$, α is continuous at every discontnuity point of f, then $f \in \mathcal{R}(\alpha)$.*

9.5.5 Continuous Function of Integrable Function. *If $f \in \mathcal{R}(\alpha)$, h is continuous real-valued function defined on the range of f, then $h(f(x)) \in \mathcal{R}(\alpha)$.*

9.5.6 Proposition (Multiplication, absolute value).

1. *if $f, g \in \mathcal{R}(\alpha)$ then $fg \in \mathcal{R}(\alpha)$*
2. *if $f \in \mathcal{R}(\alpha)$ then $|f| \in \mathcal{R}(\alpha)$.*

9.5.3 Properties of Riemann-Stieltjes Integral

9.5.7 Linearity of $\mathcal{R}(\alpha)$. *For $f, g \in \mathcal{R}(\alpha)$ and $c \in \mathbb{R}$, $f + g$ and cf are in $\mathcal{R}(\alpha)$.*

$$\int f + g \, \mathrm{d}\alpha = \int f \, \mathrm{d}\alpha + \int g \, \mathrm{d}\alpha$$
$$\int cf \, \mathrm{d}\alpha = c \int f \, \mathrm{d}\alpha$$

9.5.8 Comparison. *If $f \leq g$ on $[a, b]$ then $\int f \, \mathrm{d}\alpha \leq \int g \, \mathrm{d}\alpha$, in particular*

$$\left| \int f \, \mathrm{d}\alpha \right| \leq \int |f| \, \mathrm{d}\alpha$$

9.5.9 Partition of the Integral Domain and the Distribution.

1. *If $f \in \mathcal{R}(\alpha)$ on $[a, b]$, then it's also integrable on $[a, c]$ and $[c, b]$.*
2. *If $f \in \mathcal{R}(\alpha_1)$ and $f \in \mathcal{R}(\alpha_2)$ then $f \in \mathcal{R}(\alpha_1 + \alpha_2)$ and $f \in \mathcal{R}(c\alpha)$ for $c > 0$.*

9.5.4 Stieltjes Integral as Series and as Riemann Integral

9.5.5 Fundamental Theorem of Calculus

9.5.10 Integration as Inverse of Differentiation. *Let $f \in \mathcal{R}$ on $[a, b]$, for $a \leq x \leq b$, define:*

$$F(x) = \int_a^x f \, \mathrm{d}x$$

*Then F is **continuous** on $[a, b]$, furthermore, if f is continuous at a point x_0, then F is differentiable at x_0, with*

$$F'(x_0) = f(x_0)$$

9.5.11 Fundamental Theorem of Calculus. *If $f \in \mathcal{R}$ on $[a, b]$ and if there is a differentiable function F on $[a, b]$ such that $F' = f$, then*

$$\int_a^b f \, \mathrm{d}x = F(b) - F(a)$$

9.5.6 Change of Variable and Integration by Parts

9.5.12 Change of Variable. *Let ϕ be a **strictly increasing continuous** function that maps $[A, B]$ **onto** $[a, b]$, let α be monotonoically increasing on $[a, b]$ and $f \in \mathcal{R}(\alpha)$ on $[a, b]$, define:*

$$\beta(y) = \alpha(\phi(y)), \quad g(y) = f(\phi(y))$$

then $g \in \mathcal{R}(\beta)$ and

$$\int_A^B g \, \mathrm{d}\beta = \int_a^b f \, \mathrm{d}\alpha$$

9.5.13 Integration by Parts. *Let F, G be differentiable on $[a, b]$, $F' = f \in \mathcal{R}$ and $G' = g \in \mathcal{R}$ then*

$$\int_a^b F(x)g(x) \, \mathrm{d}x = F(b)G(b) - F(a)G(a) - \int_a^b f(x)G(x) \, \mathrm{d}x$$

9.6 Sequence of Functions and Uniform Convergence

9.6.1 Definition

Uniform Convergence of $\mathbb{R} \mapsto \mathbb{R}$. Let $f_n : E \subset \mathbb{R} \to \mathbb{R}$ be a sequence of functions, $\{f_n\}$ converges **uniformly** to f if

$$\forall \varepsilon > 0, \quad \exists N, \quad \forall n \geq N, \quad \sup_{x \in E} |f_n(x) - f(x)| \leq \varepsilon$$

that is iff $\{f_n\}$ converges to f with respect to the metric induced by the sup norm $\|f\| := \sup |f(x)|$

9.6.1 Proposition (Cauchy Criterion for Uniform Convergence). *The sequence $f_n : E \to \mathbb{R}$ converges uniformly to f if and only if for any $\varepsilon > 0$ there exists an integer N such that for $m, n \geq N$,*

$$\sup |f_m(x) - f_n(x)| \leq \varepsilon$$

9.6.2 Weierstrass-M test. *Suppose f_i is a sequence of complex-valued functions defined on a set A, and there is a sequence of positive $\{M_i\}$ such that*

1. *$|f_i(x)| \leq M$, for all i, x*
2. *$\sum_{i=1}^{\infty} M_i < \infty$*

*then the series $\sum_{i=1}^{\infty} f_i(x)$ converges **absolutely** and **uniformly**.*

Proof. For any $x \in A$, by comparison test, $|f_i| \leq M_i$ and $\sum_{i=1}^{\infty} M_i$ converges, we know $\sum_{i=1}^{\infty} |f_i(x)|$ also converges, so the series converges absolutely. $g_n(x) := \sum_{i=1}^{n} f_i(x)$, for any x, $|g_n(x) - g_m(x)| \leq \sum_{i=n}^{m} |f_i(x)| \leq \sum_{i=n}^{m} M_i < \varepsilon$ which does not depend on x, so $g_n(x)$ converges uniformly. □

9.6.2 Interchange with Limit of Functions and Continuity

9.6.3 Theorem. *If $f_n \to f$ uniformly on a set E in a metric space. Let p be a limit point of E. If*

$$\lim_{x \to p} f_n(x) \to A_n \quad \textit{for all} \quad n$$

then A_n converges and

$$\lim_{x\to p} f(x) = \lim_{n\to\infty} A_n$$

Meaning:

$$\lim_{x\to p}\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty}\lim_{x\to p} f_n(x)$$

Corollary 9.6.3.1. *If f_n are continuous on E, and $f_n \to f$ uniformly then f is continuous.*

9.6.3 Interchange Limit with Integral

9.6.4 Interchange Limit and Integral. *If α is nondecreasing on $[a, b]$, $f_n \in \mathcal{R}(\alpha)$, $f_n \to f$ uniformly on $[a, b]$. Then $f \in \mathcal{R}(\alpha)$ on $[a, b]$ and*

$$\int_a^b f \, \mathrm{d}\alpha = \lim_{n\to\infty} \int_a^b f_n \, \mathrm{d}\alpha$$

9.6.4 Interchange Limit with Differentiation

9.6.5 Interchange with Differentiation. *Let $\{f_n\}$ be a sequence of functions differntiable on $[a, b]$, $f_n(x_0)$ converges for some $x_0 \in [a, b]$. If $\left\{f_n'\right\}$ converges uniformly on $[a, b]$, then $\{f_n\}$ uniformly converges to a function f and*

$$f'(x) = \lim_{n\to\infty} f_n'(x)$$

9.6.5 Partial Fraction

9.7 Special Functions

9.7.1 Real Power Series

9.7.2 Exponential and Logarithmic Functions

9.7.3 Trigonometric Functions

9.7.4 Fourier Series

9.7.5 Gamma Function

Chapter 10

Complex Analysis

Complex analysis is very useful in many aspects. For example, it gives a method to calculate integrals that are otherwise unsolvable. It is related to Fourier analysis and the Uncertainty Principle can be related to the fact that you can't simultaneously localize a function and its fourier transformation. It's related to the probability by considering the Gamma function for example. It's also useful in the number theory by the Riemann Zeta function. By taking Riemann-Zeta function at 2 we have a proof that there are infinitely many prime numbers. Another application is analyse the moments of a random variable, the moment may not exist, but hte fourier transformation does and so we can give a proof of the central limit theorem. And the Goldbach problem can be converted to an integral problem.

As an illustration of the difference between complex and real analysis. consider the four questions:

1. If $f : \mathbb{R} \to \mathbb{R}$ is differentiable, it's **not** inifinitely differentiable. For example $x^3 \sin\left(\frac{1}{x}\right)$.

2. Differentiable $f : \mathbb{R} \to \mathbb{R}$ may **not** be given by its Taylor series expansion: consider $f(x) = \exp\left\{-\frac{1}{x^2}\right\}$ for $x \neq 0$ and $f(0) = 0$.

3. Differentiable f can be defined everywhere, bounded but **not** constant. $\sin(x)$. But for $g : \mathbb{C} \to \mathbb{C}$ it is not possible.

4. The line integral over a closed curve is zero(Green's Theorem), for real analysis it's **no**. But in complex analysis it's **yes**.

A reminder of the Green's theorem which relates line integral to the volume integral. If a function has a primitive then the integral over a closed curve will be 0. A basic integral is $\oint_{\partial D(0,r)} z^n \, \mathrm{d}z$ along the unit circle. Here $z = re^{i\theta}$ and $\mathrm{d}z = ire^{i\theta} \, \mathrm{d}\theta$, so we have

$$\frac{1}{2\pi i} \oint z^n \, \mathrm{d}z = \begin{cases} 1 & \text{if} \quad n = -1 \\ 0 & \text{otherwise} \end{cases}$$

10.1 Structures on the Complex Plane

10.1.1 Algebra of the Complex Numbers

Notice the **square root** in Ahlfors 1953.

On $\mathbb{C}$ we can define $z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$ and $z_1 z_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1)$. The conjugate of z is $\overline{z} = x - iy$ and the absolute value is $\sqrt{z\overline{z}}$. The real and imaginary parts of z is x and y respectively. We have

$$\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2} \quad \overline{z_1 z_2} = \overline{z_1}\,\overline{z_2}$$

We have that $\mathbb{C}$ equipped with the complex addition and multiplication is a **field**. Any complex number can also be expressed as

$$z = re^{i\theta}$$

where r is the **radius** and the θ is the **argument**.

10.1.2 Topology

Let $|z| := \sqrt{z\overline{z}}$. The topology on $\mathbb{C}$ is the topology induced by the norm $|\cdot|$. Furthermore, $\mathbb{C}$ is a **complete** normed vector space. The completeness inherites from that of the $\mathbb{R}$.

Now we can characterise the **compact sets** and **connected sets** or **regions** in $\mathbb{C}$. A compact subset of $\mathbb{C}$ is compact iff it's closed and bounded. A subset Ω of $\mathbb{C}$ is connected iff any two points in Ω can be joint by a curve γ that is entirely contained in Ω.

A **region** is an open connected subset.

10.1.3 Geometric Representation and Riemann Sphere

The addition is parallel addition and the multiplication is rotation and scaling.

The extended complex $\mathbb{C} \bigcup \{\infty\}$ can be identified with the Riemann Sphere.

10.2 Differential Analysis of Complex Functions

10.2.1 Holomorphic Functions and Cauchy-Riemann Equations

A complex differentiable function is called **holomorphic**, being holomorphic tells a lot more than than "real" differentiable because of the multiplication of complex numbers. In particular we have that f is holomorphic iff the **Cauchy-Riemann** equation holds for its real and imaginary parts.

Holomorphic Function. Let $f : U \to \mathbb{C}$ where U is a region(nonempty open connected subset) in the complex plane, f is **holomorphic** at $z_0 = x_0 + iy_0$ if there is a complex number $f'(z_0) \in \mathbb{C}$ such that

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \phi(z)|z - z_0|$$

If f is holomorphic at every $z \in U$, then we say f is holomorphic on U. If $U = \mathbb{C}$ then the function is called **entire**.

10.2.1 Cauchy-Riemann Equation. *Let $z = x + iy$, we identify $f : \mathbb{C} \to \mathbb{C}, z \mapsto f(z)$ with the function $F : \mathbb{R}^2 \to \mathbb{R}^2, (x, y) \mapsto (u(x, y), v(x, y))$, then f is complex differentiable at $z_0 = x_0 + iy_0$* ***if and only if*** *F is real differentiable at (x_0, y_0) and*

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Furthermore,

$$\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = |f'(z_0)|^2$$

Proof. Write down the Jacobian matrix and compare with the complex terms. □

Remark. A question is if we know f is complex differentiable and u is known, can we find the imaginary part v up to a constant.

Remark. An easy rule-of-thumb is that if $f(z, \bar{z})$ does not involve $\bar{z}$, then f is holomorphic.

10.2.2 Analytic Function

Power Series. Let (c_n) be a sequence of complex numbers and z be a complex number, the series

$$\sum_{n=0}^{\infty} c_n z^n$$

is called the power series with coefficients c_n.

Depending on the choice of z this sereis might converge or diverge.

10.2.2 Convergence of Power Series. *Given the power series $\sum_{n=0}^{\infty} c_n z^n$, define*

$$\alpha = \limsup_{n\to\infty} (|c_n|)^{\frac{1}{n}}, \quad R = \frac{1}{\alpha}$$

then $\sum c_n z^n$ converges if $|z| < R$ and diveregs if $|z| > R$. R is the ***radius of convergence****.*

Proof. By comparison to the geometric series. □

10.2.3 Differentiablity of Power Series. *If $f(z) = \sum_{n=0}^{\infty} c_n z^n$ with radius of convergence R then the series is differentiable. The derivative has the same radius of convergence and we can interchange the infinite sum and the derivative:*

$$f'(z) = \sum_{n=0}^{\infty} n c_n z^{n-1}$$

Proof. First we show that $g(z) := \sum_{n=0}^{\infty} n c_n z^{n-1}$ has the radius of convergence R by seeing that $\limsup |nc_n|^{\frac{1}{n}} = \frac{1}{R}$. Then we show that $f' = g$ by splitting the $f = S_N + E_N = \sum_{n=0}^{N} c_n z^n + \sum_{N+1}^{\infty} c_n z^n$. And use the fact that we can interchange derivative and **finite sum**. □

Analytic Function. If a function $f = \sum_{n=0}^{\infty} c_n z_n$, $c_n \in \mathbb{C}$, it's analytic for $|z| < R$ where R is the radius of convergence.

10.3 Integral Analysis of Complex Functions

10.3.1 Integral Along Smooth Curves

Remark. A motivation of the complex analysis and the most important results of Cauchy and Residual theorem can be motivated from the Green theorem. If Ω is nice region with nice boundary $\partial\Omega$ and $\vec{F} = (P(x, y), Q(x, y))$ then

$$\oint \vec{F} \cdot \mathrm{d}\vec{s} = \oint_{\partial\Omega} P\,\mathrm{d}x + Q\,\mathrm{d}y = \iint_{\Omega} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathrm{d}x\,\mathrm{d}y$$

When applied to complex analysis, this implies that the line integral complex differentiable functions over a smooth curve is zero. When there are **poles,** we can use the Green theorem and reduce the volume integral over a region with poles to the line integral around the poles.

Integral over a Curve in the Complex Plane. Define the line integral over a parametrized curve $\gamma : t \in [a, b] \to z(t)$ to be:

$$\int_{\gamma} f(z)\,\mathrm{d}z := \int_{t=a}^{b} f(z(t))z'(t)\,\mathrm{d}t$$

10.3.2 Goursat's Lemma and Cauchy Integral Theorems

10.3.1 Lemma (Goursat's Lemma). *Let $f : \mathbb{C} \to \mathbb{C}$ be holomorphic on an open set containing a triangle T, then $\oint_T f = 0$.*

Proof. We divide the triangle T_0 into four similar smaller tirangles by bisecting the edges of T_0, and find that

$$\oint_{T_0} f\,\mathrm{d}z = \sum_{j=1}^{4} \oint_{T_{1j}} f\,\mathrm{d}z \implies \left| \oint_{T_0} f\,\mathrm{d}z \right| \leq \sum_{j=1}^{4} \left| \oint_{T_{1j}} f\,\mathrm{d}z \right|$$

Pick the part with the largest integral value and repeat we have

$$\left| \oint_{T_0} f\,\mathrm{d}z \right| \leq 4^n \left| \oint_{T_n} f\,\mathrm{d}z \right|$$

And the triangles converge to a point z_0 by the intersection of compact sets. Because f is holomorphic it's approximately $f(z_0) + f'(z_0)(z - z_0) + \phi(z)(z - z_0)$, the first two components have primitives and when integrated over a closed curve give 0. The last part is at most $\sup_{z \in T_n} |\phi(z)| \operatorname{diam}(T_n)$ times the length of T_n which is smaller than $\sup |\phi(z)| \frac{1}{4^n}$. □

Corollary 10.3.1.1. *Goursat for rectangles.*

A consequence of the Goursat's Lemma is that 1. every holomorphic function on a disc has a primitive and 2. the integral of the holomorphic function over a closed curve is 0.

10.3.2 Holomorphic Function has a Primitive. *Let $f : D \to \mathbb{C}$ be holomorphic on an open disc D, then there is a primitive function $F : D \to \mathbb{C}$ such that $f = F'$.*

Proof. We define a function $F(z)$ to be the integral of f first along the real axis and then along the imaginary axis. □

10.3.3 Cauchy's Integral Theorem for a disc. *Let $f : D \to \mathbb{C}$ be holomorphic in a disc, then along a closed curve $\gamma \subset D$, the integral $\oint_\gamma f(z)\,\mathrm{d}z = 0$.*

10.3.4 Cauchy's Integral Theorem. *If a function f is holomorphic in an open set that contains a simple closed piecewise-smooth curve Γ and its interior, then*

$$\oint_\Gamma f = 0$$

Remark. The Cauchy's theorem can be extended. See Jordan's theorem for piecewise-smooth curves.

10.3.5 Morera's Theorem.

Example. Show that $e^{-\pi\xi} = \int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi i x \xi}\,\mathrm{d}x$.

Example. Evaluate the integral $\int_{-\infty}^{\infty} \frac{1-\cos(x)}{x^2}\,\mathrm{d}x$. The answer is π.

10.3.6 Cauchy's Formula. *Let f be holomorphic in an open set that contains the closure of a disc D, if C is the boundary circle of D with the positive orientation, then*

$$f(z) = \frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{\zeta - z}\,\mathrm{d}\zeta \quad \textit{for any} \quad z \in D$$

Proof. Use the Keyhole method. The function $\frac{f(\zeta)}{\zeta - z}$ only blows up at z, so

$$\begin{aligned}\oint_C \frac{f(\zeta)}{\zeta - z}\,\mathrm{d}\zeta &= \oint_{C_\varepsilon} \frac{f(\zeta)}{\zeta - z} \\ &= \oint_{C_\varepsilon} \frac{f(\zeta) - f(z) + f(z)}{\zeta - z}\,\mathrm{d}\zeta\end{aligned}$$

The part $\frac{f(\zeta)-f(z)}{\zeta - z}$ is Lipschitz and converges to 0 as $\varepsilon \to 0$. What's left is $f(z)$. □

Remark. The nice thing about the Cauchy Formula is that it tranfers the dependence of f on z to an integral representation where z appears in a nice form $\frac{1}{\zeta - z}$, which leads to much easier analysis of $f(z)$.

Corollary 10.3.6.1. *If f is holomorphic in an open set Ω, then it's infinitely complex differentiable in Ω. Moreover, if $C \subset \Omega$ is a circle whose interior is also contained in Ω, then*

$$f^{(n)}(z) = \frac{n!}{2\pi i} \oint_C \frac{f(\zeta)}{(\zeta - z)^{n+1}}\,\mathrm{d}\zeta \quad \textit{for all} \quad z \in \operatorname{int}(C)$$

Proof. We use the definition of derivatives and induction. □

Corollary 10.3.6.2 (Cauchy Inequalities)**.** *If f is holomorphic in an open set that contains the closure of a disc D centered at z_0, and of radius R then*

$$\left|f^{(n)}(z_0)\right| \leq \frac{n!\|f\|_C}{R^n}$$

where $\|f\|_C = \sup_{z \in C} |f(z)|$.

10.3.7 Analyticity of Holomorphic Functions. *Let f be holomorphic in an open set Ω, let D be a disc centered at z_0 and its closure is contained in Ω, then f is* ***analytic*** *at z_0, that is it has a power series expansion at z_0:*

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n \quad \textit{for all} \quad z \in D$$

and the coefficients are

$$a_n = \frac{1}{n!} f^{(n)}(z_0) \quad \textit{for all} \quad n \geq 0$$

Proof. □

Corollary 10.3.7.1 (Liouville's Theorem)**.** *Let f be entire and bounded, then f is constant.*

10.3.8 Accumulating zeros. *Let f is a holomorphic function in a region Ω such that $f(w_k) = 0$ for a sequence of distinct points $(w_k)_{k=1}^{\infty}$ that has a limit point in Ω, then $f \equiv 0$.*

10.3.3 Singularities and Residue Theorem

Singularities, zeros and poles. f has a **pole** of order N if $\frac{1}{f}$ has a zero of order $N \in \mathbb{N}$. Essential singularities are all other cases when f blows up for example: $f(z) = \exp\{\frac{1}{z}\}$, there is no way to extend the function to $z = 0$ that's continuous, it has infinitely many negative powers in the Taylor expansion at the singularity point.

Example. What happens when $z \to 0$ with $e^{\frac{1}{z}}$. Solve $e^{\frac{1}{z}} = \omega$.

Residue.

10.3.9 Residue Theorem.

Remark. Follow the steps: 1. complexify the function; 2. find appropriate contour to integrate; 3. find the residue and use the formula.

Example. Evaluate that for Cauchy distribution $f(x) = \frac{1}{\pi}\frac{1}{1+x^2}$ the integral $\int f(x) = 1$.

Example. Evaluate $\int_{-\infty}^{\infty} \frac{\sin x}{x}\,\mathrm{d}x$ and $\int_{-\infty}^{\infty} \left(\frac{\sin}{x}\right)^2 \mathrm{d}x$

Proof. $f(z) = \frac{e^{iz}}{iz}$, $\frac{1}{2\pi i}\oint_\gamma \frac{e^{iz}}{iz}\,\mathrm{d}z = \mathrm{Res}_f(0)$ □

Example. Evaluate $\int_0^{\infty} \frac{\mathrm{d}x}{1+x^3}$.

10.3.10 Riemann's Theorem on Removable Singularities. *Let f be holomorphic on an open set $\Omega \setminus \{z_0\}$ bounded on Ω, then if z_0 is a singularity for f it's removable.*

Corollary 10.3.10.1. *If f has an isolated singularity at the point z_0, then z_0 is a pole of f iff $|f(z)| \to \infty$ as $z \to z_0$.*

Example. $e^{\frac{1}{z}}$ is not a pole. Take $z = iy$ and $z = x$, along the imaginary axis it's bounded, along the real axis it is explosive.

10.3.11 Casorati-Weierstrass. *Suppose f is holomorphic in the punctured disc $D_r(z_0) \setminus \{z_0\}$ and has an essential singularity at z_0, then $f(z) : D_r(z_0) \setminus \{z_0\} \to \mathbb{C}$ is dense in the complex plane.*

Remark. Compare this to the Picard's Little Theorem: any entire function that misses two values must be constant.

Chapter 11

Fourier Analysis

This follows from the EE 261 Stanford Online Course. Fourier Series concerns with the mathematical analysis of **periodic phenomenon** and **Fourier Transfromation** as a limiting case of Fourier Series deals with aperiodic phenomena. The purposes in engineering of Fourier analysis is to

1. **Analysis**: break up a signal/function into constituent parts.
2. **Synthesis**: reassemble the signal/function from constituent parts.

Both analysis and synthesis are accomplished by **linear operations**, i.e., **integral and series**. Need to check out the course EE 263 on linear system and EE 278.

11.1 Fourier Series of Periodic Functions

11.1.1 Introduction: Modelling Periodic Phenomena

Periodic phenomena are common and important, they are the regularly repeating patterns. Usually the repeating patterns happen in

1. **Periodic in Time**: e.g. harmonic motion, the pendulum
2. **Periodic in Space**: some measurement of physical quantity distributed over a region with **symmetry**. The periodicity arises because of the symmetry, that's why the **group theory** provides a general framework for studying Fourier analysis.

Example (The distribution of heat on a circular ring)**.** The circular ring is symmetric in space and the temperature function is periodic.

Mathematical descriptions of periodicity usually talk about **frequency** in time: that is the number of repetitions of pattern over time; and **period** in space, that's how "long" or "big" the pattern is that repeats.

Periodicity in time and space come together for example in **wave motion**. And then there is a relation between measurement of two periodicity: let λ be the **wavelength**(we fix the time and measure the wave in space), and ν be the frequency(we fix a position and let the wave comes to us), then

$$\lambda\nu = v \quad \text{the constant velocity of wave}$$

So there is a reciprocal relation between the two. We can analyse the periodic phenomena because there are simple functions that exhibit periodicity: $\cos(t)$ and $\sin(t)$, the periodicity comes from the symmetry of the unit circle.

We can model periodic signal or functions with the sin and cos functions. A function f is periodic with period 1 iff $f(t+1) = f(t), \forall t$. If we know $f(t)$ in any interval of length 1, we know the full function/signal. We can model such functions by modifying the amplitude, period and phase of the function $\sin(2\pi t)$ and then add them up:

$$g(t) = \sum_{k=1}^{n} a_k \sin(2\pi kt + \phi_k)$$

By using trignometric algebra and complex exponential, we can rewrite the sum as

$$\begin{aligned} g(t) &= \sum_{k=1}^{n} a_k \sin(2\pi kt)\cos(\phi_k) + a_k \sin(\phi_k)\cos(2\pi kt) \\ &= \sum_{k=-n}^{n} c_k e^{2\pi ikt}, \quad c_k \in \mathbb{C} \end{aligned}$$

If $g(t)$ is a real function, we know c_0 is real and $c_{-k} = \overline{c_k}$, the complex conjugate. This is a nice symmetry.

The fundamental question is, given a periodic function f of period 1, can we write

it as a linear combination.

$$f(t) = \sum_{k=-n}^{n} c_k e^{2\pi ikt} \tag{11.1}$$

11.1.1 Proposition (Finding Fourier Coefficients). *If $f(t)$ can be written as in Equation 11.1, then for any $m \in \mathbb{Z}$, the Fourier coefficient is given by*

$$c_m = \int_0^1 e^{-2\pi imt} f(t)\,\mathrm{d}t$$

Proof. Due to the fact that

$$\int_0^1 e^{2\pi i(k-m)t}\,\mathrm{d}t = \begin{cases} 1 & \text{if} \quad k = m \\ 0 & \text{otherwise} \end{cases}$$

□

11.1.2 Fourier Series

Let f be a complex-valued **integrable** function defined either periodically on $[a, b]$ or on the unit circle $e^{i\theta} : \theta \in [0, 2\pi]$.

Let the Fourier coefficients and Fourier series be defines as:

$$\hat{f}(n) := \frac{1}{b-a} \int_a^b f(x) e^{-i\frac{2\pi}{b-a}nx}\,\mathrm{d}x$$

$$f \sim \sum_{i=-\infty}^{\infty} \widehat{f}(n) e^{i\frac{2\pi}{b-a}nx}$$

11.1.2 Uniqueness of Fourier Series. *Let f be an integrable function on the circle with $\widehat{f}(n) = 0$ for all $n \in \mathbb{Z}$, then $f(\theta_0) = 0$ if f is continuous at θ_0.*

Proof. The idea is to do **analysis**: choose a sequence of trigonometric polynomials that peak at θ_0 and let the peak go to infinity. Assume $\theta_0 = 0$ and $f(0) > 0$. We construct

$$p(\theta) = \varepsilon + \cos(\theta) \quad \text{and} \quad p_k(\theta) = [p(\theta)]^k$$

By the continuity of f, we can choose $\delta : 0 < \delta < \frac{\pi}{2}$, such that □

This theorem has a lot of important results:

Corollary 11.1.2.1. *If f is continuous on the circle, and $\hat{f}(n) = 0$ for all n then $f = 0$.*

11.1.3 Theorem. *Let f be a continuous function on the circle and the Fourier series converges absolutely $\sum_{n=-\infty}^{\infty} \left|\widehat{f}(n)\right| < \infty$, then the Fourier series of f converges **uniformly** to f.*

$$\lim_{N\to\infty} S_N(f)(\theta) = f(\theta) \quad \textit{uniformly}$$

Proof. By the Weierstrass M-test. □

Remark. The question now is what conditions should f satisfy in order for the absolute convergence of Fourier series to hold. It is related to the smoothness of f.

Corollary 11.1.3.1. *If f is twice continuously differentiable on the circle then*

$$\widehat{f}(n) = O\left(\frac{1}{|n|^2}\right) \quad \textit{as} \quad |n| \to \infty$$

In particular, $S_N(f)$ converges to f absolutely and uniformly.

11.1.3 Convolution and Good Kernel

Convolution provides a way to express the partial sum of Fourier series and is a powerful tool. It's like a weighted avergae.

Convolution. Let f, g be integrable, **periodic** on unit circle, define $f * g : [-\pi, \pi] \to \mathbb{C}$ as

$$(f * g)(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) g(x - y) \, \mathrm{d}y$$

Good Kernel. A family of kernels $\{K_n(x)\}_{n=1}^{\infty}$ on the unit circle is **good kernels** if

1. $$\frac{1}{2\pi} \int_{-\pi}^{\pi} K_n(x) \, \mathrm{d}x = 1 \quad \text{for all} \quad n \geq 1$$

2. There exists $M > 0$ and for all $n \geq 1$:

$$\int_{-\pi}^{\pi} |K_n(x)| \, \mathrm{d}x \leq M$$

3. For all $\delta > 0$:
$$\int_{\delta \le |x| \le \pi} |K_n(x)| \, \mathrm{d}x \to 0 \quad \text{as} \quad n \to \infty$$

11.1.4 Approximation to Identity. *If $\{K_n\}$ is a family of good kernels, and f is integrable on the unit circle, then at every point of continuity of f we have*

$$\lim_n (f * K_n)(x) = f(x)$$

If f is continuous, then the convergence is uniform.

Remark. This is a pointwise convergence result for all the points where f is continuous. Unfortunately, the Dirichlet kernel is not a good kernel.

11.1.4 Cesaro and Abel Summation, Fejer's Theorem

Cesaro Mean. Let (c_n) be a sequence of complex numbers, then the $N-$th Cesaro Mean is

$$\sigma_N := \frac{1}{N}\left(\sum_{i=1}^{1} c_i + \sum_{i=1}^{2} c_i + \cdots + \sum_{i=1}^{N} c_i\right)$$

Fejer's Kernel. Define N-th Fejer Kernel as

$$F_N(x) := \frac{1}{N}(D_0(x) + \cdots + D_N(x))$$

where $D_i(x)$ is the Dirichlet kernel. So that the Cesaro Mean of the Fourier series is

$$\sigma_N(f)(x) = (f * F_n)(x)$$

11.1.5 Lemma. *We have F_N is a family of good kernels.*

$$F_N(x) = \frac{1}{N}\frac{\sin^2(Nx/2)}{\sin^2(x/2)}$$

We can apply the Approximation to Identity theorem to F_N.
A corollary is

Corollary 11.1.5.1. *Continuous functions f on the unit circle can be* ***uniformly*** *approximated by trigonometric polynomials such that for all $\varepsilon > 0$ there is a trig-polynomials P such that*

$$|f(x) - P(x)| < \varepsilon \quad \textit{for all} \quad -\pi \leq x \leq \pi$$

Remark. It means that trigonometric polynomials are dense in the space of continuous functions on unit circle with the sup-norm.

11.1.5 Convergence in L^2

11.1.6 Riemann Localization Principle

11.1.6 Local Derivative determines Fourier Coefficient. *If $f : \mathbb{T} \to \mathbb{C}$ is differentiable at a point θ_0, t*

11.2 Fourier Transformation

11.2.1 Schwartz Space and Functions of Moderate Decrease

11.2.2 Fourier Transformation and Inversion

11.2.3 Radon Transformation

11.3 Applications of Fourier Analysis

Chapter 12

Vector Analysis

12.1 Euclidean Space

12.1.1 Definition

Euclidean $\mathbb{R}^n$. Let n be a positive integer, then the vector space $\mathbb{R}^n$ over $\mathbb{R}$ equipped with the usual vector addition, scalar multiplication and the inner product and norm is called the n-dimensional Euclidean space.

$$\mathbf{a} + \mathbf{b}, \quad c \cdot \mathbf{a}, \quad \langle \mathbf{a} | \mathbf{b} \rangle = \sum_{i=1}^{n} a_i b_i, \quad |\mathbf{a}| = \sqrt{\langle \mathbf{a} | \mathbf{a} \rangle}$$

The dot product can be used to calculate the angle θ between two vectors $\mathbf{a}$, $\mathbf{b}$ because

$$\langle \mathbf{a} | \mathbf{b} \rangle = |\mathbf{a}||\mathbf{b}| \cos(\theta)$$

and $\langle \mathbf{a} | \mathbf{b} \rangle = 0$ iff $\mathbf{a} \perp \mathbf{b}$.

Euclidean Norm and Sup Norm. The Euclidean and Sup norms are defined and

connected by

$$|x| := \sqrt{\sum_{i=1}^{n} x_i^2}$$
$$\|x\|_\infty := \max_i |x_i|$$
$$\|x\|_\infty \leq |x| \leq \sqrt{n}\|x\|_\infty$$

As a result, they give the same topology.

12.1.1 Proposition (Completeness of $\mathbb{R}^n$). *It's easy to see from the metric induced by the norm.*

12.1.2 Compactness in $\mathbb{R}^n$. *For any subset of $\mathbb{R}^n$, it's compact iff it's closed and bounded.*

12.1.3 Proposition (Compactness of n-cell). *Every n-cell $I_n := \left\{\prod_{i=1}^{n}[a_i, b_i] : a_i < b_i\right\}$ is compact.*

12.1.2 Vector Algebra

Addition, Dot, Outer, Cross and Triple Products. See Matthews. Scalar and Vector Fields as a function of the position.

Dot Product. $\mathbf{a} \cdot \mathbf{b} := \sum_i a_i b_i$.

We can use the dot product to find the angle between $\mathbf{a}$ and $\mathbf{b}$, to determine if $\mathbf{a} \perp \mathbf{b}$ and to determine the component of $\mathbf{a}$ along a unit vector $\widehat{u}$ which is $(\mathbf{a} \cdot \widehat{u})\widehat{u}$.

In two dimensions, the determinant of $\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}$ is the signed area of the parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$. And $\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$ is the volume of the 3-dimensional parallelopide.

Cross Product.

$$\mathbf{a} \times \mathbf{b} := \begin{vmatrix} \widehat{i} & \widehat{j} & \widehat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

which is a vector with length equal to the area of the parallelogram of $\mathbf{a}, \mathbf{b}$ and the direction following the right-hand rule.

So it's easy to see that the **triple product** below also gives the volume.

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

12.2 Vector Differential

12.2.1 Differential and Operations

Differentiable and Derivative. Given an open subset U of $\mathbb{R}^n$, a map $\mathbf{f} : U \to \mathbb{R}^m$ and a point $a \in U$, $\mathbf{f}$ is differentiable at a vector $\mathbf{x}$ if there exists a **linear transformation** $A : \mathbb{R}^n \to \mathbb{R}^m$ such that for $\mathbf{h} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$,

$$\frac{|\mathbf{f}(\mathbf{x}+\mathbf{h}) - \mathbf{f}(\mathbf{x}) - A\mathbf{h}|_{\mathbb{R}^m}}{|\mathbf{h}|_{\mathbb{R}^n}} \to 0 \quad \text{as} \quad \mathbf{h} \to \mathbf{0}.$$

and define the **(total) derivative or differential** of $\mathbf{f}$ at $\mathbf{x}$ to be

$$D\mathbf{f}(\mathbf{x}) = A$$

Remark. It's equivalent: $\mathbf{f}(\mathbf{x}+\mathbf{h}) - \mathbf{f}(\mathbf{x}) = \mathbf{f}'(\mathbf{x})\mathbf{h} + \mathbf{r}(|\mathbf{h}|)$ where

$$\lim_{\mathbf{h}\to\mathbf{0}} \frac{|\mathbf{r}(\mathbf{h})|}{\mathbf{h}} = 0$$

12.2.1 Proposition (Uniqueness of Total Derivative). *If A exists, it's unique.*

12.2.2 Chain Rule. *Let U be an open set in $\mathbb{R}^n$, let $\mathbf{f} : U \to \mathbb{R}^n$ and $\mathbf{f}$ be differentiable at $\mathbf{x}_0 \in U$. Let $\mathbf{g}$ maps an open set containing $\mathbf{f}(U)$ to $\mathbb{R}^m$ and g is differentiable at $\mathbf{f}(\mathbf{x}_0)$, then $\mathbf{h} : U \to \mathbb{R}^m$ where $\mathbf{h}(\mathbf{x}) = \mathbf{g}(\mathbf{f}(\mathbf{x}))$ is differentiable at $\mathbf{x}_0$ with the differential:*

$$\mathbf{h}'(\mathbf{x}_0) = \mathbf{g}'(\mathbf{f}(\mathbf{x}_0))\mathbf{f}'(\mathbf{x}_0)$$

12.2.2 Partial Derivatives

Partial Derivatives. Let $\mathbf{f} : U \subset \mathbb{R}^n \to \mathbb{R}^m$, where U is open, let $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ and $\{\mathbf{u}_1, \ldots, \mathbf{u}_m\}$ be bases, then we can define $f_1, \ldots, f_m$ as $f_i(\mathbf{x}) = \langle \mathbf{f}(\mathbf{x}) | \mathbf{u}_i \rangle$ so that

$\mathbf{f} = \sum_{i=1}^{m} f_i \mathbf{u}_i$. Define, if the limit exists,

$$\frac{\partial f_i}{\partial x_j} := \big(D_j f_i\big)(\mathbf{x}) = \lim_{t \to 0} \frac{f_i(\mathbf{x} + t\mathbf{e}_i) - f_i(\mathbf{x})}{t}$$

12.2.3 Contraction Principle

12.2.4 Inverse Function Theorem

12.2.3 Inverse Function Theorem. *Let $\mathbf{f} \in C^1(E, \mathbb{R}^n)$, where E is an open subset of $\mathbb{R}^n$. $\mathbf{f}'(\mathbf{a})$ is invertible for some $\mathbf{a} \in E$ and $\mathbf{b} = \mathbf{f}(\mathbf{a})$, then*

1. *there are open sets $U, V \in \mathbb{R}^n$ such that $\mathbf{a} \in U, \mathbf{b} \in V$ and $\mathbf{f}$ is one-to-one on U and $\mathbf{f}(U) = V$, hence it has an inverse.*
2. *if $\mathbf{g}$ is the inverse of $\mathbf{f}$, $\mathbf{g} : V \to U$, $\mathbf{g}(\mathbf{f}(\mathbf{x})) = \mathbf{x}$ then $\mathbf{g} \in C^1(V)$.*

12.2.5 Implicit Function Theorem

12.2.6 The Rank Theorem

12.2.7 Gradient, Divergence and Curl

12.3 Integral in Euclidean Space

12.3.1 Rectifiable Curve

Curve, Arc and Closed Curve. A curve in $\mathbb{R}^n$ is a continuous mapping $\gamma : [a, b] \subset \mathbb{R} \to \mathbb{R}^n$. If γ is **one-to-one** then it's an **arc**, if $\gamma(a) = \gamma(b)$ then it's **closed**.

Rectifiable Curve. Let γ be a curve and $P = \{x_0, x_1, \ldots, x_k\}$ a partition on $[a, b]$, define the **length** of γ as

$$\Lambda(\gamma) := \sup_P \Lambda(P, \gamma) \quad \text{where} \quad \Lambda(P, \gamma) := \sum_{i=1}^{k} |\gamma(x_i) - \gamma(x_{i-1})|$$

A curve with finite length is called **rectifiable**.

12.3.1 Continuously differentiable curve is rectifiable. *If γ' is continuous on $[a, b]$ then γ is rectifiable and*

$$\Lambda(\gamma) = \int_a^b |\gamma'(t)| \, \mathrm{d}t$$

12.3.2 Line, Surface and Volume Integral

12.4 Tensor Product

Chapter 13

Measure and Lebesgue Theory

13.1 Abstract Measure Space

13.1.1 Measurable Space and Completion

Measure is the extension of such notions as length, volume, etc. A simple idea is that they must be "additive", but uncountable additive is too much and gives rise to unusual situations(for example, $\sum_{x\in[0,1]} \mu(x) = 1$ while $\mu(x) = 0$). So we instead impose countable additivity. Even then, we can't apply measure to the power set(for a counterexample, think of Vitali sets in $\mathbb{R}$). In fact, even finitely additive measure is inconsistent if we let the domain be the power set, we need to restrict to $\sigma-$algebra.

Sigma-Algebra and Measure

Ring, Semiring, Algebra, $\sigma-$algebra. Here are some structures of subsets of a set

1. A **semiring** $\mathcal{D}$ is a collection of subsets such that $\varnothing \in \mathcal{D}$ and for any A and B in $\mathcal{D}$, $A \bigcap B \in \mathcal{D}$ and $A \setminus B = \bigcup_{i=1}^{n} C_i$ for finite n and disjoint C_i

2. A **ring** $\mathcal{R}$ is a collection of subsets of a set X such that $\varnothing \in \mathcal{R}$ and for any $A, B \in \mathcal{R}$, $A \bigcup B$ and $B \setminus A \in \mathcal{R}$.

3. An **algebra** $\mathcal{A}$ is a ring such that $X \in \mathcal{A}$.

4. A $\sigma-$**algebra** $\mathcal{E}$ is an algebra such that $\bigcup_{i=1}^{n} A_i \in \mathcal{E}$, closed under countable set union.

σ-algebra. Given a set X, σ-algebra is a subset $\mathcal{E}$ of 2^X such that $X \in \mathcal{E}$, $\mathcal{E}$ is closed under set complement and **countable** union.

1. $A \in \mathcal{E} \implies X \setminus A \in \mathcal{E}$
2. $A_1, A_2, \cdots \in \mathcal{E} \implies \bigcup_n A_n \in \mathcal{E}$

13.1.1 Proposition (Generating σ-algebra). *Arbitrary intersection of a family of σ-algebras is still a σ-algebra, and hence for a collection $\mathcal{C}$ of subsets of X, there is a smallest σ-algebra that includes $\mathcal{C}$, $\sigma(\mathcal{C})$: the σ-algebra generated by $\mathcal{C}$.*

Measure. Let $(X, \mathcal{E})$ be a measurable space, a **measure** is a mapping $\mu : \mathcal{E} \to \overline{\mathbb{R}}_+ := [0, \infty]$ such that:

1. $\mu(\varnothing) = 0$
2. μ is countably additive, that is $\mu(\sqcup_n A_n) = \sum_n \mu(A_n)$ for every sequence $(A_n) \subset \mathcal{E}$ that are disjoint.

Example. Here are some examples,

1. **Dirac Measure** sitting at $x \in X$ is $\delta_x(A) = \mathbf{1}_{x \in A}$
2. **Counting Measure**: $\mu(A) = \text{card}(A)$ if A is finite, and ∞ if A is infinite, for $A \subset X$. Let D be a fixed countable subset of X, $\nu(A) = \sum_{x \in D} \delta_x(A)$
3. **Discrete Measure**: Let D be a countable subset of X and for each $x \in D$, there is a positive $m(x)$: $\mu(A) := \sum_{x \in D} m(x)\delta_x(A)$.

13.1.2 Proposition (Properties of Measure). *Let μ be a measure on $(X, \mathcal{E})$, then the following properties hold for $A, B, \{A_n\} \in \mathcal{E}$,*

1. ***Monotonicity**: $A \subset B \implies \mu A \leq \mu B$*
2. ***Boole's Inequality, Subadditivity** $\mu(\bigcup A_n) \leq \sum_n \mu(A_n)$*
3. ***Sequential Continuity from below**: $A_n \uparrow A \implies \mu(A_n) \uparrow \mu(A)$.*
4. ***Sequential Continuity from above**: $A_n \downarrow A, \mu(A_1) < \infty \implies \mu(A_n) \downarrow \mu(A)$*

Finiteness of Measure. Let μ be a measure on $(X, \mathcal{E})$,

1. μ is **finite** if $\mu(X) < \infty$
2. μ is σ**-finite** if there exists a measurable partition $(A_n)_{n=1}^{\infty}$ of X such that $\mu(A_n) < \infty$ for all n.
3. μ is Σ**-finite** if there exists a sequence of finite measures μ_n such that $\mu = \sum_n \mu_n$

And we have finiteness $\implies$ σ-finite $\implies$ Σ-finite.

Negligible Sets and Completion

Negligible Sets. If $(X, \mathcal{E}, \mu)$ is a measure space, then every $E \in \mathcal{E}$ such that $\mu E = 0$ is called a negligible or null set. A measure whose domain includes **all subsets of negligible sets** is called **complete**.

13.1.3 Completion of Measure Space. *Let $(X, \mathcal{E}, \mu)$ be a measure space, let $\mathcal{N} := \{N \in \mathcal{E} : \mu(N) = 0\}$ and*

$$\bar{\mathcal{E}} := \left\{E \bigcup F : E \in \mathcal{E} \quad \textit{and} \quad F \subset N \quad \textit{for some} \quad N \in \mathcal{N}\right\}$$

Then $\bar{\mathcal{E}}$ is a σ-algebra and there is a unique extension of $\bar{\mu}$ of μ which is a complete measure on $\bar{\mathcal{E}}$.

Proof. See Folland Question C1.5. □

There are some useful facts about negligible sets.

Measurable Function

Measurable Mappings. A mapping $f : (X, \mathcal{E}) \to (Y, \mathcal{F})$ where X, Y are measurable spaces is measurable if for any $B \in \mathcal{F}$, $f^{-1}(B) \in \mathcal{E}$.

Consider the identity mapping from $(X, \mathcal{E}_1) \to (X, \mathcal{E}_2)$, then to make identity mapping measurable we need $\mathcal{E}_2 \subset \mathcal{E}_1$.

13.1.4 Criterion for Measurability. *$f : (X, \mathcal{E}) \to (Y, \mathcal{F})$, if there is a family of subsets $\mathcal{G}$ in Y that generates $\mathcal{F}$, f is measurable iff for any $G \in \mathcal{G}$, $f^{-1}(G) \in \mathcal{E}$.*

Measurability on Subset. $f : (X, \mathcal{M}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is **measurable on a subset** $E \in \mathcal{M}$, if for all $B \in \mathcal{B}(\mathbb{R})$, $f^{-1}(B) \bigcap E \in \mathcal{M}$.

In particular, for the Borel σ-algebra we can take the topology as the set generating the σ-algebra.

13.1.2 Outer Measure and Caratheodory's Theorem

See Folland Chapter 1.3 - 1.5.

The idea goes as this: we have Caratheodory theorem that asserts that the restriction of an outer measure μ^* to a σ-algebra is a complete measure. A premeasure on an algebra can induce an outer measure and hence a measure. First we make the following definitions.

Outer Measure and Outer Measurable. Given a space X, an **outer measure** is $\mu^* : 2^X \to [0, \infty]$, such that

1. $\mu^* \varnothing = 0$
2. if $E \subset F \subset X$, then $\mu^*(E) \leq \mu^*(F)$
3. μ^* is coutably subaddtive: $\mu^*\left(\bigcup_{j=1}^{\infty} A_j\right) \leq \sum_j \mu^*(A_j)$

A set $E \subset X$ is μ^*-measurable, or $E \in \mathcal{M}^*$ if for any $A \subset X$:

$$\mu^*(A) = \mu^*(A \bigcap E) + \mu^*(A \bigcap E^c).$$

Premeasure on Algebra. Let X be a space with an **algebra** $\mathcal{A}$, a **premeasure** is $\mu_0 : \mathcal{A} \to [0, \infty]$ such that

1. $\mu_0 \varnothing = 0$,
2. if **disjoint** $(A_j)_{j=1}^{\infty} \subset \mathcal{A}, \sqcup_j A_j \in \mathcal{A}$, then $\mu_0(\sqcup_j A_j) = \sum_j \mu_0 A_j$.

13.1.5 Caratheodory Outer Measure Theorem. *If μ^* is an outer measure on X, then $\mathcal{M}^*$ of outer measurable sets is a σ-algebra and $\mu^*|\mathcal{M}^*$ is a **complete** measure.*

13.1.6 Proposition (Outer Measure Induced by ρ on $\mathcal{S} \subset 2^X$). *Let $\mathcal{S} \subset 2^X$ and $\rho : \mathcal{S} \to [0, \infty]$ be such that $\varnothing, X \in \mathcal{S}$ and $\rho(\varnothing) = 0$, then for $E \subset X$ define*

$$\mu^*(E) := \inf \left\{ \sum_{i=1}^{\infty} \rho(A_i) : A_i \in \mathcal{S}, E \subset \bigcup A_i \right\}$$

then μ^* is an outer measure.

13.1.7 Proposition (Outer Measure Induced by Premeasure). *Let μ_0 be a premeasure on an algebra $\mathcal{A}$, then define on 2^X*

$$\mu^*(E) := \inf\left\{\sum_{j=1}^{\infty}\mu_0(A_j) : A_j \in \mathcal{A}, E \subset \bigcup_{j=1}^{\infty} A_j\right\}$$

*called the **outer measure induced by the premeasure** μ_0, then*

1. *$\mathcal{A} \subset \mathcal{M}(\mu^*)$, i.e. every $A \in \mathcal{A}$ is μ^*-measurable.*
2. *$\mu^*|\mathcal{A} = \mu_0$.*

13.1.8 Hahn–Kolmogorov Extension Theorem. *There are three statements:*

1. *Given a space X with an algebra $\mathcal{A}$ and a premeasure μ_0 on $\mathcal{A}$, we can extend μ_0 to a measure μ on σ-algebra $\mathcal{M} = \sigma(\mathcal{A})$ generated by $\mathcal{A}$, such that $\mu|\mathcal{A} = \mu_0$.*
2. *If ν is another measure on $\mathcal{M}$, then $\nu(E) \leq \mu(E), \forall E \in \mathcal{M}$ with equality if $\mu(E) < \infty$.*
3. *If μ_0 is σ-finite then μ is the unique extension of μ_0.*

Let $\mathcal{C} := \{(a, b] : a, b \in \mathbb{R}\}$ and let $\lambda : \mathcal{C} \to [-\infty, \infty]$ be defined as $\lambda(a, b] = b - a$. We first prove that λ and generalised length $\mu(a, b] = G(b) - G(a)$ is countably additive. Then we extend it from semiring $\mathcal{C}$ to the σ-algebra.

The **outer measure** has the property of **continuity from below** even for possibly nonmeasurable sets.

See Folland Q22 in Chapter 1 for details on the σ-algebras involved(that the unique extension is the completion).

13.1.3 Borel and Lebesgue-Stieltjes Measure

Lebesgue Measure is natural in calculating the volume of a subset of $\mathbb{R}^n$. Here we first construct the Lebesgue measure on $\mathbb{R}$. Starting with the length of intervals, we extend the usual sense of length to a larger family of subsets (the σ-algebra).

Borel σ-algebra. Let X be topological, then the σ-algebra generated by all the open sets of X is called the Borel σ-algebra($\mathcal{B}_E$ or $\mathcal{B}(E)$). Its elements are called the **Borel sets**. A measure on the Borel σ-algebra $\mathcal{B}(\mathbb{R})$ is called **Borel Measure**

Borel Functions. Let $(X, \mathcal{T})$ be a topological space, let $\mathcal{E} := \sigma(\mathcal{T}) = \mathcal{B}(X)$, then the $\mathcal{E}$-measurable functions are called **Borel Functions**.

13.1.9 Proposition (Borel σ-algebra on $\mathbb{R}$). *$\mathcal{B}_{\mathbb{R}}$ is generated by each of (1) open intervals; (2) closed intervals; (3) half-open intervals; (4) open rays and closed rays.*

An Algebra of h-intervals. **H-intervals** are a member of either $\{(a, \infty)\}$, $\varnothing$ or

$$\{(a, b] : -\infty \leq a < b < \infty\}.$$

13.1.10 Proposition (An Algebra and Premeasure on $\mathbb{R}$). *We can give an algebra and a premeasure on $\mathbb{R}$:*

1. *Let $\mathcal{A}$ be **finite disjoint union** of h-intervals then $\mathcal{A}$ is an algebra. $\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{A})$.*
2. *Let $F : \mathbb{R} \to \mathbb{R}$ be increasing and right continuous, for disjoint h-intervals $(a_j, b_j], j \leq n$. Define*

$$\mu_0\left\{\bigcup_{j=1}^{n}(a_j, b_j]\right\} = \sum_{j=1}^{n}[F(b_j) - F(a_j)]$$

then μ_0 is a premeasure on $\mathcal{A}$.

13.1.11 Construction of Lebesgue-Stieltjes Measure. *We can now construct the **Lebesgue - Stieltjes measure** associated with F.*

1. *For increasing, right continuous $F : \mathbb{R} \to \mathbb{R}$, there is a unique **Borel measure** μ_F such that $\mu_F((a, b]) = F(b) - F(a), \forall a, b$.*
2. *If G is another such function, then $\mu_F = \mu_G$ iff $F - G$ is a constant.*
3. *Conversely, if μ is a Borel measure on $\mathbb{R}$ that is finite on all bounded Borel sets, then F is increasing and right continuous, $\mu = \mu_F$ where F is defined:*

$$F(x) = \begin{cases} \mu(0, x] & \text{if} \quad x > 0 \\ 0 & \text{if} \quad x = 0 \\ -\mu(x, 0] & \text{if} \quad x < 0 \end{cases}$$

*The **Lebesgue-Stieltjes measure** is the completion of μ_F that is given by the outer measure induced by the premeasure restricted on the outer measurable sets.*

13.1.12 Lemma (Replace H-Intervals with Open Intervals). *For the Lebesgue-Stieltjes measure μ on $\mathbb{R}$, for every set $E \in \mathcal{M}_\mu$,*

$$\begin{aligned}\mu(E) &:= \inf\left\{\sum_{j=1}^{\infty}\mu(a_j, b_j] : E \subset \bigcup_{j=1}^{\infty}(a_j, b_j]\right\}\\ &= \inf\left\{\sum_{j=1}^{\infty}\mu(a_j, b_j) : E \subset \bigcup_{j=1}^{\infty}(a_j, b_j)\right\}\end{aligned}$$

13.1.13 Properties of Lebesgue Measurable Sets. *Let μ be a complete Lebesgue-Stieltjes measure on $\mathbb{R}$ associated with some F, $\mathcal{M}_\mu$ be its domain, we have the following characterisations,*

1. ***Regularity:*** *we can approximate measurable sets by open and compact sets. If $E \in \mathcal{M}_\mu$*

$$\begin{aligned}\mu(E) &= \inf\{\mu(U) : E \subset U, U \quad \text{is open} \quad \}\\ &= \sup\{\mu(K) : K \subset E, K \quad \text{is compact} \quad \}.\end{aligned}$$

2. ***Criterion For Measurability:*** *Let $E \subset \mathbb{R}$, $E \in \mathcal{M}_\mu \iff E = V \setminus N_1$ where V is a G_δ set and $\mu(N_1) = 0 \iff E = H \bigcup N_2$ where H is a F_σ set and $\mu(N_2) = 0$.*

3. ***Littlewood's First Principle:*** *If $E \in \mathcal{M}_\mu$ and $\mu(E) < \infty$, then for any fixed ε there is a **finite** union of intervals F such that*

$$\mu(E \triangle F) < \varepsilon$$

13.1.14 Lemma (Q18 Lemma). *Let $\mathcal{A}$ be an algebra on X and $\mathcal{A}_\sigma$ be the set of all countable union of members of $\mathcal{A}$ and $\mathcal{A}_{\sigma\delta}$ the set of all countable intersection of members of $\mathcal{A}_\sigma$. Let μ_0 be a premeasure on $\mathcal{A}$ and μ^* be induced by μ_0, then*

1. *For any $E \subset X$ and $\varepsilon > 0$ there exists $A \in \mathcal{A}_\sigma$ such that $E \subset A$ and $\mu^*(A) \leq \mu^*(E) + \varepsilon$.*

2. *If $\mu^*(E) < \infty$ then E is μ^*- measurable iff there exists $B \in \mathcal{A}_{\sigma\delta}$ such that $E \subset B$ and $\mu^*(B \backslash E) = 0$.*

3. *If μ_0 is σ-finite, then the condition $\mu^*(E) < \infty$ in (2) is not needed.*

Lebesgue Measure. The complete measure μ_F associated with $F(x) = x$ is the **Lebesgue measure**, its domain which includes the $\mathcal{B}_{\mathbb{R}}$ is the **Lebesgue measurable** sets denoted by $\mathcal{L}$. Then it enjoys the properties:

1. **Measurability:** If $E \in \mathcal{L}$ then $E + s \in \mathcal{L}$ and $rE \in \mathcal{L}$ for $s, r \in \mathbb{R}$.

2. **Invariance:** $m(E + s) = m(E)$ and $m(rE) = |r|m(E)$.

13.1.4 Product Measure

Product sigma-algebra

Product σ-algebra. Let $(E_\alpha, \mathcal{E}_\alpha), \alpha \in A$ be a family of measurable spaces, then the **product σ-algebra** is defined as

$$\otimes_{\alpha \in A} \mathcal{E}_\alpha := \sigma\left\{\pi_\alpha^{-1}(S) : S \in \mathcal{E}_\alpha, \alpha \in A\right\}$$

that is the σ-algebra generated by the sets above.

13.1.15 Proposition (Finite and Countable Product σ-algebra). *If the index set A is countable, then*

$$\otimes_{\alpha \in A} \mathcal{E}_\alpha = \sigma\left\{\prod_{\alpha \in A} S_\alpha : S_\alpha \in \mathcal{E}_\alpha\right\}.$$

*where the sets are called **measurable rectangles**. For example on $E_1 \times E_2$ the product σ-algebra $\mathcal{E}_1 \otimes \mathcal{E}_2$ is the σ-algebra generated by $\{A \times B : A \in \mathcal{E}_1, B \in \mathcal{E}_2\}$.*

13.1.16 Proposition (Measurability wrt the Product σ-algebra). *$X : (\Omega, \mathcal{H}) \to (E, \mathcal{E}), Y : (\Omega, \mathcal{H}) \to (F, \mathcal{F})$ are measurable iff $Z = (X, Y) : (\Omega, \mathcal{H}) \to (E \times F, \mathcal{E} \otimes \mathcal{F})$ is measurable.*

Proof. $\implies$. The set $\{A \times B : A \in \mathcal{E}, B \in \mathcal{F}\}$ generates the σ-algebra $\mathcal{E} \otimes \mathcal{F}$, and for any member of this set $A_1 \times B_1$, $Z^{-1}(A_1 \times B_1) = X^{-1}(A_1) \bigcap Y^{-1}(B_1) \in \mathcal{H}$. So Z is measurable. □

$\Longleftarrow$. Let Z be measurable, then $Z^{-1}(A \times F) = X^{-1}(A) \in \mathcal{H}$ for any $A \in \mathcal{E}$, so X is measurable. □

Q.E.D □

Remark. Notice that product σ-algebra preserves the measurability of the product function, just like product topology preserves the continuity.

13.1.17 Proposition (Measurability in Infinite Products). *Let $(E, \mathcal{E})$ be a measurable space, $(F, \mathcal{F}) := \otimes_{t \in T}(G_t, \mathcal{G}_t)$, for each $t \in T$, let f_t be a mapping from E to G_t, for each $\omega \in E$, define $f(\omega)$ as the point $(f_t(\omega))_{t \in T} \in F$.*

Then mapping $f : E \to F$ is measurable iff $f_t : E \to G_t$ is measurable for every t.

13.1.18 Proposition (Measurability of Sections). *If $E \in \mathcal{E} \otimes \mathcal{F}$ then $E_x := \{y \in Y : (x, y) \in E\} \in \mathcal{F}$ for all x. If f is $\mathcal{E} \otimes \mathcal{F}$-measurable, then $f_x(y) = f(x, y)$ is $\mathcal{F}$-measurable for all x.*

13.1.19 Borel σ-algebra of product topology and product σ-algebra of Borel σ-algebras. *The theorem has two parts. Let $(X_1, \mathcal{T}_1)$ and $(X_2, \mathcal{T}_2)$ be two topological spaces.*

1. *the Borel σ-algebra of product topology includes the product σ-algebra of Borel σ-algebras.*

2. *If both X_1 and X_2 are **second countable** then the aforementioned two σ-algebras are the same.*

Proof. **Part 1:** The product σ-algebra of $\mathcal{B}(X_1)$ and $\mathcal{B}(X_2)$ is the σ-algebra generated by $\{A \times B : A \in \mathcal{B}(X_1), B \in \mathcal{B}(X_2)\}$, because $\sigma(C_1) \subset \sigma(C_2)$ iff $C_1 \subset \sigma(C_2)$, we need to show that

$$\mathcal{B}(X_1) \times \mathcal{B}(X_2) \subset \mathcal{B}(X_1 \times X_2) \quad \text{with the product topology} \quad =: \mathcal{B}$$

Let $\mathcal{U} := \{A \subset X_1 : A \times X_2 \in \mathcal{B}(X_1 \times X_2)\}$, then first $X_1 \in \mathcal{U}$ and for any contable union of $A_n \in \mathcal{U}$, $\bigcup_n A_n \times X_2 = \bigcup_n (A_n \times X_2) \in \mathcal{B}$, so $\bigcup_n A_n \in \mathcal{U}$, and $A^c \times X_2 = (X_1 \times X_2) \setminus (A \times X_2) \in \mathcal{B}$, so $A^c \in \mathcal{U}$. So $\mathcal{U}$ is a σ-algebra. $\mathcal{T}_1 \subset \mathcal{U}$, so $\mathcal{B}(X_1, \mathcal{T}_1) \subset \mathcal{U}$. For any $A \in \mathcal{B}(X_1, \mathcal{T}_1)$, $A \times X_2 \in \mathcal{B}$. Similarly we can find $\mathcal{V}$ such that for any $B \in \mathcal{B}(X_2)$, $X_1 \times B \in \mathcal{B}$.

As a result for any $A \times B \in \mathcal{B}(X_1, \mathcal{T}_1) \times \mathcal{B}(X_2, \mathcal{T}_2)$, $A \times B = A \times X_2 \bigcap X_1 \times B \in \mathcal{B}$.

Part 2: Let $\mathcal{W}$ be the countable base of $X_1 \times X_2$ consisting of $A \times B$ where A, B are from the countable bases of X_1, X_2, then $\sigma(\mathcal{W})$ is the σ-algebra of the product topology(because the base is only countable, any open set can be written as the countable union of members of the base). It's inluded in the product σ-algebra(for any $W_1 \times W_2 \in \mathcal{W}$, $W_1 \in \mathcal{T}_1 \subset \mathcal{B}(X_1, \mathcal{T}_1)$). □

Product Measure

Let $(X, \mathcal{E}, \mu)$ and $(Y, \mathcal{F}, \nu)$ be two measure spaces, then we construct a **product measure** on $(X \times Y, \mathcal{E} \otimes \mathcal{F})$.

Product Measure. A product measure $\mu \times \nu$ on $\mathcal{E} \otimes \mathcal{F}$ is a measure such that for all $A \in \mathcal{E}$ and $B \in \mathcal{F}$,

$$\mu \times \nu(A \times B) = \mu(A) \times \nu(B)$$

13.1.20 Existence and Uniqueness of Product Measure. *The following statements are true:*

1. *Given $(X, \mathcal{E}, \mu)$ and $(Y, \mathcal{F}, \nu)$, there exists a product measure.*
2. *If $(X, \mathcal{E}, \mu)$ and $(Y, \mathcal{F}, \nu)$ are σ-finite, then the product measure is unique.*

Proof. *1.* Let $\pi : \mathcal{E} \times \mathcal{F} \to [0, \infty]$ be defined as

$$\pi(A \times B) = \mu(A) \times \nu(B)$$

Then we want to verify the conditions in the Theorem 13.1.8: 1. $\mathcal{E} \times \mathcal{F}$ can be extended to an algebra; 2. π can be extended to a premeasure on that algebra and 3. $\sigma(\mathcal{E} \times \mathcal{F}) = \mathcal{E} \otimes \mathcal{F}$. In that case we can extend the measure π to a measure on $\mathcal{E} \otimes \mathcal{F}$. □

2. For uniqueness, we want to verify that if μ and ν are σ-finite, then π is σ-finite and hence the extension is unique. □

□

13.1.5 Transition Kernel

Kernel. Given measurable $(E, \mathcal{E})$ and $(F, \mathcal{F})$, Let $K : E \times \mathcal{F} \to [0, \infty]$, K is called a **transition kernel** from $(E, \mathcal{E})$ to $(F, \mathcal{F})$ if

1. $x \mapsto K(x, B)$ is $\mathcal{E}$-measurable for every $B \in \mathcal{F}$.

2. $B \mapsto K(x, B)$ is a measure on $(F, \mathcal{F})$ for every $x \in E$.

13.1.21 Proposition (Measure-Kernel-Function). *Let K be a trasition kernel from $(E, \mathcal{E})$ to $(F, \mathcal{F})$, then*

$$Kf(x) := \int_F f(y)K(x, \mathrm{d}y), \quad x \in E$$

defines a function $Kf \in \mathcal{E}_+$ for every $f \in \mathcal{F}_+$.

$$\mu K(B) = \int_E K(x, B)\,\mathrm{d}\mu$$

defines a measure μK on $(F, \mathcal{F})$ for every measure μ on $(E, \mathcal{E})$;

$$(\mu K)f = \mu(Kf) = \int_E \mu(\mathrm{d}x) \int_F f(y)K(x, \mathrm{d}y)$$

Proof. See Çınlar 2011. □

The next theorem is a general method to construct measure on product space $(E \times F, \mathcal{E} \otimes \mathcal{F})$:

13.1.22 Construction of Product Measure. *Let μ be a measure on $(E, \mathcal{E})$ and K be a Σ-finite transition kernel from $(E, \mathcal{E})$ to $(F, \mathcal{F})$, for $f \in (\mathcal{E} \otimes \mathcal{F})_+$,*

$$\pi f := \int_E \mu(\mathrm{d}x) \int_F f(x, y)K(x, \mathrm{d}y)$$

defines a measure π on $(E \times F, \mathcal{E} \otimes \mathcal{F})$.

Furthermore, if μ is σ-finite and K is σ-finite, then π is σ-finite, and is the unique measure on $(E \times F)$ such that

$$\pi(A \times B) = \int_A K(x, B)\mu(\mathrm{d}x), \quad A \in \mathcal{E}, B \in \mathcal{F}.$$

13.1.6 Kolmogorov Extension Theorem

Theorem 10 (Kolmogorov extension theorem) Let the situation be as in Problem 8. If all the measurable spaces $(\Omega_i, \mathcal{F}_i)$ are standard Borel, then there exists probability measure μ_A solving the extension problem (which is then unique, thanks to Exercise 9).

13.2 Integration of Functions

13.2.1 Numberical Functions and Measurability

Measurable Numerical Functions. For the following mappings:

1. **numerical function** $f : (X, \mathcal{E}) \to [-\infty, +\infty] =: \overline{\mathbb{R}}$
2. **real-valued function** $f : (X, \mathcal{E}) \to \mathbb{R} := (-\infty, +\infty)$
3. **nonnegative function** $f : (X, \mathcal{E}) \to \overline{\mathbb{R}}_+ := [0, \infty]$

the function is $\mathcal{E}$-measurable if it's measurable relative to $\mathcal{E}$ and the Borel σ-algebra on the codomain. The set of $\mathcal{E}$-measurable numerical functions is also denoted $\mathcal{E}$, if they are nonnegative, it's $\mathcal{E}_+$.

Measurable Complex Function. $f : (X, \mathcal{E}) \to \mathbb{C}$ is measurable iff $\operatorname{Re} f$ and $\operatorname{Im} f$ are both $\mathcal{E}$-measurable.

13.2.1 Proposition (Continuity). *A continuous function between metric spaces $f : (X, \mathcal{B}(X)) \to (Y, \mathcal{B}(Y))$ is Borel measurable.*

Remark. Be careful that $\mathcal{L}$(Lebesgue measurable) is strictly larger than $\mathcal{B}$. So composition might not be Lebesgue measurable.

13.2.2 Proposition (Measurability Criterion). *For a numerical function $f : (E, \mathcal{E}) \to \overline{\mathbb{R}}$ it is measurable iff for any $r \in \mathbb{R}$, $f^{-1}[-\infty, r] \in \mathcal{E}$ iff $f^{-1}[-\infty, r) \in \mathcal{E}$.*

13.2.3 Proposition (Measurability of Operations). *For measurable functions $f, g : (X, \mathcal{E}) \to \overline{\mathbb{R}}$ we have*

1. $\max(f, g)$ *and* $\min(f, g)$ *are measurable.*
2. *Let (f_n) be a sequence of $\mathcal{E}$-measurable numberical functions, then*

$$\inf f_n \quad \sup f_n \quad \liminf f_n \quad \limsup f_n$$

are $\mathcal{E}$-measurable, if $\lim f_n$ *exists for every $x \in X$, it's measurable too.*

For measurable $f, g : (X, \mathcal{E}) \to \mathbb{C}$ or $\mathbb{R}$ functions:

1. *$(f+g)$ is $\mathcal{E}$-measurable.*

2. *fg is measurable*

3. *cf is measurable, for any $c \in \mathbb{C}$ or $\mathbb{R}$.*

So the space $\mathcal{L}^0(X, \mathcal{E})$ is a vector space over $\mathbb{R}$.

Proof of Proposition 13.2.3. $f+g$. □

□

Simple Function. A simple function $\phi : (X, \mathcal{E}) \to \mathbb{C}$ is a **finite sum**:

$$\phi = \sum_{i=1}^{n} a_i \mathbf{1}_{E_i}$$

where $a_i \in \mathbb{C}$ and $E_i \in \mathcal{E}$. Simple functions are **measurable**. The **standard representation** is

$$\phi = \sum_{i=1}^{n} a_i \mathbf{1}_{E_i} \quad \text{where} \quad E_i = f^{-1}(\{a_i\}) \quad \text{and} \quad \phi : X \to \{a_1, \ldots, a_n\}$$

13.2.4 Proposition (Simple Functions Operations). *If f and g are simple, then the following are simple:*

$$f+g \quad f-g \quad fg \quad f/g \quad f \vee g \quad f \wedge g$$

13.2.5 Proposition (Simple Function Approximation). *Let $(X, \mathcal{E})$ be a measurable space.*

1. *If $f : X \to \overline{\mathbb{R}}_+$ is measurable, then there is a sequence of simple functions $\{\phi_n\}$ such that $0 \le \phi_1 \le \cdots \le f$, $\phi_n \to f$ pointwise and $\phi_n \to f$ **uniformly** on any set on which f is bounded.*

2. *If $f : X \to \mathbb{C}$, then there is an increasing sequence of simple functions $\{\phi_n\}$, $\phi_n \to f$ pointwise and $\phi_n \to f$ uniformly on any set on which f is bounded.*

13.2.2 Integral of Measurable Functions

Lebesgue integral is a completion of Riemann integral just like real numbers are completion of rational numbers.

Let $(X, \mathcal{E}, \mu)$ be **measure space**, then the measure is a mapping $\mu : \mathcal{E} \to \overline{\mathbb{R}}_+ = [0, \infty]$, it will be seen that by defining **integration**, we extend the domain of μ to $\mathcal{E}_+ := \{f : X \to [0, \infty], f \text{ is measurable }\}$(and beyond), that is $\mu : \mathcal{E}_+ \to \overline{\mathbb{R}}_+$. Notice that the σ-algebra $\mathcal{E}$ is identified with the indicator functions.

We can take integration of a measurable numerical function $f : (X, \mathcal{E}) \to [-\infty, +\infty]$. The definition starts from simple functions and we define the integral in the following procedure:

1. Define the integral of **nonnegative simple functions** wrt μ: $\mu f := \int f \, \mathrm{d}\mu := \sum_{i=1}^{n} a_i \mu(B_1)$, prove it is well-defined(equals to the value as we write f in canonical form).

2. For nonnegative simple function, the integral is linear: $\int f + g \, \mathrm{d}\mu = \int f \, \mathrm{d}\mu + \int g \, \mathrm{d}\mu$ and $\int cf \, \mathrm{d}\mu = c \int f \, \mathrm{d}\mu$.

3. For nonnegative simple functions, if $f(x) \leq g(x)$for all x then $\int f \, \mathrm{d}\mu \leq \int g \, \mathrm{d}\mu$.

4. For any nonnegative measurable(wrt to the Borel σ-algebra) function $f : (X, \mathcal{E}, \mu) \to ([0, \infty], \mathcal{B})$, define the integral as

$$\int f \, \mathrm{d}\mu := \sup \left\{ \int g \, \mathrm{d}\mu : 0 \leq g \leq f, g \text{ is simple} \right\}$$

   and establish Monotone Convergence.

5. Then for any **measurable** numerical function $f = f^+ - f^-$, define $\mu f = \mu f^+ - \mu f^-$ if not both terms are infinite.

Integral of nonnegative simple functions. If f is a nonnegative simple function, then we define:

$$\int f d\mu := \sum_{i=1}^{n} a_i \mu(B_i). \tag{13.1}$$

It's well-defined(if f can be written in two different forms, the integral will remain the same, in particular, it's the same as if we write f in its canonical form, see Dudley 2018)

13.2.6 Integral of nonnegative measurable functions. *For any nonnegative measurable $f : (X, \mathcal{E}) \to [0, \infty]$, define:*

$$\int f \, d\mu := \sup \left\{ \int g \, d\mu : 0 \leq g \leq f, \textit{where } g \textit{ is simple.} \right\} \tag{13.2}$$

and there is a sequence of simple functions f_n, pointwisely $0 \leq f_n \uparrow f$ and for any such sequence f_n

$$\int f_n d\mu \uparrow \int f d\mu \tag{13.3}$$

Proof. Take $E_{nj} = f^{-1}(\frac{j}{2^n}, \frac{j+1}{2^n}]$ and $E_n = f^{-1}(n, \infty]$ and consider

$$f_n = n\mathbf{1}_{E_n} + \sum_{j=1}^{2^n n-1} \frac{j}{2^n} \mathbf{1}_{E_{nj}}$$

□

13.2.3 Properties of Integral

We provide a characterisation of integral as linear functional. In fact we can start with this as definition and back out the measure.

13.2.7 Characterisation of Integral. *For $(X, \mathcal{E}, \mu)$ there exists a unique functional $\tilde{\mu} : \mathcal{M}^+(X) \to [0, \infty]$ such that:*

1. *$\tilde{\mu}(_A) = \mu(A)$ for all $A \in \mathcal{E}$.*
2. *$\tilde{\mu}(0) = 0$.*
3. *For nonnegative real numbers a, b and $f, g \in \mathcal{M}^+$:*

$$\tilde{\mu}(af + bg) = a\tilde{\mu}(f) + b\tilde{\mu}(g)$$

4. *if $f, g \in \mathcal{M}^+$ and $f \leq g$ a.e. then $\tilde{\mu}(f) \leq \tilde{\mu}(g)$*
5. *if $f_1, f_2, \ldots$ is a sequence in $\mathcal{M}^+$ such that $0 \leq f_1 \leq \ldots$ and for every x, $f_n \to f$ for some f then $\tilde{\mu}(f_n) \uparrow \tilde{\mu}(f)$*

Integrable Functions. $L^0(X, \mathcal{E}, \mu)$ is the set of all measurable complex-valued functions. A measurable complex-valued function f is **integrable** if $\int |f| d\mu < \infty$. The set of all integrable functions is $L^1(X, \mathcal{E}, \mu)$.

13.2.8 Positivity and Monotonicity of Integral. *Let f, g be nonnegative measurable numerical functions, then $\mu f \geq 0$ and if $f \leq g$ then $\mu f \leq \mu g$.*

13.2.9 Linearity: nonnegative functions in L^0. *For any measurable space $(X, \mathcal{E}, \mu)$ and any two **nonnegative** measurable functions $f, g : X \to [0, \infty]$, for any $a, b \in \mathbb{R}_+$*

$$\mu(af + bg) = a\mu f + b\mu g$$

13.2.10 Linearity: integrable functions in L^1. *On the space of $L^1(X, \mathcal{E}, \mu)$, the mapping*

$$\begin{aligned} L^1(X, \mathcal{E}, \mu) &\to \mathbb{R} \\ f &\mapsto \int f \, d\mu \end{aligned}$$

is a linear functional, that is for any $f, g \in L^1$ and $c \in \mathbb{R}$, $\int cf \, \mathrm{d}\mu = c \int f \, \mathrm{d}\mu$ and $\int (f + g) \, \mathrm{d}\mu = \int f \, \mathrm{d}\mu + \int g \, \mathrm{d}\mu$

Proof. Here we need to first test the measurability of $f + g$ which is a mapping $\mathbb{R} \times \mathbb{R} \to \mathbb{R}$. □

These are three extremely useful theorems. All of them consider the interchange of $\int$ and lim. The dominated convergence is very easy to apply and we can see that the monotone convergence is one of the defining feature of the **integral operator or integral functional**.

13.2.11 Monotone Convergence Theorem. *Let f_n be measurable functions from $(X, \mathcal{E}, \mu)$ into $[-\infty, +\infty]$ such that $f_n(x) \uparrow f(x)$ pointwise a.e. and $\mu f_1 > -\infty$, then $\mu f_n \uparrow \mu f$.*

Proof. First we show that f is measurable, because for any $c \in \mathbb{R}$, $f^{-1}(c, \infty] = \bigcup_n f_n^{-1}(c, \infty]$.

Then we show that if f is nonnegative then the result holds. Choose simple functions $f_{nm} \uparrow f_n$ for all n, let $g_n = \max(f_{1n}, f_{2n}, \ldots, f_{nn})$ then g_n is simple and $0 \leq g_n \uparrow f$. So $\int g_n \, d\mu \uparrow \int f \, d\mu$ and $g_n \leq f_n \uparrow f$. □

The following Fatou's Lemma shows a weaker limit property but doesn't require convergence or monotonicity.

13.2.12 Fatou's Lemma. *Let f_n be any nonnegative measurable functions on X, then*

$$\mu\left(\liminf_{n\to\infty} f_n\right) \leq \liminf_{n\to\infty} \mu(f_n) \tag{13.4}$$

Proof. Define $g_n := \inf_{m\geq n} f_m$, then $g_n \leq f_n$ for every n and $\{g_n\}$ increases monotonely to $\liminf f_n$. By monotone convergence $\mu g_n \uparrow \mu(\liminf f_n)$. Because $\mu g_n \leq \mu f_n$,

$$\liminf \mu g_n = \mu(\liminf f_n) \leq \liminf f_n$$

□

13.2.13 Dominated Convergence Theorem. *Let $\{f_n\}$ and g be integrable, if $|f_n(x)| \leq g(x)$ for a.e. x and for all n, and if $\lim f_n(x)$ exists a.e. then*

$$f := \lim f_n \in L^1 \quad \textit{and} \quad \lim_{n\to\infty} \mu f_n \to \mu f$$

Proof. Firstly, $|f| \leq g$ for all x because $|f_n(x)|$ is bounded for all n, so $\mu|f| \leq \mu g$ and hence $f \in L^1$.

Secondly, we apply the Fatou's Lemma to $\{g + f_n\}$ and $\{g - f_n\}$, and use the fact that $\liminf(-A) = -\limsup A$ to deduce that

$$\limsup \mu f_n \leq \mu f \leq \liminf \mu f_n$$

□

13.2.14 Proposition (Negligible Sets). *For every measure μ:*

1. *if g is nonnegative and measurable function, $\mu g < \infty$, then $g < \infty, a.e.$ and $\{x : g(x) > 0\}$ is σ-finite. If $f \in L^1$, then $\{x : f(x) \neq 0\}$ is σ-finite.*
2. *if g, h are nonnegative measurable, and $g = h, a.e.$ then $\mu g = \mu h$.*
3. *if g is nonnegative and measurable and $\mu g = 0$ then $g = 0, a.e.$*
4. *If $f, g \in L^1$, then $\int_E f = \int_E g$ for all $E \in \mathcal{E}$ iff $\int |f - g| = 0$ iff $f = g$ a.e.*

Proof. For statement (4): if $g \geq 0$ and $\mu g = 0$, then $\{g > 0\} = \bigcup_n \{g > \frac{1}{n}\}$, if it has a positive measure, then for at least one n, $\mu\{g > \frac{1}{n}\} > 0$ but then $\mu g \geq \frac{1}{n}\mu\{g > \frac{1}{n}\} > 0$. □

Question 13.2.1. *Try using Dominated Convergence to conclude that*

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_0^1 e^{xt}\,\mathrm{d}x = \int_0^1 \frac{\mathrm{d}}{\mathrm{d}t}e^{xt}\,\mathrm{d}x$$

Corollary 13.2.14.1 (Bounded Convergence Theorem)**.** *Let $(f_n) \subset \mathcal{E}$ suppose that (f_n) is bounded and μ is finite, if* $\lim f_n$ *exists then it's a bounded integrable function and*

$$\mu \lim f_n = \lim \mu f_n$$

Corollary 13.2.14.2 (Series of Functions)**.** *Suppose $\{f_i\}$ is a sequence of integrable functions and $\sum_{i=1}^{\infty} \mu|f_i| < \infty$ then $\sum_{i=1}^{\infty} f_i$ converges a.e. to an* ***integrable*** *function and*

$$\int \sum_{i=1}^{\infty} f_i\,\mathrm{d}\mu = \sum_{i=1}^{\infty} \int f_i\,\mathrm{d}\mu$$

13.2.15 Proposition (Simple Inequality). *If $f : X \to \mathbb{C}$ is in L^1, then $|\mu f| \leq \mu|f|$*

13.2.16 Interchange Derivative and Integral. *See Folland.*

13.2.17 Lebesgue and Riemann Integral. *Let f be a bounded real-valued function on $[a, b]$, then*

1. *if f is Riemann integrable then it's Lebesgue integrable and $\int_a^b f(x)\,\mathrm{d}x = \int_{[a,b]} f\,\mathrm{d}m$*
2. *f is Riemann integrable iff $\{x \in [a, b] : f$ is discontinuous at $x\}$ has Lebesgue measure* 0.

So the **proper** Riemann integral is subsumed in Lebesgue integral, for improper we may still need to take limits.

13.2.4 Regularities and Convergence of Measurable Functions

13.2.18 Density of Simple Functions in L^1. *The following are true:*

1. *If $f \in L^1(\mu)$ and $\varepsilon > 0$,then there is an integrable simple function $\phi = \sum_{i=1}^n a_i E_i$ such that $\int |f - \phi| \, \mathrm{d}\mu < \varepsilon$.*

2. *If μ is a Lebesgue-Stieltjes measure on $\mathbb{R}$, we can choose E_i to be finite unions of open intervals and there is a continuous function g such that $g \equiv 0$ outside a bounded interval such that $\mu(|f - g|) < \varepsilon$.*

Convergence Mode

Convergence Modes. Convergence in L^1.

Example. Consider the following functions:

$$f_{1,n} = \frac{1}{n}\mathbf{1}\{(0,n)\} \quad f_{2,n} = \mathbf{1}\{(n,n+1)\} \quad f_{3,n} = n\mathbf{1}\left\{[0,\frac{1}{n}]\right\} \quad f_{4,n} = \mathbf{1}\left\{[\frac{j}{2^k},\frac{j+1}{2^k}]\right\}$$

13.2.19 Almost Everywhere Convergence and Convergence in Measure. *If $f_n \to f$ in measure, then there is a subsequence $\{f_{n_j}\}$ that converges to f a.e.*

There are different concepts for a sequence of complex-valued functions f_n to converge to f, we have uniform convergence, pointwise convergence, a.e. convergence, convergence in L^1 and convergence in measure.

Remark. In general convergence a.e. does **not** imply convergence in measure. But on finite measure space a much stronger result holds.

13.2.20 Egoroff's Theorem. *If $\mu(X) < \infty$ and (f_n), f are complex-valued measurable functions on X such that $f_n \to f$ a.e. Then for every $\varepsilon > 0$, there is $E \subset X$ such that $\mu(E) < \varepsilon$ and $f_n \to f$ **uniformly** on E^C*

Remark. The condition $\mu(X) < \infty$ can be replaced by $|f_n| \le g \in L^1, \forall n$.

Proof. First observe the sets of all convergence point of f_n is

$$\mathcal{C} = \bigcap_{k=1}^{\infty} \bigcup_{n\in\mathbb{N}} \bigcap_{m\ge n} \left\{x : |f_m(x) - f(x)| < \frac{1}{k}\right\}$$

Write $E_n(k) := \bigcup_{m \geq n} \left\{ |f_m(x) - f(x)| \geq \frac{1}{k} \right\}$. Then we have for any fixed k, $\bigcap_n E_n(k) \subset \mathcal{C}^C$ which has measure 0. Because $\mu(X) < \infty$, by continuity of measure:

$$\mu\left(\bigcap_n E_n(k)\right) = \lim_n \mu(E_n(k)) = 0$$

For any $\varepsilon > 0$ and any k, there is $n(k)$ such that $\mu\big(E_{n(k)}(k)\big) < \varepsilon 2^{-k}$. Then $E := \big(\bigcup_k E_{n(k)}(k)\big)$, we have $\mu E < \varepsilon$ and on E^c, the f_n converges uniformly. □

Question 13.2.2 (Folland Q2.39)**.** *Show that almost unifrom convergence implies convergence a.e. and in measure.*

Question 13.2.3 (Folland Q2.43)**.** *This is an extension of Egoroff's Theorem to a continuous index: suppose that $\mu X < \infty$ and $f : X \times [0,1] \to \mathbb{C}$ such that $f(\cdot, t)$ is measurable for all t and $f(x, \cdot)$ is continuous for all $x \in X$, then for any $\varepsilon > 0$ there is a set $E \subset X$ such that $\mu E < \varepsilon$ and on E^c, $f(x,t)$ converges to $f(x,0)$ uniformly as $t \to 0$.*

13.2.21 Lusin's Theorem. *If f is a complex measurable function on X such that $f(x) = 0$ for $x \notin A$, $\mu A < \infty$, for any $\varepsilon > 0$, there exists a $g \in C_c(X)$, the set of all compactly supported continuous functions, such that*

$$\mu\{x : f(x) \neq g(x)\} < \varepsilon$$

and we can arrange it so that

$$\sup_{x \in X} |g(x)| \leq \sup_{x \in X} |f(x)|$$

Remark. "There are three principles, roughly expressible in the following terms: Every (measurable) set is nearly a finite sum of intervals; every function (of class Lp) is nearly continuous; every convergent sequence of functions is nearly uniformly convergent." - Littlewood.

13.2.22 Vitali-Caratheodory Covering Theorem. *Let $f \in L^1(\mu)$, f is real-valued and $\varepsilon > 0$, then there exist functions u, v on X, such that $u \leq f \leq v$, u is upper semicontinuous and bounded above, v is lower semicontinuous and bounded below such that:*

$$\int_X (v - u)\, \mathrm{d}\mu < \varepsilon$$

13.2.5 Fubini-Tonelli

Recall the definition of **sections**, for $E \subset X \times Y$, $E_x = \{y : (x, y) \in E\}$ and $E^y = \{x : (x, y) \in E\}$. For functions we define: $f_x(y) = f^y(x) = f(x, y)$.

13.2.23 Theorem. *Let $(X, \mathcal{E}, \mu)$ and $(Y, \mathcal{F}, \nu)$ be two σ-**finite** measurable spaces, if $E \in \mathcal{E} \otimes \mathcal{F}$ then the functions $x \mapsto \nu(E_x)$ and $y \mapsto \mu(E^y)$ are measurable on X and Y and*

$$\mu \times \nu(E) = \int \nu(E_x)\,\mathrm{d}\mu(x) = \int \mu(E^y)\,\mathrm{d}\nu(y)$$

This theorem is about indicator functions and we can extend them to a wider class. The proof uses Monotone Class Argument.

13.2.24 Fubini-Tonelli Theorem. *Let $(X, \mathcal{E}, \mu)$ and $(Y, \mathcal{F}, \nu)$ be two σ-**finite** measurable spaces, then*

1. ***Tonelli:** If $f \in L^+(X \times Y)$ then $g(x) = \int f_x\,\mathrm{d}\nu$ and $h(y) = \int f^y\,\mathrm{d}\mu$ are in $L^+(X)$ and $L^+(Y)$ and*

$$\int f\,\mathrm{d}(\mu \times \nu) = \iint f(x, y)\,\mathrm{d}\mu(x)\,\mathrm{d}\nu(y) = \iint f(x, y)\,\mathrm{d}\nu(y)\,\mathrm{d}\mu(x) \tag{13.5}$$

2. ***Fubini:** If $f \in L^1(\mu \times \nu)$, then $f_x \in L^1(\nu)$ for a.e. $x \in X$, $f^y \in L^1(\mu)$ for a.e. $y \in Y$, and hence the defined $g(x)$ and $h(y)$ are in $L^1(\mu)$ and $L^1(\nu)$ and Equation 13.5 holds.*

Remark. These two theormes are usually used in tandem: use Tonelli to show $\iint |f| < \infty$ and then use Fubini to interchange order.

13.2.6 Integration in Euclidean Space

Integration in Standard Coordinates

n-dimensional Lebesgue Measure. m^n or simply m on $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n})$ or on $\mathcal{L}^n$ is the completion of the product of Lebesgue measure.

$GL(n, \mathbb{R})$. The general linear group $GL(n, \mathbb{R})$ is defined as the group of invertible linear transformations of $\mathbb{R}^n$. Any $T \in GL$ can be written as finite product of the three elementary matrices.

Diffeomorphism.

13.2.25 Regularity of m. *We have the following regularities of m on $\mathcal{L}^n$. For $E \in \mathcal{L}^n$*

1. $m(E) = \inf\{m(U) : E \subset U, \text{ and } U \text{ is open}\} = \sup\{m(K) : K \subset U, K \text{ is compact.}\}$
2. $E = A_1 \bigcup N_1 = A_2 \setminus N_2$ *where* $A_1 \in F_\sigma$ *and* $A_2 \in G_\delta$, $m(N_1) = m(N_2) = 0$.
3. *If* $m(E) < \infty$, *for any* $\varepsilon > 0$, *there is a finite collection* $\{R_j\}_{j=1}^N$ *of disjoint rectangles whose sides are intervals and* $m(E \triangle \bigcup_j R_j) < \varepsilon$.

13.2.26 Density of Simple Functions and $C_c(\mathbb{R}^n)$. *If $f \in L^1(m)$ and $\varepsilon > 0$, there is a simple function $\phi = \sum_{i=1}^N a_i \mathbf{1}_{R_i}$ where R_i is a product of intervals, such that $\int |f - \phi| < \varepsilon$ and there is a continuous function g such that $g = 0$ outside a bounded set such that $|f - g| < \varepsilon$.*

13.2.27 Translation Invariance of Lebesgue Measure. *For $a \in \mathbb{R}^n$, define $\tau_a(x) = x + a$, then*

1. *If $E \in \mathcal{L}^n$ then $\tau_a(E) \in \mathcal{L}^n$ and $m(\tau_a(E)) = m(E)$.*
2. *If $f : \mathbb{R}^n \to \mathbb{C}$ is Lebesgue measurable, then $f \circ \tau_a$ is measurable. If $f \geq 0$ or $f \in L^1(m)$ then $\int (f \circ \tau_a)\,\mathrm{d}m = \int f\,\mathrm{d}m$.*

13.2.28 Lebesgue Integral under Linear Transformation. *Suppose $T \in GL(n, \mathbb{R})$, then*

1. *If f is Lebesgue measurable on $\mathbb{R}^n$ then $f \circ T$ is also measurable. If $f \geq 0$ or $f \in L^1$ then*
$$\int f(x)\,\mathrm{d}x = |\det(T)| \int f \circ T(x)\,\mathrm{d}x$$
2. *If $E \in \mathcal{L}^n$ then $T(E) \in \mathcal{L}^n$ and $m(T(E)) = |\det(T)| m(E)$.*

Corollary 13.2.28.1 (Invariance of Rotation)**.** *A rotation is $T \in GL(n, \mathbb{R})$ such that $TT^* = I$.*

13.2.29 Lebesgue Integral under Differentiable Maps.

Integration in Polar Coordinates

13.3 Signed and Complex Measures

13.3.1 Signed Measure

Signed Measure. Let $(X, \mathcal{E})$ be a measruable space, a **signed measure** is a function $\nu : \mathcal{E} \to [-\infty, \infty]$ satisfying

1. $\nu(\varnothing) = 0$;
2. ν can take at most one of $\pm\infty$.
3. For a disjoint sequence of measurable sets $\{E_j\}$, $\nu(\bigcup_j E_j) = \sum_{j=1}^{\infty} \nu E_j$ and the latter sum converges absolutely if $\nu(\bigcup_j E_j) < \infty$.

Remark. The only two examples of signed measures are

1. $\mu_1 - \mu_2$: the difference between two measures.
2. For $f : X \to [-\infty, \infty]$ such that at least one of $\mu f^+, \mu f^-$ are finite, then define $\nu(E) := \mu(f\mathbf{1}_E) = \int_E f \, \mathrm{d}\mu$.

Remark. Again we have the continuity from below of ν and continuity from above if $|\nu(E_1)| < \infty$.

Positive and Negative Set. A set $E \in \mathcal{E}$ is **positive** for ν if for all $F \in \mathcal{E}$ and $F \subset E$, $\nu(F) \geq 0$. A set is **null** if for such F, $\nu(F) = 0$.

13.3.2 Hahn and Jordan Decomposition

13.3.1 The Hahn Decomposition Theorem. *If ν is a signed measure on $(X, \mathcal{E})$, there is a positive P and a negative N for ν such that $P \bigcup N = X$ and $P \bigcap N = \varnothing$. If P', N' is another partition, then $P \triangle P' = N \triangle N'$ is null for ν.*

Singular. Two sigend measures μ and ν are **mutually singular** if there exist $E, F \in \mathcal{E}$ such that $E \bigcap F = \varnothing$, $E \bigcup F = X$ and E is null for μ and F is null for ν, written as $\mu \perp \nu$.

13.3.2 The Jordan Decomposition. *If ν is a signed measure, there exist unique positive measures ν^+ and ν^- such that $\nu = \nu^+ - \nu^-$ and $\nu^+ \perp \nu^-$, called the **positvie** and **negative variations** of ν.*

Total Variation. Given the Jordan decomposition define the **total variation** as $|\nu| = \nu^+ + \nu^-$. A signed measure is **finite** if $|\nu|$ is finite, similarly for σ-finite.

$L^1(\nu)$. The set of **integrable** function is $L^1(\nu) = L^1(\nu^+) \bigcap L^1(\nu^-)$. We define the integration with respect to signed measure ν as

$$\nu(f) = \nu^+(f) - \nu^-(f)$$

13.3.3 Proposition. $L^1(\nu) = L^1(|\nu|)$. *If $f \in L^1(\nu)$, $\left|\int f \, \mathrm{d}\nu\right| \leq \int |f| \, \mathrm{d}|\nu|$. For $E \in \mathcal{E}$, $|\nu|(E) = \sup \left\{ \left| \int_E f \, \mathrm{d}\nu \right| : |f| \leq 1 \right\}$.*

Remark. From the definition of total variation $|\nu| = \nu^+ + \nu^-$, and for a Hahn decomposition $P \sqcup N = X$,

$$\nu(E) = \int_E f \, \mathrm{d}\mu \quad \text{where} \quad f = \mathbf{1}_P - \mathbf{1}_N \quad \text{and} \quad \mu = |\nu|$$

13.3.3 Absolute Continuity and Lebesgue-Radon-Nikodym

Absolute Continuity. Let $(X, \mathcal{E})$ be a measurable space, a signed measure ν is **absolutely continuous** with respect to a positive measure μ written as $\nu \ll \mu$ if $\nu(E) = 0$ for every $E \in \mathcal{E}$ such that $\mu(E) = 0$.

The Signed Measure induced by Integrable Function. For f an extended μ-integrable function, i.e., at least one of μf^+ and μf^- is finite, we can define a signed measure $\nu(E) = \int_E f \, \mathrm{d}\mu$ for all $E \in \mathcal{E}$. We write this relationship as

$$\mathrm{d}\nu = f \, \mathrm{d}\mu$$

13.3.4 $\varepsilon - \delta$ Characterisation When ν is Finite. *A **finite** signed measure ν is absolutely continuous wrt a positive μ on $(X, \mathcal{E})$ iff for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for all E that $\mu(E) < \delta$, $|\nu(E)| < \varepsilon$.*

Corollary 13.3.4.1. *If $f \in L^1(\mu)$ and $\mathrm{d}\nu = f\,\mathrm{d}\mu$, ν is finite, then for every $\varepsilon > 0$ there is a $\delta > 0$ such that $\left|\int_E f\,\mathrm{d}\mu\right| < \varepsilon$ when $\mu(E) < \delta$.*

13.3.5 Lebesgue-Radon-Nikodym. *Let ν be a σ-finite signed measure and μ be a σ-finite positive measure on $(X, \mathcal{E})$. There exist unique σ-finite signed measures λ, ρ such that*

$$\lambda \perp \mu \qquad \rho \ll \mu \qquad \nu = \lambda + \rho$$

Furthermore, there is an extended μ-integrable function $f : X \to \mathbb{R}$ such that $\mathrm{d}\rho = f\,\mathrm{d}\mu$ and any two such functions are equal μ-a.e.

Radon-Nikodym Derivative. Let ν be a σ-finite signed measure and μ be a σ-finite positive measure on $(X, \mathcal{E})$ if $\nu \ll \mu$, there exists an extended μ-integrable function called the **Radon-Nikodym derivative** $\frac{\mathrm{d}\nu}{\mathrm{d}\mu}$ such that

$$\mathrm{d}\nu = \frac{\mathrm{d}\nu}{\mathrm{d}\mu}\,\mathrm{d}\mu$$

13.3.6 Proposition (Calculus with Radon-Nikodym Derivative). *Let ν be a σ-finite signed measure and μ, λ be σ-finite positive measures on $(X, \mathcal{E})$ such that $\nu \ll \mu \ll \lambda$.*

1. *If $g \in L^1(\nu)$, then $g\left(\frac{\mathrm{d}\nu}{\mathrm{d}\mu}\right) \in L^1(\mu)$ and*

$$\int g\,\mathrm{d}\nu = \int g\frac{\mathrm{d}\nu}{\mathrm{d}\mu}\,\mathrm{d}\mu$$

2. *$\nu \ll \lambda$ and*

$$\frac{\mathrm{d}\nu}{\mathrm{d}\lambda} = \frac{\mathrm{d}\nu}{\mathrm{d}\mu}\frac{\mathrm{d}\mu}{\mathrm{d}\lambda} \quad \lambda - a.e.$$

13.3.4 Complex Measure

Complex Measure. A **complex measure** is a map $\nu : \mathcal{E} \to \mathbb{C}$ such that

1. $\nu(\varnothing) = 0$
2. if $\{E_j\}$ is a sequence of disjoint sets in $\mathcal{E}$ then $\nu(\bigcup E_j) = \sum \nu(E_j)$ where the series converges absolutely.

13.3.5 Lebesgue Differentiation Theorem

Here we consider the differentiation on $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n})$.

Locally Integrable. A measurable function $f : \mathbb{R}^n \to \mathbb{C}$ is **locally integrable** if for every bounded measurable set $K \subset \mathbb{R}^n$, $\int_K |f| < \infty$. The space of locally integrable functions is L^1_{loc}.

For $x \in \mathbb{R}^n$ and $r > 0$, define the average value of f on $B(x, r)$ to be

$$A_r f(x) = \frac{1}{m(B(x,r))} \int_{B(x,r)} f(y)\,\mathrm{d}y$$

Hardy-Littlewood Maximal Function. If $f \in L^1_{\text{loc}}$, the **Hardy-Littlewood maximal function** Hf is

$$Hf(x) = \sup_{r>0} A_r|f|(x) = \sup_{r>0} \frac{1}{m(B(x,r))} \int_{B(x,r)} |f(y)|\,\mathrm{d}y$$

The following theorem shows that the maximal function $Hf \in \text{weak} - L^1$ for integrable $f \in L^1$. This is related to the Chebyshev inequality. The space weak-L^1 consists of measurable functions such that there exists C such that for every $t \in (0, \infty)$, $\mu\{x : |f(x)| > t\} \leq \frac{C}{t}$.

13.3.7 Maximal Theorem. *There is a constant $C > 0$ such that for all $f \in L^1$ and $\alpha > 0$,*

$$m(\{x : Hf(x) > \alpha\}) \leq \frac{C}{\alpha} \int |f(x)|\,\mathrm{d}x$$

Proof. First we have two lemmas:

13.3.8 Lemma. *Let $\mathcal{C}$ be a family of open balls in $\mathbb{R}^n$ and $U = \bigcup_{B \in \mathcal{C}} B$, if $m(U) > c$ for a $c \in \mathbb{R}$, then there exist disjoint $B_1, \ldots, B_k \in \mathcal{C}$ such that $\sum_{j=1}^n mB_j > 3^{-n}c$.*

Proof. Use regularity from inside by compact set, get finite cover, then choose suitable $B_1, \ldots, B_k$. □

13.3.9 Lemma. *For $f \in L^1_{loc}$, $A_r f(x)$ is jointly continuous in $r, x \in (0, \infty) \times \mathbb{R}^n$.*

Proof. Notice that $m(B(x,r)) = B(0,1)r^n$ and apart from $S(x_0, r_0)$ which has measure 0, $\mathbf{1}_{B_{x,r}} \to \mathbf{1}_{B_{x_0,r_0}}$ a dominated convergence shows that $\int_{B(x,r)} |f(y)|\,\mathrm{d}y$ is continuous and so is $A_r f(x) = c^{-1} r^{-n} \int_{B(x,r)} |f(y)|\,\mathrm{d}y$. □

Lemma 13.3.9 shows that Hf is measurable. For any α, let $E_\alpha = \{x : Hf(x) > \alpha\}$, for each $x \in E_\alpha$ we can choose r_x and $E_\alpha \subset B(x, r_x)$, then we choose $c \to m(E_\alpha)$ and finite (B_j), to show

$$c \leq \frac{3^n}{\alpha} \int_{\mathbb{R}^n} |f(y)| \, \mathrm{d}y$$

By the first lemma. □

Lebesgue Set. The **Lebesgue set** L_f of f is

$$L_f = \left\{ x : \lim_{r \to 0} \frac{1}{m(B(x,r))} \int_{B(x,r)} |f(y) - f(x)| \, \mathrm{d}y = 0 \right\}$$

13.3.10 Lebesgue Set. *If $f \in L^1_{loc}$, then $m\left(\left(L_f\right)^c\right) = 0$.*

Nice Shrink. A family $E_r : r > 0$ of Borel subsets of $\mathbb{R}^n$ is said to **shrink nicely** to $x \in \mathbb{R}^n$ if

1. $E_r \subset B(x, r)$ for each r;
2. there is a constant $\alpha > 0$, independent of r such that $m(E_r) > \alpha m(B(x, r))$.

13.3.11 Differentiation Theorem. *Suppose $f \in L^1_{loc}$. For every x in the Lebesgue set of f(in particular for a.e. x), we have*

$$\lim_{r \to 0} \frac{1}{m(E_r)} \int_{E_r} |f(y) - f(x)| \, \mathrm{d}y = 0$$

and for every family $\{E_r\}_{r>0}$ that shrinks nicely to x

$$\lim_{r \to 0} \frac{1}{m(E_r)} \int_{E_r} f(y) \, \mathrm{d}y = f(x)$$

13.3.6 Fundamental Theorem of Calculus

Functions of Bounded Variations.

Absolute Continuity.

13.3.12 Fundamental Theorem of Calculus for Lebesgue Integrals. *If $-\infty < a < b < \infty$, and $F : [a, b] \to \mathbb{C}$, then TFAE:*

1. *F is absolutely continuious on $[a, b]$.*

2. *$F(x) - F(a) = \int_a^x f(t)\,\mathrm{d}t$ for some $f \in L^1([a, b], m)$.*

3. *F is differentiable a.e. on $[a, b]$, $F' \in L^1([a, b], m)$ and $F(x) - F(a) = \int_a^x F'(t)\,\mathrm{d}t$.*

13.3.13 Integration by Parts.

Chapter 14

Functional Analysis

14.1 The Lebesgue Space

14.2 Sobolev Space

14.3 Analytic Form of Hahn-Banach Theorem

Subliner functional. $p : X \to K$, a functional on a vector space over field K satisfies

$$p(\alpha x) = \alpha p(x) \qquad \text{for all} \quad x \in X, \alpha > 0$$
$$p(x+y) \leq p(x) + p(y) \qquad \text{for all} \quad x, y \in X$$

The following fundamental result depends on the Axiom of Choice.

14.3.1 Hahn-Banach. *Let X be a **real** vector space and let p be a **sublinear** functional on X, and let Y be a **subspace** of X and $l : Y \to \mathbb{R}$ that satisfies:*

$$l(y) \leq p(y) \quad \textit{for all} \quad y \in Y$$

Then there exists a linear functional $\tilde{l} : X \to \mathbb{R}$ that for all $y \in Y$, $x \in X$

$$\tilde{l}(y) = l(y) \quad \textit{and} \quad \tilde{l}(x) \leq p(x)$$

Proof. First show that there is $f : D_f \to \mathbb{R}$ whre $D_f = \{(\alpha x_0 + y) \in X : \alpha \in \mathbb{R}, y \in Y\}$ satisfying the conditions. Then we use **Zorn's Lemma** to show that there is $D_{\tilde{l}} = X$. □

Note. Note that the norm and seminorm are special cases of sublinear functionals.

14.3.2 Proposition (Uniqueness of Extension).

Part III

On the Probability Space

Chapter 15

Probability

15.1 Probability Space

15.1.1 Probability Concepts

Probability Space. The triple of a measurable space $(\Omega, \mathcal{F}, P)$ is a probability space if the measure $P(\Omega) = 1$. The elements of $\mathcal{F}$ are called **events**.

A simple example is when Ω is finite and we have the **uniform** measure on 2^{Ω}.

15.1.2 Random Variable and Stochastic Process

Random variables are the mappings defined on measurable spaces.

Random Variable. Let $(\Omega, \mathcal{H}, P)$ be a probability space, and $(Y, \mathcal{F})$ be a measurable space. Then a random variable taking value in $(Y, \mathcal{F})$ is a measurable function $X : \Omega \to Y$.

Depending on the space Y, the random variables have additional structures, if $(Y, \mathcal{F}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ we call X **real-valued**, if $(Y, \mathcal{F}) = (\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$ we call X **random vector** and if Y is a space of functions, we call X **random functions**.

Stochastic Process. Let $(E, \mathcal{E})$ be a measurable space, let T be an arbitrary set(countable or uncountable). The collection $\{X_t : t \in T\}$ is called a **stochastic process** where X_t are random variables on $(\Omega, \mathcal{H}, P)$ taking values in **state space** $(E, \mathcal{E})$ for every t in **parameter set** T.

Law of a random variable. The **law** of a random variable $X : (\Omega, \mathcal{H}, P) \to (Y, \mathcal{F})$ is the **image measure** $P \circ X^{-1}$ defined on $\mathcal{F}$.

Remark. We can regard the $\{X_t : t \in T\}$ as a random variable from $(\Omega, \mathcal{H})$ to $(F, \mathcal{F}) := (E^T, \mathcal{E}^T)$, the product space. The **law** of the stochastic process is the image measure $P \circ X^{-1}$ on $(F, \mathcal{F})$.

15.1.3 Measurability, Information and Filtration

On a probability space $(\Omega, \mathcal{H}, P)$, a random variable X from $\Omega \to (E, \mathcal{E})$ contains "information" and the information is reflected in the σ-algebra generated by X.

Information Contained in Random Variable

σ-algebra generated by a random variable. Let $X : (\Omega, \mathcal{H}) \to (E, \mathcal{E})$ be a random variable(measurable mapping), then the σ-algebra generated by X is defined as

$$\sigma X := \left\{X^{-1}A : A \in \mathcal{E}\right\} \subset \mathcal{H}$$

It is the smallest σ-algebra relative to which X is measurable.

15.1.1 Determinability. *Let X be a random variable from $(\Omega, \mathcal{H})$ to $(E, \mathcal{E})$, a **numerical function** $V : \Omega \to \overline{\mathbb{R}}$ is σX-measurable iff*

$$V = f \circ X$$

*for some **deterministic function** $f : (E, \mathcal{E}) \to \overline{\mathbb{R}}$.*

Information Contained in Stochastic Process

Remember a stochastic process is $\{X_t : t \in T\}$ for an index set T such that X_t is a random variable on $(\Omega, \mathcal{H})$ taking values in $(E_t, \mathcal{E}_t)$. We can regard $\{X_t\}$ as a random variable $X : \Omega \to (\times_{t\in T} E_t, \otimes_{t\in T} \mathcal{E}_t)$ such that $X(\omega) = (X_t(\omega))_{t\in T}$.

15.1.2 Proposition. *If X be defined in that way, $\sigma X = \sigma\{X_t : t \in T\}$ the latter being the σ-algebra generated by the union of σX_t on Ω.*

15.1.3 Proposition (When $T = \mathbb{N}_+$).

15.1.4 Proposition (When T is arbitrary). *Let T be arbitrary, for each t in T, let X_t be a random variable taking values in some measurable space $(E_t, \mathcal{E}_t)$. Then numerical $V : \Omega \to \overline{\mathbb{R}}$ is measurable relative to $\sigma\{X_t\}$ iff there exists a sequence $(t_n) \subset T$ and a function $f \in \otimes_n \mathcal{E}_{t_n}$*

$$V = f\left(X_{t_1}, X_{t_2}, \ldots\right)$$

15.1.4 Distance and Affinity between Probability Measures

Total variantion, Hellinger and Relative Entropy.

15.2 Independence and Conditioning

15.2.1 Independence

The independence condition justifies certain factorization of expectation or integration.

Independent. 1. Let $\{\mathcal{G}_i\}_{i=1}^n$ be sub-σ-algebras of $(\Omega, \mathcal{H}, P)$, then the collection $\{\mathcal{G}_i\}_{i=1}^n$ is **independent** if

$$P(G_1 \ldots G_n) = PG_1 \ldots PG_n \quad \text{for all} \quad G_i \in \mathcal{G}_i, i = 1, \ldots, n$$

2. For arbitrary T,$\{\mathcal{G}_t : t \in T\}$ is an independency if every **finite** subset of it is an independency.

3. Events, random variable, stochastic processes are independent, if the σ-algebra generated by them are independent. For example, family of measurable functions $\{X_i : (\Omega, \mathcal{H}) \to (\mathcal{X}_i, \mathcal{E}_i)\}_{i \in I}$ are independent if

$$P\left(\bigcap_{i \in S} \{X_i \in E_i\}\right) = \prod_{i \in S} P\{X_i \in E_i\}$$

for all finite subset $S \subset I$ and all choices $E_i \in \mathcal{E}_i$.

15.2.1 Proposition (Criterion for independence of σ-algebra). *Let $(\mathcal{G}_i)_{i=1}^n$ be sub-σ-algebras of $\mathcal{H}$ for each $i \leq n$, let $\mathcal{P}_i$ be a p-system that generates $\mathcal{G}_i$, then $\{\mathcal{G}_1, \ldots, \mathcal{G}_n\}$ are independent iff*

$$P\left(H_1 \bigcap \ldots \bigcap H_n\right) = P(H_1) \ldots P(H_n)$$

For all $H_i \in \mathcal{P}_i \bigcup \{\Omega\}$

15.2.2 Factorization of Integration. *Random variables $(X_i)_{i=1}^n$ are independent iff*

1. *For all nonnegative measurable functions $f_1 \ldots f_n$ in $\mathcal{E}_1, \ldots, \mathcal{E}_n$ or for integrable functions,*

$$E(f_1 \circ X_1) \ldots (f_n \circ X_n) = E(f_1 \circ X_1) \ldots E(f_n \circ X_n)$$

2. *Their joint distribution is the product of their marginal distribution.*

As an interesting and important result of independence we have the Kolmogorov 0-1 law.

Tail σ-algebra. Let $\mathcal{F}_1, \mathcal{F}_2, \ldots$ be a sequence of sub-σ-algebra of $\mathcal{H}$, $\mathcal{T}$ is the **tail σ-algebra** defined as :

$$\mathcal{T}_n = \sigma\left(\bigcup_{m \geq n} \mathcal{F}_m\right); \quad \mathcal{T} := \bigcap_n \mathcal{T}_n$$

15.2.3 Kolmogorov 0-1 law. *Let $\mathcal{F}_1, \mathcal{F}_2 \ldots$ be independent, then $P(H) = 0$ or 1 for every $H \in \mathcal{T}$.*

Proof. We want to show that $\mathcal{T}$ is independent of itself. □

15.2.2 Conditional Distribution

15.2.3 Conditional Expectation

Conditional Expectation. Let $(\Omega, \mathcal{H}, P)$ be a probability space, let $\mathcal{F}$ be a sub-σ-algebra of $\mathcal{H}$, we define the conditional expectation of $X \in \mathcal{H}_+$ as any $\hat{X} = E(X|\mathcal{F})$ satisfying the conditions:

1. **measurability**: $\hat{X} \in \mathcal{F}_+$

2. **projection property**: $EVX = EV\hat{X}$ for all $V \in \mathcal{F}_+$.

The definition is unique up to almost-surely equivalence.

For $X \in \mathcal{H}$ if EX exists, then we define:

$$E(X|\mathcal{F}) := E\left(X^+|\mathcal{F}\right) - E(X^-|\mathcal{F})$$

15.2.4 Existence of Conditional Expectation. *Let $X \in \mathcal{H}^+$, let $\mathcal{F}$ be a sub-σ-algebra of $\mathcal{H}$, then $E(X \mid \mathcal{F})$ exists and is unique up to equivalence.*

Proof. Use Radon-Nikodym theorem. □

15.2.5 Properties inherited from Expectation.

15.2.6 Special Properties. *Let $\mathcal{F}$ and $\mathcal{G}$ be sub-σ-algebras of $\mathcal{H}$, $W, X \in \mathcal{H}$ such that EX and EWX exists, then*

1. ***Conditional determinism**: $W \in \mathcal{F} \implies E(WX \mid \mathcal{F}) = WE(X \mid \mathcal{F})$*
2. ***Repeated Conditioning**: $\mathcal{F} \subset \mathcal{G} \implies E_{\mathcal{F}}E_{\mathcal{G}}X = E_{\mathcal{G}}E_{\mathcal{F}}X = E_{\mathcal{F}}X$*

15.3 Stochastic Convergence

15.3.1 Modes of Convergence

When we talk about "convergence" the first thing is to think about the topology/metric and when "random" or "stochastic" is mentioned, think about measurablity. Similarly, we talk about "orthogonal" in inner product space, "add" and "scalar multiplication" in vector spaces. Always bear in mind the structure of the mathematical subjects we are investigating.

Almost surely Convergence. Let $X_n : (\Omega, \mathcal{E}, P) \to (Y, \mathcal{T})$ be a sequence of random variables taking values in a topological space(with Borel $\sigma-$algebra), then the sequence converges **almost surely** to a random variable X if $P(X_n \to X) = 1$.

Converge in Probability in Separable Metric Spaces. Let $X, X_n : (\Omega, \mathcal{E}, P) \to (S, d)$ be random variables, where (S, d) is a **separable** metric space, then $X_n \xrightarrow{p} X$ if for any $\varepsilon > 0$

$$\lim_{n\to\infty} P(d(X_n, X) > \varepsilon) = 0 \tag{15.1}$$

Note. The reason we consider the separable metric space is that here $d(X_n, X)$ is a proper random variable. [see Dudley].

Converge in Law. Let $(S, \mathcal{T})$ be a **topological space**, let P_n be a sequence of laws(probability measures) defined on the **Borel** σ-algebra generated by $\mathcal{T}$, let $C_b(S)$ be the set of all **bounded, continuous, real-valued** functions on S, the laws P_n converges to a law P, if and only if

$$\int f \, \mathrm{d}P_n \to \int f \, \mathrm{d}P, \quad \forall f \in C_b(S) \quad \text{as} \quad n \to \infty,$$

Random variables $X_n : \Omega \to S$ converge **in law** to a random variable X if $\mathcal{L}(X_n) \to \mathcal{L}(X)$ in law.

This is a stronger convergence concept than convergence in law.

Converge in Total Variation. For $(P_n), P$ defined on the Borel σ-algebra generated by the topology $\mathcal{T}$ on X, $P_n \to P_0$ in total variation iff as $n \to \infty$, $|P_n - P_0|(S) \to 0$, where $|\mu| = \mu^+ + \mu^- \geq 0$ is the total variation measure.

Convergence in p-th Mean. Let $(X_n), X$ be random variables in L^p, X_n converge in p-th mean to X if

$$E|X_n - X|^p \to 0 \quad \text{as} \quad n \to \infty$$

15.3.2 Features of Almost Sure Convergence

15.3.1 Almost sure convergence of subsubsequence. *For any random variables $X_n, X : (\Omega, \mathcal{H}, P) \to (S, d)$ where (S, d) is a **separable metric space**, $X_n \to X$ in probability iff for every subsequence $X_{n(k)}$ of X_n, there is a further sub-sub-sequence $X_{n(k(r))}$ that $X_{n(k(r))} \to X$ almost surely.*

15.3.2 Proposition ("Cauchy" type criterion for almost sure convergence). *Let (S, d) be a **complete separable** metric space and $(\Omega, \mathcal{H}, P)$ a proability space. Let $Y_1, Y_2 \ldots$ be random variables from $\Omega \to S$ s.t.*

$$\forall \varepsilon > 0, \quad P\left\{\sup_{k \geq n} d(Y_n, Y_k) \geq \varepsilon\right\} \to 0 \quad \textit{as} \quad n \to \infty$$

then for some random variable Y, $Y_n \to Y$ almost surely.

15.3.3 Proposition (Borel-Cantelli Type Criterion). *If for every $\varepsilon > 0$, $\sum_{n=1}^{\infty} \{|X_n - X| > \varepsilon\} < \infty$, then $X_n \to X$ a.s.*

15.3.3 Features of Weak Convergence

15.3.4 Portmanteau Theorem: Characterisation of Weak Convergence. *For laws P_n and P on a metric space (S, d), TFAE:*

1. *$P_n \rightsquigarrow P$*
2. *For all open $U \subset S$, $\liminf_n P_n(U) \geq P(U)$;*
3. *For all closed F, $\limsup P_n(F) \leq P(F)$;*
4. *For all **continuity set** A of P(such that $P(\partial A) = 0$), $\lim_n P_n(A) = P(A)$.*

Proof. (1) $\Longrightarrow$ (2). Let U be open, $F = U^c$, fix $\varepsilon > 0$, define $d(x, F) := \inf_{y \in F} d(x, y)$. Define $f_m(x) := \min(1, md(x, F))$ so that $0 \leq f_m \uparrow \mathbf{1}_U$. Let $F_m := f_m^{-1}(\{1\})$, then for m large, □

□

15.3.5 Helly-Bray Theorem.

Uniformly Tight. A set of random vectors $\{X_\alpha\}$ is uniformly tight if there is a constant M such that

$$\sup_\alpha P(\|X_\alpha\| > M) < \varepsilon$$

15.3.6 Prohorov's Theorem. *Let X_n be a random vector in $\mathbb{R}^k$,*

1. *If $X_n \rightsquigarrow X$, then $\{X_n : n \in \mathbb{N}\}$ is uniformly tight;*
2. *If X_n is uniformly tight, then there exists a subsequence of X_n such that $X_{n(k)} \rightsquigarrow X$ as $k \to \infty$ for some X.*

15.3.4 Relationship between Modes of Convergences

15.3.7 Continuous Mapping Theorem. *Let $g : \mathbb{R}^k \to \mathbb{R}^m$ be continuous at every point of a set C such that $P(X \in C) = 1$, then*

1. *if* $X_n \rightsquigarrow X$, *then* $g(X_n) \rightsquigarrow g(X)$;
2. *if* $X_n \xrightarrow{p} X$ *then* $g(X_n) \xrightarrow{p} g(X)$;
3. *if* $X_n \to X$ *almost surely, then* $g(X_n) \to g(X)$ *almost surely.*

15.3.5 Uniformly Integrable

Uniform Integrability. A family of random variables $\{X_t : t \in T\}$ is **uniformly integrable** if

$$\sup_{t \in T} P(|Z_t|\{|Z_t| > M\}) \to 0 \quad \text{as} \quad M \to \infty$$

15.3.8 Lemma. *A family* $\{X_t : t \in T\}$ *is uniformly integrable iff*

1. $\sup_{t \in T} P|Z_t| < \infty$
2. *for each* $\varepsilon > 0$ *there exists* $\delta > 0$ *such that* $\sup_{t \in T} P(|Z_t F|) \leq \varepsilon$ *for every* $F \in \mathcal{E}$ *with* $PF < \delta$.

15.3.9 Uniform Integrability, Convergence in Probability and L^1. *Let* $\{X_n : n \in \mathbb{N}\}$ *be a sequence of integrable random variables, TFAE:*

1. $\{X_n\}$ *is uniformly integrable and* $X_n \xrightarrow{p} X_\infty$ *for some random variable* X_∞ *which is necessarily integrable.*
2. X_n *converges in* L^1 *norm, with a limit* X_∞ *that is necessarily integrable,*

$$P|X_n - X_\infty| \to 0.$$

15.4 Metrization of Convergence in Probability and in Law

15.4.1 Convolution; Characteristic Function

The sum of **independent** random variables can be characterised by the convolution of laws as well as the **product of characteristic functions**.

Convolution. Let μ and ν be two probability measures on the Euclidean $\mathbb{R}^k$, define the convolution between two measures $(\mu * \nu)(A) := \int_{\mathbb{R}^k} \nu(A - x)\, d\mu(x)$

15.4.1 Sum of Independent Random Variables. *Let X, Y be independent random variables in Euclidean $\mathbb{R}^k$, with laws $L(X) = \mu$ and $L(Y) = \nu$, then $L(X + Y) = \mu * \nu$.*

Characteristic Function. If $X : \Omega \to \mathbb{R}^k$, define the characterisation function of X to be

$$f_P(t) := E e^{i\langle x,t\rangle} = \int e^{i\langle x,t\rangle}\, dP(x)$$

for all $t \in \mathbb{R}^k$, and $\langle \cdot, \cdot \rangle$ is the inner product on $\mathbb{R}^k$.

15.4.2 Independence and Product of Characteristic Functions. *For independent $(X_1, X_2, \ldots, X_n)$ taking values in $\mathbb{R}^k$, the characteristic function of $\sum_{i=1}^n X_i$ is the product of individual Characteristic functions:*

$$E e^{i\langle \sum_{i=1}^n X_i, t\rangle} = \prod_{j=1}^n E e^{i\langle X_j, t\rangle}$$

15.4.3 Proposition (Global Moment and Local Differentiability). *If $E|X|^r < \infty$ then $E e^{iXt}$ is C^r in t. see Dudley*

15.4.4 Uniqueness of Characteristic Functions.

Chapter 16

Random Vectors

16.1 Numerical Random Variables

16.1.1 Expectation of Numerical Random Variables

Expectation. When X is a numerical random variable define the expectation as

$$EX := \int X \, \mathrm{d}P$$

when this integral exists, that is EX^+ and EX^- are not both infinite.

16.1.1 Expectation of Function of Random Variable. *Let X be a random variable on $(\Omega, \mathcal{H}, P)$ taking value in measurable space $(E, \mathcal{E})$, if μ is the law of X, then for every $f \in \mathcal{E}_+$*

$$Ef \circ X = \mu f$$

16.1.2 Stochastic Inequalities

16.1.2 Proposition (Chebyshev Inequality). *For any real random variables X and $t > 0$,*

$$P(|X| \geq t) \leq \frac{EX^2}{t^2}$$

Proof. We have $E(X^2) \geq E\big(X^2 \mathbf{1}\{|X| \geq t\}\big) \geq t^2 E(\mathbf{1}\{|X| \geq t\}) = t^2 P(|X| \geq t)$. □

16.2 Convergence of Real-Valued Series

16.2.1 Kolmogorov's Inequality

Let $(\Omega, \mathcal{H}, P)$ be a probability space and (X_n) be a sequence of **indpendent real-valued** random variables, The object of interest is the partial sum and the series of (X_n), denoted as $S_n = \sum_{i=1}^{n} X_i$.

16.2.1 Lemma (Kolmogorov's Inequality). *Suppose that X_n are independent and have mean 0 then for every $a \in (0, \infty)$,*

$$a^2 P\left(\max_{k \leq n} |S_k| < a\right) \leq \operatorname{var} S_n$$

16.2.2 Convergence of Series and Variance

16.2.3 Kolmogorov Three Series Theorem

16.3 Sum of Random Variables and Random Vectors

16.3.1 Law of Large Numbers

Law of large numbers and central limit theorems are of central importance to the probability theory, they both concern the **asymptotic** behaviour of

$$c_n \sum_{i=1}^{n} w_i X_i$$

where X_i are randome variables that may not be independent.

16.3.1 A Weak Law of Large Numbers. *Let $X_1, \ldots, X_n$ be defined on the same probability space with $EX_i = 0$, $EX_i^2 = 1$ and $EX_iX_j = 0$ for all i, j.then*

$$\frac{1}{n} S_n := \frac{1}{n} \sum_{i=1}^{n} X_i \xrightarrow{p} c = EX_i = 0$$

Notice here the X_i are orthonormal in the Hilber Space $L^2(\Omega, P)$

Proof. Let $S_n = \sum_{i=1}^n X_i$, then we have $E((\frac{S_n}{n})^2) = \frac{1}{n}$. By Propsition 16.1.2, we have

$$P(\left|\frac{S_n}{n}\right| > \varepsilon) \leq \frac{1}{\varepsilon^2}\frac{1}{n} \to 0, \quad \text{as} \quad n \to \infty$$

□

16.3.2 A Strong Law of Large Numbers.

16.3.2 Central Limit Theorems

16.3.3 Lindeberg-Feller CLT. *See Billingsley 2008.*

16.3.4 Liapunov CLT with Third Moment Condition. *See Cinlar and the proof. It's very interesting idea.*

16.3.5 Liapunov Central Limit. *Let $\{Z_{n,i}\}$ be a double array of **independent** random variables such that $E(Z_{n,i}) = \mu_{n,i}$ and $\operatorname{var}(Z_{n,i}) = \sigma_{n,i}^2$ with $E\left|Z_{n,i}\right|^{2+\delta} < \infty$ for some $\delta > 0$. Let $S_n = \sum_{i=1}^n Z_{n,i}$ and let $\sigma_n^2 = \operatorname{var}(S_n) = \sum_{i=1}^n \sigma_{n,i}^2$ if $\sigma_n^2 = \sigma^2 + o(1)$ and*

$$\lim_{n\to\infty} \sum_{i=1}^n E\left|(Z_{n,i} - \mu_{n,i})\right|^{2+\delta} = 0 \quad \textit{for some} \quad \delta > 0$$

Then

$$\sigma_n^{-1}(S_n - E(S_n)) = \sigma_n^{-1} \sum_{i=1}^n \left[Z_{n,i} - E(Z_{n,i})\right] \rightsquigarrow N(0,1)$$

16.3.6 Multivariate Central Limit Theorem.

16.3.7 Lindeberg Central Limit. *For each $n \in \mathbb{N}$, let $Y_{n,1}, \ldots, Y_{n,n}$ be independent random vectors with finite covariance matrices such that*

$$\frac{1}{n}\sum_{i=1}^n \operatorname{cov} Y_{n,i} \to \Sigma$$

$$\frac{1}{n}\sum_{i=1}^n E\left\|Y_{n,i}\right\|^2 \mathbf{1}\{\left\|Y_{n,i}\right\| > \varepsilon\sqrt{n}\} \to 0, \quad \textit{for every} \quad \varepsilon > 0$$

Then the sequence

$$\frac{1}{\sqrt{n}}\sum_{i=1}^n (Y_{n,i} - EY_{n,i}) \rightsquigarrow N(0,\Sigma)$$

16.3.8 Classic CLT for Finitely Dependent Sequence.

An interesting question in A. W. Van der Vaart n.d. Exercise 4.1 where we can apply Cesaro's Lemma in a concrete example.

16.3.9 CLT for Linear Processes. *See A. W. Van der Vaart n.d.*

16.3.10 CLT for Strong Mixing Sequence.

Corollary 16.3.10.1 (Ibragimov)**.** *If X_t is a strictly stationary time series with mean 0, $E|X_0|^r < \infty$ and $\sum_h \alpha(h)^{1-2/r} < \infty$ for some $r > 2$ then $v = \sum_h \gamma_X(h)$ converges absolutely and $\sqrt{n}\bar{X}_n \rightsquigarrow N(0, v)$.*

16.3.11 Martingale Difference Central Limit. *If X_t is a **martingale difference** sequence relative to the filtration $\mathcal{F}_t$ such that $\frac{1}{n}\sum_{t=1}^n E\left(X_n^2 \mid \mathcal{F}_{t-1}\right) \xrightarrow{p} v$ for a positive constant v and such that $\frac{1}{n}\sum_{t=1}^n E\left(X_t^2 \mathbf{1}\left\{\left|X_t > \varepsilon\sqrt{n}\right| \mid \mathcal{F}_{t-1}\right\}\right) \xrightarrow{p} 0$ for every $\varepsilon > 0$, then $\sqrt{n}\bar{X}_n \rightsquigarrow N(0, v)$.*

16.3.3 Cramer–Wold Device

Chapter 17

Random Matrix

17.1 Introduction

Random matrix theory deals mostly with the following four questions.

Firstly, given a sequence of matrices A_n with increasing dimensions, each with elements that are random (i.i.d, can be extended to allow for weak correlation), find the **limiting spectral distribution** or LSD. The common method is to use the **Stieltjes transformation** of distribution function G

$$S_G \mathbb{C}^+ \to \mathbb{C} : z \mapsto S_G(z) = \int \frac{1}{\lambda - z} \, \mathrm{d}G(\lambda)$$

If $G = F^A$ the **empirical distribution** of the eigenvalues, then

$$S_{F^A}(z) = \mathrm{tr}(A - zI)^{-1}$$

Two results then help us find the limiting spectral distribution $\lim F^{A_n}(x)$: 1. if $\lim_n F^{A_n}$ exists, then it has Stieltjes transformation $S_{F^A} = \lim_{n\to\infty} S_{F^{A_n}}$; 2. We have an inversion function $S_G \to G$ that can give the limiting distribution.

Known results including: for symmetric matrix, the LSD has **semi-circular** density. For sample covariance matrix, LSD follows **M-P** law. For canonical correlation matrix, the LSD has **Wachter distribution**. For $S_{xx}^{-1} S_{yy}$ it has **F- distribution**.

When we calculate the S_{F^A}, where $A = \frac{1}{n} XX'$ there is a method from **perturbation**

theory:

$$\begin{aligned} S_{F^A}(z) &= \operatorname{tr}\left(\frac{1}{n}\sum X_i X_i' - zI\right)^{-1} \\ &= \operatorname{tr}\left(\frac{1}{n}\sum_{i\neq k} X_i X_i' - zI\right)^{-1} \qquad \text{we eliminate one component} \\ &= f(S_{F^{A_{n\setminus k}}}) \end{aligned}$$

When n is large, in the limit, $f(S_{F^{A_{n\setminus k}}}) \approx f(S_{F^{A_n}})$ so

$$S_{F^A} \approx f(S_{F^A})$$

Using this equation we can find S_{F^A}.

As an application see Onatski and Wang 2018a, where they explain why the test for cointegration when n is large has the over-rejection problem.

Second question is the pointwise limits of the extreme eigenvalues

$$\lim_{n\to\infty} \lambda_1 \qquad \lim_{n\to\infty} \lambda_n$$

Third question is derive the limiting distribution of

$$\lim_{n\to\infty} g(n)(\lambda_n - b)$$

where b is the largest endpoint of the support of LSD

Fourth question is the CLT, for analytic function f

$$\sum f(\lambda_i)$$

For nonsymmetric matrix, there is the **circular law**, the complex eigenvalues are uniformly distributed on the unit circle of the complex plane.

17.2 Important Methods

17.2.1 Moment Convergence Theorem

17.2.2 Stieltjes Transformation

17.2.3 Orthogonal Polynomial Decomposition

17.2.4 Free Probability Method

17.3 Limiting Spectral Distribution

17.3.1 Wigner Matrix and Semicircular Law

Wigner Matrix. A Wigner is a Hermitian matrix such that the dianonal elements are i.i.d. real random variables, and the upper off-diagonal elements are i.i.d. complex random variables with finite variance σ^2.

17.3.1 Semicircular Law. *As $n \to \infty$, the empirical spectral distribution $F^{n^{-1/2}W_n}$ converges almost surely to a distribution with density given by the semicircular law.*

17.3.2 Lemma (Rank Inequality). *For $n \times n$ Hermitian matrix A, B, we have*

$$\left\|F^A - F^B\right\| \leq \frac{1}{n} \operatorname{rank}(A - B)$$

17.3.3 Lemma (Difference Inequality). *Let Levy distance between two probability CDF functions F, G be*

$$L(F, G) := \inf \{\varepsilon : F(x - \varepsilon) - \varepsilon \leq G(x) \leq F(x + \varepsilon) + \varepsilon, \forall x \in \mathbb{R}\}$$

then we have the following inequality for Hermitian A, B

$$L^3(F^A, F^B) \leq \frac{1}{n} \operatorname{tr}\left[(A - B)^2\right]$$

Proof. We first remove the diagonal elements, then remove the mean of the off-diagonal elements, and truncate the matrix. Then we use moment covergence theorem to show the convergence. □

17.3.2 Sample Convariance Matrix

17.3.3 Product of Random Matrices

Chapter 18

Empirical Process

18.1 Outer Measure and Majorant

The measurability condition for usual weak convergence concept easily fails, so we want to expand the definition.

Outer Measure. For any $T : (\Omega, \mathcal{A}, P) \to \overline{\mathbb{R}}$, define the **outer integral** as

$$P^*T := \inf \{PU : U \geq T, U \in \mathcal{A}, PU \text{ exists}\}$$

Weak Convergence. For any $X_n : (\Omega_n, \mathcal{A}_n, P_n) \to (\mathbb{D}, d)$, where $\mathbb{D}$ is a metric space equipped with the Borel σ-algebra, and if X is a Borel measurable function, then $X_n \rightsquigarrow X$ iff

$$P^* f(X_n) \to Pf(X) \quad \text{for all} \quad f \in C_b(\mathbb{D})$$

where $C_b(\mathbb{D})$ is the set of all bounded continuous function from $\mathbb{D}$ to $\mathbb{R}$.

Majorant. For any T, there is a **majorant** or **measurable cover function**, or **measurable envelop function** T^* such that:

1. T^* is measurable;
2. $T^* \geq T$
3. $T^* \leq U$ a.s. for every measurable $U : \Omega \to \overline{\mathbb{R}}$ such that $T \leq U$ a.s.

If PT^* exists, then $P^*T = PT^*$.

18.2 Entropy, Glivenko-Cantelli and Donsker Class

Part IV

Stochastic Processes over Time

Chapter 19

Information over Time

19.1 Filtration and Stopping Time

Filtration enbodies the flow of information. Let $(\Omega, \mathcal{H}, P)$ be a probability space, let $\mathbf{T}$ be an index set and it's either $\mathbb{N}$ or a subset of $\overline{\mathbb{R}}$, think of it as time(discrete or continuous), then a filtration defined on this probability space is

Filtration.

1. A filtration $\mathcal{F} := (\mathcal{F}_t)_{t\in\mathbf{T}}$ is an increasing family of sub-σ-algebras of $\mathcal{H}$ indexed by $\mathbf{T}$.

2. If we consider continuous time, that is $\mathbf{T} = [0, \infty]$, then we usually assume two conditons on the $\mathcal{F}$ of $\mathbf{T}$ that are (a) **right continuity**: $\mathcal{F}_t = \bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}$ for all $t \in \mathbf{T}$. (b) the filtration $\mathcal{F}$ is **augumented**: $(\Omega, \mathcal{H}, P)$ is complete and that $\mathcal{F}_0$ includes all negligible events in $\mathcal{H}$.

Adapted Stochastic Process. A stochastic process $(X_t)_{t\in\mathbf{T}}$ taking values in $(E, \mathcal{E})$ is adapted to a filtration $(\mathcal{F}_t)_{t\in\mathbf{T}}$ iff for any t, X_t is measurable relative to $\mathcal{F}_t$ and $\mathcal{E}$.

Stopping Time. Let $\mathcal{F}$ be a filtration on $\mathbf{T}$, a random time $T : \Omega \to \mathbf{T} \bigcup \{\infty\}$ is a **stopping time** of $\mathcal{F}$ if $\{\omega : T(\omega) \leq t\} \in \mathcal{F}_t$ for each $t \in \mathbf{T}$.

End of Time. Because a stopping time T can take value ∞, $T^{-1}(\infty)$ may not be in any $\mathcal{F}_t$, we extend $\mathcal{F}$ to $\mathcal{F} \bigcup \{\mathcal{F}_\infty\}$ where $\mathcal{F}_\infty := \sigma(\bigcup_{t\in\mathbf{T}} \mathcal{F}_t)$.

Past Until a Stopping Time. Let T be a stopping time, it's randoom, we define the σ-algebra $\mathcal{F}_T$ to embody the information accumulated until T to be

$$\mathcal{F}_T := \left\{ H \subset \mathcal{H} : H \bigcap \{T \leq t\} \in \mathcal{F}_t \quad \text{for all} \quad t \in \mathbf{T} \right\}$$

Then we can show that $\mathcal{F}_T$ is a σ-algebra, T is $\mathcal{F}_T$-measurable and $\mathcal{F}_T \subset \mathcal{F}_\infty \subset \mathcal{H}$.

19.1.1 Numerical $\mathcal{F}_T$-measurable Function. *A numerical random variable V is $\mathcal{F}_T$-measurable iff*

$$V\mathbf{1}_{\{T \leq t\}} \in \mathcal{F}_t \quad \textit{for all} \quad t \in \mathbf{T} \bigcup \{\infty\}$$

19.2 Comparing Past Information

Let S and T be two stopping times, then $\mathcal{F}_S$ and $\mathcal{F}_T$ are the information accumulated by T, if $S \leq T$ for all $\omega \in \Omega$, then we should expect $\mathcal{F}_S \subset \mathcal{F}_T$.

19.2.1 Proposition ($S \wedge T$ and $S \vee T$). *If S and T are stopping times with respect to a filtration $(\mathcal{F}_t)_{t \in \mathbf{T}}$, then $S \wedge T$ is a stopping time as well. So is $S \vee T$.*

Proof. For all $t \in \mathbf{T}$, $\{\omega : S \wedge T(\omega) \leq t\} = \{S \leq t\} \bigcup \{T \leq t\} \in \mathcal{F}_t$. □

19.2.2 Comparing Pasts. *Let S, T be stopping times, then the following holds*

1. $\mathcal{F}_{S \wedge T} = \mathcal{F}_S \bigcap \mathcal{F}_T$
2. *If $S \leq T$ for all ω then $\mathcal{F}_S \subset \mathcal{F}_T$.*
3. *if $V \in \mathcal{F}_S$ then the following are in $\mathcal{F}_{S \wedge T}$:*

$$V\mathbf{1}\{S \leq T\} \quad V\mathbf{1}\{S = T\} \quad V\mathbf{1}\{S < T\}$$

Proof. See Çınlar 2011 □

19.3 Approximation by Discrete Time

We can show that by defining $d_n(t) = \frac{k+1}{2^n}$ for $\frac{k}{2^n} \leq t < \frac{k+1}{2^n}$, we can show that $T_n = d_n \circ T$ is a sequence of stopping time decreasing to T.

19.4 Conditioning on Stopping Time

Chapter 20

Martingales

20.1 Martingale Definition

Martingale. A **real-valued** stochastic process $X = (X_t)_{t\in\mathbf{T}}$ is called an $\mathcal{F}$-**submartingale** if X is adapted to $\mathcal{F}$, X_t is **integrable** for all t and

$$E(X_t - X_s \mid \mathcal{F}_s) \geq 0 \quad \text{for all} \quad s < t$$

It is a supermartingale if $-X$ is a submartingale and it's a **martingale** if it's both a super- and a submartingale.

20.2 Discrete-time Martingale

20.2.1 Doob's Decomposition and Martingale Transformation

Let $\mathbf{T} = \mathbb{N}$, then a predictable process can be defined as

Predictable Process. Let $(A_n)_{n\in\mathbb{N}}$ a stochastic process, then (A_n) is predictable if $A_{n+1} \in \mathcal{F}_n$ for all n and $A_0 \in \mathcal{F}_0$.

20.2.1 Doob's Decompostion. *Let X_n be **adapted** and **integrable**, then it can be decomposed as*

$$X_n = X_0 + M_n + A_n, \quad \textit{for all} \quad n \in \mathbb{N}$$

where M_n is a martingale starting at 0 and A_n is a $\mathcal{F}$-predictable process starting at

0. *This decomposition is unique up to equivalence. If A_n is increasing then X_n is a submartingale.*

Integration in Discrete Time. Let (D_n) and (F_n) be real-valued stochastic processes, then the integral of F wrt D, or the **transform** of D by F is

$$X = \int F \, \mathrm{d}D := D_0 F_0 + (D_1 - D_0)F_1 + \cdots + (D_n - D_{n-1})F_n \quad \text{for all} \quad n \in \mathbb{N}$$

Remark. It's difficult to define this **Lebesgue-Stieltjes** integral in continuous time, because the restriction of bounded variation is usually violated.

20.2.2 Transform of Martingale. *Let F be a **bounded predictable** process, and $X = \int F \, \mathrm{d}M$ where M is a martingale then so is X. If M is a submartingale and F is **nonnegative** then X is a submartingale as well.*

Process Stopped. Let T be a random time, given a stochastic process $(D_n)_{n\in\mathbb{N}}$, define the process D stopped at T be

$$X_n(\omega) := D_{n\wedge T(\omega)}(\omega)$$

Corollary 20.2.2.1 (Martingale Stopped)**.** *If T is a **stopping time**, X is a martingale M stopped at T, then X is a martingale. If M is a submartingale, then X is a submartingale.*

Proof. $X = \int F \, \mathrm{d}M$ where $F = \mathbf{1}_{[0,T]}$ which is bounded, predictable, nonnegative. □

20.2.2 Doob's Stopping, Upcrossing, Maxima

20.2.3 Doob's Stopping Theorem. *Let M be **adapted** to $\mathcal{F}$, then TFAE:*

1. *M is a submartingale*
2. *For every **bounded** stopping times S, T with $S \leq T$, the random variables M_S and M_T are integrable and*
$$E_S(M_T - M_S) \geq 0$$
3. *For every **bounded** stopping times S, T with $S \leq T$, the random variables M_S and M_T are integrable and*
$$E(M_T - M_S) \geq 0$$

Number of Upcrossings. Let M be adapted to $\mathcal{F}$, fix $a, b \in \mathbb{R}$ with $a < b$, for each integer $k \geq 1$ define

$$S_k = \inf\{n > T_{k-1} : M_n \leq a\}, \quad T_k = \inf\{n > S_k : M_n > b\}$$

where $S_1, S_2, \ldots$ is the **downcrossing times**, $T_1, T_2, \ldots$ are **upcrossing times**, they are stopping times. The number of upcrossings during $[0, n]$ is

$$U_n(a, b) = \sum_{k=1}^{n} \mathbf{1}_{(0,n]} \circ T_k$$

20.2.4 Upcrossing Inequality. *Suppose M is a submartingale then*

$$(b - a)EU_n(a, b) \leq E\left[(M_n - a)^+ - (M_0 - a)^+\right]$$

20.2.5 Maxima Inequality.

20.2.3 Martingale Convergence

20.2.6 Martingale Convergence. *Let X be a submartingale, suppose that*

$$\sup_n EX_n^+ < \infty \tag{20.1}$$

*then the sequence $(X_n)_{n\in\mathbb{N}}$ converges **almost surely** to an integrable random variable.*

20.2.7 Convergence and Uniform Integrability. *Let X be a submartingale, X converges **almost surely and in** L^1 iff it's **uniformly integrable**. Moreover, if it is so, setting $X_\infty = \lim X_n$ extends X to a submartingale $\overline{X} = (X_n)_{n\in\overline{\mathbb{N}}}$.*

20.2.8 Hunt's Dominated Convergence Theorem. *Let (X_n) be dominated by an integrable random variable and suppose that $X_\infty = \lim X_n$ exists almost surely, then the sequence $(E_n X_n)$ converges to $E_\infty X_\infty$ almost surely and in L^1.*

20.3 Continuous Martingale

20.3.1 Definitions

Doob's Property. The process X is said to have the Doob property for (S, T) if S, T are stopping times with $S \leq T$, and X_S and X_T be well-defined and integrable and

$$X_S = E_S X_T$$

Doob Martingale. The process X is said to be a Doob Martingale on $[0, \iota]$ if ι is a stopping time and X has the Doob property for (S, T) for all stopping times S, T such that $0 \leq S \leq T \leq \iota$

20.3.2 Optional Stopping

20.3.3 Uniform Integrability

20.3.4 Semimartingale and Local Martingale

20.4 Martingale Characterisation

20.4.1 Of Wiener Process

Example. Sum of independent mean 0 random variables.

Product of independent mean 1 nonnegative random variables.

20.4.1 Martingale Characterisation of Markov Chain.

20.4.2 Martingale Characterisation of Wiener Process.

20.4.3 Martingale Characterisation of Poisson Pure-jump.

20.4.2 Of Poisson

Chapter 21

Gaussian Family

21.1 Gaussian Random Variable and Random Vectors

Gaussian Random Vector. A standard Gaussian Random Vector is a measurable function X from $(\Omega, \mathcal{H}, P) \to \mathbb{R}^d$, where $\mathbb{R}^d$ with inner product $\langle \cdot | \cdot \rangle$, if for any $u \in E$, $\langle u | X \rangle$ is a real-valued Gaussian random variable.

21.1.1 Proposition (PDF and c.f.). *Probability density function of $X \sim N(\mu, \Sigma)$ is*

$$f(x) = (2\pi)^{-\frac{d}{2}} \det(\Sigma)^{-\frac{1}{2}} e^{-\frac{1}{2}(x-\mu)'\Sigma^{-1}(x-\mu)}.$$

The characteristic function is

$$E\left[e^{i\langle u|X\rangle}\right] = e^{i\langle u|m_X\rangle} - \frac{1}{2} q_X(u)$$

21.1.2 Proposition (Conditional Distribution). *Let* $\begin{bmatrix} x \\ y \end{bmatrix} \sim N\left(\begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}, \begin{bmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy} \end{bmatrix}\right)$, *then*

$$x \mid y \sim N(\mu_{x|y}, \Sigma_{xx|y})$$

where $\mu_{x|y} = \mu_x + \Sigma_{xy}\Sigma_{yy}^{-1}(y - \mu_y)$ *and* $\Sigma_{xx|y} = \Sigma_{xx} - \Sigma_{xy}\Sigma_{yy}^{-1}\Sigma_{yx}$.

Chapter 22

Brownian Motion

Chapter 23

Poisson Random Measure

Chapter 24

Stochastic Calculus

Part V

Statistics and Econometrics

Chapter 25

Asymptotic Statistics

25.1 M-estimator and Z-estimator

25.1.1 Consistency

25.1.2 Asymptotic Normality

25.2 Limits of Experiments

Chapter 26

Time Series

26.1 Stationary Time Series

First we have the definition of complex-vector time series and stationarity.

Time Series, Stationarity. A time series is a doubly infinite sequence $(X_t : t \in \mathbb{Z})$, for each t, $X_t : (\Omega, \mathcal{H}, P) \to \mathbb{C}^d$. (X_t) is **stationary** if EX_t and $EX_{t+h}X_t$ are finite and independent of t, for all $h \in \mathbb{N}$. X_t is **strictly stationary** if the distribution of $(X_t, \ldots, X_{t+h})$ is independent of time t for every $h \in \mathbb{N}$.

Linear Filter. A **linear filter** is an operation $T : (X_t) \mapsto Y_t = \sum_{j \in \mathbb{Z}} \psi_j X_{t-j}$.

26.1.1 Dependence and Mixing Conditions

See Fan Local Polynomial book Lemma 6.1. Basically with α-mixing condition, the characteristic function of the dependent sample is close to independent sample.

26.1.2 Asymptotics for Stationary Time Series

26.1.3 Estimation of Time Series

26.1.4 Spectral Analysis

26.1.5 Time Series Regression

26.2 Nonstationary Time Series

26.3 Hidden Markov Model and State Space Form

26.3.1 State Space Form

In a state space model, we have unobserved underlying states $x_1, \ldots, x_t$ and associated obsrevations $y_1, \ldots, y_t$. If it's *linear*, then

$$y_t = A_t x_t + e_t$$
$$x_{t+1} = B_t x_t + \eta_t$$

The goal is to predict x_t and use the information to predict y_t. The most successful method is the **Kalman Filter** which consists of two steps. It's closely related to Gaussianity of the y_t.

26.3.2 Kalman Filter

26.3.3 Dynamic Conditional Score Model

Chapter 27

Regression

27.1 Least Squares

27.2 Kernel Smoothing and Local Polynomial Regression

27.3 Sieve Regression

27.4 LASSO

Part VI

Finance

This is my note on the **Asset Pricing**, **Empirical Finance** and related issues. The goal is to create a coherent framework of theoretical and empirical models. The main references are Duffie 2010 and Campbell et al. 1997.

Chapter 28

Asset Pricing

28.1 Discrete Time Asset Pricing

28.2 Continuous Time Asset Pricing

Chapter 29

Empirical Finance

29.1 Test the Efficient Market Hypothesis

29.2 Market Microstructure

See Easley and O'Hara 2003.

29.3 Testing Asset Pricing Models

29.4 Volatility Models

Volatility of return is important in portfolio and risk management. There are different ways to measure and model Volatility. It can be

1. Historical volatility calculated by squared or absolute returns.
2. Fitting parametric models suh as GARCH.
3. Implied volatility from option prices.
4. Realized volatility. Andersen, Bollerslev, Diebold, and Labys 2001, Andersen, Bollerslev, Diebold, and Ebens 2001

It is widely agreed that although daily and monthly financial asset returns are approximately unpredictable, return volatility is highly predictable, a phenomenon with important implications for financial economics and risk management.

There are ex ante and ex post measure of volatility.

Part VII

Toolbox

Chapter 30

Inequalities

30.1 Basic Inequalities of Numbers

30.1.1 Proposition (c_r-inequality).

$$E(|X+Y|^r) \leq c_r[E(|X|^r) + E(|Y|^r)]$$

where $c_r = 1$ *if* $0 < r \leq 1$, $c_r = 2^{r-1}$ *if* $r > 1$.

Proof. $f(x) = x^r$ is concave when $0 < r \leq 1$ and convex when $r > 1$. □

30.1.2 Arithmetic-Geometric Mean. *For any nonnegative numbers* $(x_i)_{i=1}^n$,

$$(\prod_{i=1}^n x_i)^{\frac{1}{n}} \leq \frac{1}{n}\sum_{i=1}^n x_i$$

30.2 Convex and Concave Inequalities

30.2.1 Proposition (Jensen's Inequality). *Let* $f : \mathbb{R} \to \mathbb{R}$ *be a convex function on* $[a, b]$, *then for* $t \in [0, 1]$:

$$f(ta + (1-t)b) \leq tf(a) + (1-t)f(b)$$

30.2.2 Proposition ($f(a+b)$). *The following holds*

1. *If* f *is concave and* $f(0) = 0$ *then* $f(a+b) \leq f(a) + f(b)$.

2. *If f is convex then $f(\frac{a+b}{2}) \leq \frac{1}{2}[f(a) + f(b)]$*

Proof. The first one can be shown by the Mean Value Theorem 9.4.5. □

30.3 Integration Inequalities

There are some important inequalities(hard analysis) about Lebesgue integral.

30.3.1 Proposition (Absolute value). *For an integrable function from measurable space $(X, \mathcal{E}, \mu)$ to $\mathbb{R}$ or $\mathbb{C}$,*

$$\left|\int f d\mu\right| \leq \int |f| d\mu$$

30.3.2 Holder Inequality. *If $1 < p < \infty$, $p^{-1} + q^{-1} = 1$ and $f \in L^p(X, \mathcal{E}, \mu)$ and $g \in L^q(X, \mathcal{E}, \mu)$, then $fg \in L^1(X, \mathcal{E}, \mu)$ and*

$$\left|\int fg \, \mathrm{d}\mu\right| \leq \int |fg| \, \mathrm{d}\mu \leq \|f\|_p \|g\|_q \tag{30.1}$$

30.3.3 Cauchy - Schwarz. *For any $f, g \in L^2(X, \mathcal{E}, \mu)$, we have $fg \in L^1$ and $\left|\int fg \, \mathrm{d}\mu\right| \leq \|f\|_2 \|g\|_2$*

30.3.4 Minkowski- Riesz. *For $1 < p \leq \infty$, if f and g are in L^p then $f + g \in L^p$ and $\|f + g\| \leq \|f\|_p + \|g\|_q$*

A very useful result is the **Jensen's inequality**:

30.3.5 Jensen's Inequality. *Let $\Phi : \mathbb{R} \to \mathbb{R}$ be a convex function, then Φ is **measurable** and for measurable $X : (\Omega, \mathcal{E}, P) \to \mathbb{R}$, if X is **integrable**, then*

$$P\Phi(X) \geq \Phi(PX)$$

30.4 Matrix Inequalities

30.4.1 Hadamard's Inequality. *This inequality relates the determinants to the product of sums of rows. If B is nonsingular real square $n \times n$ matrix,*

$$\left|B\right|^2 \leq \prod_{i=1}^{n} \left(\sum_{k=1}^{n} b_{ik}^2 \right)$$

30.4.2 Positive Semi-Definitedness of Moment Matrix. *Let μ_r be the rth moment of a random variable, then the matrix (μ_{i+j-2}) is positive semidefinite. This condition is necessary but not sufficient.*

Proof. See Rao et al. 1973. □

30.4.1 Extreme Values of Quadratic Forms

30.4.3 Supreme of Quadratic Form.

30.4.2 Eigenvalue Inequalities

30.4.4 Lemma (Sturmian Separation Theorem).

30.4.5 Lemma (Poincare Separation Theorem).

30.4.6 Lemma (Interlacing Inequality).

30.4.3 Minimization of Trace

30.5 Probability Inequalities

30.5.1 Information Theory Inequality

See Rao et al. 1973 p.58.

30.5.2 Moment Inequality

30.5.1 Bukholder's Inequality.

30.5.3 Concentration Inequalities

30.5.2 Lemma (Bernstein's Inequality). *Let $X_1, \ldots, X_n$ be independent random variables, with* $\mathrm{var}(X_i) = \sigma_i^2$ *and* $|X_i - E(X_i)| \leq M_i$.

$$M = \max(M_i) \quad S = \sum_{i=1}^{n} X_i \quad \textit{and} \quad \sigma^2 = \sum_{i=1}^{n} \sigma_i^2$$

For $t > 0$, wlog assume $E(X_i) = 0$, we have

$$P(S \geq t\sigma) < \exp\left(-\frac{t^2}{2 + \frac{2}{3}\frac{M}{\sigma}t}\right)$$

Proof. See Bennett 1962. We observe the following:

[1]. $E\left[e^{cX_i}\right] \leq e^{\frac{1}{2}c^2\sigma_i^2 F_i}$ where $F_i = \sum_{r=2}^{\infty} \frac{c^{r-2}EX_i^r}{\frac{1}{2}r!\sigma_i^2}$.

[2]. $P(\sum X_i \geq t\sigma) \leq \exp\left(\frac{1}{2}c^2\sigma^2 F - ct\sigma\right)$ where $F = \max F_i$. We choose $c = \frac{t}{\sigma F}$ to minimise the RHS.

[3]. If $|X_i| \leq M_i$ then the absolute moment $\nu_r = E(|X_i|^r) \leq \sigma_i^2 M^{r-2}$, and hence

$$F_i = \sum_{r=2}^{\infty} \frac{c^{r-2}EX_i^r}{\frac{1}{2}r!\sigma_i^2} \leq \sum_{r=2}^{\infty} \frac{c^{r-2}M^{r-2}}{\frac{1}{2}r!} \leq \sum_{r=2}^{\infty} c^{r-2}\left(\frac{M}{3}\right)^{r-2} = \frac{1}{1 - cM/3}$$

provided that $cM/3 < 1$ and use $c = \frac{t}{\sigma F}$, solve for c. Plug in to the original expresion. □

30.6 Approximations

30.6.1 Proposition (Stirling's Approximation to Factorials). *In the limit case:*

$$\lim_{n\to\infty} \frac{n!}{\sqrt{2\pi n}n^n e^{-n}} = 1$$

In finite sample, for some $\omega(n) : (n + \frac{1}{2})^{-1} < \omega(n) < n^{-1}$, we have

$$n! = \sqrt{2\pi n}n^n e^{-n} e^{\omega(n)/12}$$

Chapter 31

Matrix Lemmas

31.1 Eigenstructures of Special Matrices

31.1.1 Proposition (Tri-diagonal Matrix). *Let A be a three-diagonal matrix with a on the main diagonal and b on the sub- and super-diagonal, then the eigenvalues and eigenvectors are:*

Proof. See Onatski and Wang 2018b appendix 1. □

31.1.2 Proposition (Skew-symmetric Matrix). *See Bai 2010 random matrix book.*

Chapter 32

Useful Arguments

32.1 Arguments in Probability

32.1.1 Monotone Class Argument

Monotone Class Theorem provides a way to show certain collections of subsets are σ-algebra.

π-system. A collection Π of subsets of E is a π-system if it's closed under finite intersections.

λ-system. A collection $\mathcal{D}$ of subsets of E is a λ-system on E if

1. $E \in \mathcal{D}$
2. $A, B \in \mathcal{D}$ and $A \supset B \implies A \setminus B \in \mathcal{D}$
3. $(A_n) \subset \mathcal{D}$ and $A_n \uparrow A \implies A \in \mathcal{D}$

Monotone Class. $\mathcal{C} \subset 2^X$ is a **monotone class** if it's closed under countable increasing unions and countable decreasing intersections.

32.1.1 Monotone Class Lemma. *If $\mathcal{A}$ is an algebra on X, then the monotone class $\mathcal{C}$ generated by $\mathcal{A}$ is the σ-algebra $\mathcal{E}$ generated by $\mathcal{A}$.*

32.1.2 Proposition (Characterisation of σ-algebra). *A collection of subsets $\mathcal{E}$ is a σ-algebra on E iff it's both a π-system and a λ-system on E.*

32.1.3 **$\pi - \lambda$ Theorem.** *If $\mathcal{D}$ is a λ-system, $\mathcal{E}$ is a π-system and $\mathcal{E} \subset \mathcal{D}$, then $\sigma(\mathcal{E}) \subset \mathcal{D}$.*

Proof. First we prove the lemma:

32.1.4 **Lemma.** *Let $\mathcal{D}$ be a λ-system on X, fix $D \in \mathcal{D}$,*

$$\hat{\mathcal{D}} := \left\{A \in \mathcal{D} : A \bigcap D \in \mathcal{D}\right\}$$

is a λ-system

proof of lemma. We check the conditions of λ-system:

1. $E \in \hat{\mathcal{D}}$ because $E \in \mathcal{D}$ and $E \bigcap D = D \in \mathcal{D}$.
2. Let $A, B \in \hat{\mathcal{D}}$, $B \subset A$, then $(A \backslash B) \in \mathcal{D}$ and $(A \backslash B) \bigcap D = (A \bigcap D) \backslash (B \bigcap D) \in \mathcal{D}$.
3. $\bigcup A_n \bigcap D = \bigcup (A_n \bigcap D) \in \mathcal{D}$.

Hence $\hat{\mathcal{D}}$ is a λ-system □

With this lemma, let Π be a π-system, and $\mathcal{D}$ be the smallest λ-system that includes Π. We want to show that it's also a π-system, hence includes the smallest σ-algebra $\sigma(\Pi)$.

We claim: $A \bigcap B \in \mathcal{D}$ for any $A, B \in \mathcal{D}$. For any $A \in \mathcal{D}$, $\hat{\mathcal{D}}_A := \{B \in \mathcal{D} : A \bigcap B \in \mathcal{D}\}$ is a λ-system, if it includes Π, it includes $\mathcal{D}$, and hence for any $B \in \mathcal{D}$, $A \bigcap B \in \mathcal{D}$.

So it suffice to show $\Pi \subset \hat{\mathcal{D}}_A$, that is for any $A \in \mathcal{D}$ and $P \in \Pi$, $A \bigcap P \in \mathcal{D}$. For any $A \in \mathcal{D}$. For any $P \in \Pi$, $P \in \mathcal{D}$, and $Q \bigcap P \in \mathcal{D}$ for any $Q \in \Pi$. $\hat{\mathcal{D}}_P$ must include Π and hence $\mathcal{D}$. □

32.1.5 **Monotone Class Theorem for Algebra.**

32.1.2 Monotone Class of Numerical Functions

Monotone Class. A collection $\mathcal{M}$ of numerical functions on $(E, \mathcal{E})$ is a **monotone class** if

1. The constant function $1 \in \mathcal{M}$
2. $f, g \in \mathcal{M}_b$ and $a, b \in \mathbb{R} \implies af + bg \in \mathcal{M}$

3. $(f_n) \subset \mathcal{M}_+, f_n \uparrow f \implies f \in \mathcal{M}$

32.1.6 Monotone Class Theorem. *Let $\mathcal{M}$ be a monotone class of functions on $(E, \mathcal{E})$, if for some p-system $\mathcal{P}$ generating $\mathcal{E}$, that $1_A \in \mathcal{M}$ for every $A \in \mathcal{P}$ then $\mathcal{M}$ includes all* ***nonnegative*** *$\mathcal{E}$-measurable functions and all* ***bounded*** *$\mathcal{E}$-measurable functions.*

Proof. Consider $\mathcal{D} = \{A \in \mathcal{E} : 1_A \in \mathcal{M}\}$, show it's a d-system. □

32.1.3 Borel-Cantelli Argument for Almost Sure Convergence

These results are useful when dealing with **independence and almost surely convergence**. In particular, it's used to prove the strong law of large numbers.

32.1.7 Lemma. *If $0 \le p_n < 1$ for all n, then*

$$\prod_n (1 - p_n) = 0 \iff \sum_n p_n = \infty$$

32.1.8 Borel-Cantelli Lemma. *We have two results:*

1. *If A_n are any events, then*

$$\sum_n P(A_n) < \infty \implies P(\limsup A_n) = 0 \quad \textit{that is} \quad \sum_n^\infty A_n < \infty \quad a.s.$$

2. *If A_n are independent, then*

$$\sum_n P(A_n) = +\infty \implies P(\limsup A_n) = 1 \quad \textit{that is} \quad \sum_n^\infty A_n = \infty \quad a.s.$$

Proof. For the first part, for any $m > 0$, we have

$$P(\limsup A_n) \le P\left(\bigcup_{n \ge m} A_n\right) \le \sum_{n \ge m} P(A_n)$$

And we also have

$$\lim_m \sum_{n \ge m} P(A_n) = 0 \impliedby \sum_n P(A_n) < \infty$$

□

32.1.9 Bonferroni Theorem. *For any events $A_i := A(i)$,*

$$P\left(\bigcup_{i=1}^{n} A_i\right) \geq \sum_{i=1}^{n} P(A_i) - \sum_{1 \leq i < j \leq n} P(A_i \bigcap A_j)$$

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